# CHAPTER 98 — WIRING DIAGRAMS

## CONTENTS — MAINTENANCE PROCEDURES


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# WIRING DIAGRAMS

WIRING DIAGRAMS

## 98-1. WIRING DIAGRAMS.

This chapter contains electrical wiring diagrams and component replacement charts. This information is provided to assist maintenance personnel in understanding components and circuit operations, to perform maintenance, and troubleshooting of electrical and avionics systems installed in the SAR helicopter.

## 98-2. WIRING IDENTIFICATION.

All wires on the wiring diagrams are identified by codes exactly as they are marked in the helicopter. The code or wire number indicates the circuit function, wire number, wire segment letter and the wire size of the conductor. Identification of wires may be accomplished by using the wiring identification code figure in figure 98-1 .

## 98-3. USE OF UNDERLINED CAPITAL LETTERS.

Many letters have similar upper case and lower case forms. To eliminate this possible source of error in identifying connector pin designations, underlined upper case (Capital) letters (A, B, C, etc) are used to signify the lower case pin letters (a, b, c, etc).

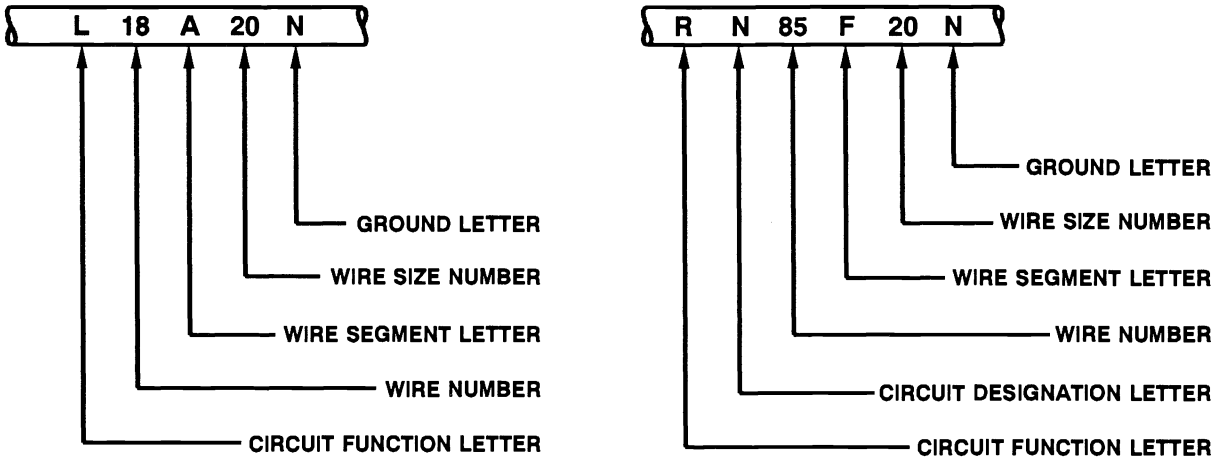
## 98-4. USE OF ZONES.

Some wiring diagrams are several sheets long. To facilitate tracking wires between the sheets, zone numbers and section letters (A, B, C, and D) have been added to the sheets. Wires going to another sheet are identified ie., Z 123B.

## 98-5. COMPONENT REPLACEMENT DIAGRAMS.

Some components are not shown complete on some figures. If a component has to be replaced, all wire numbers and connections are required. The component replacement diagram is used to show a complete component with the wire numbers and connections.

## WIRING IDENTIFICATION CODE




## CIRCUIT FUNCTIONS

| CODE | NOMENCLATURE | CODE | NOMENCLATURE |
|---|---|---|---|
| A | ARMAMENT | RC | COMMAND |
| C | CONTROL SURFACES | RF | VHF LIAISON |
| D | INSTRUMENTS (OTHER THAN FLIGHT OR ENGINE) | RL | LIAISON |
| E | SINGLE INSTRUMENTS | RM | MARKER BEACON |
| F | FLIGHT INSTRUMENTS | RN | NAVIGATION |
| H | HEATING, VENTILATING AND DE-ICING | RU | UHF COMMAND |
| J | IGNITION | RV | VHF COMMAND |
| K | ENGINE CONTROL | RZ | INTERPHONE AND HEADPHONE |
| L | LIGHTING | S | RADAR |
| M | MISCELLANEOUS ELECTRIC | SX | RECOGNITION (IFF) |
| N | GROUND | TN | TRACKING NAVIGATION |
| P | DC POWER | V | DC POWER AND DC CONTROL CABLES FOR AC SYSTEM |
| Q | FUEL AND OIL | W | WARNING AND EMERGENCY |
| R | RADIO (NAVIGATION AND COMMUNICATION) | X | AC POWER |

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COOOO

**Figure 98-1. Wiring identification**

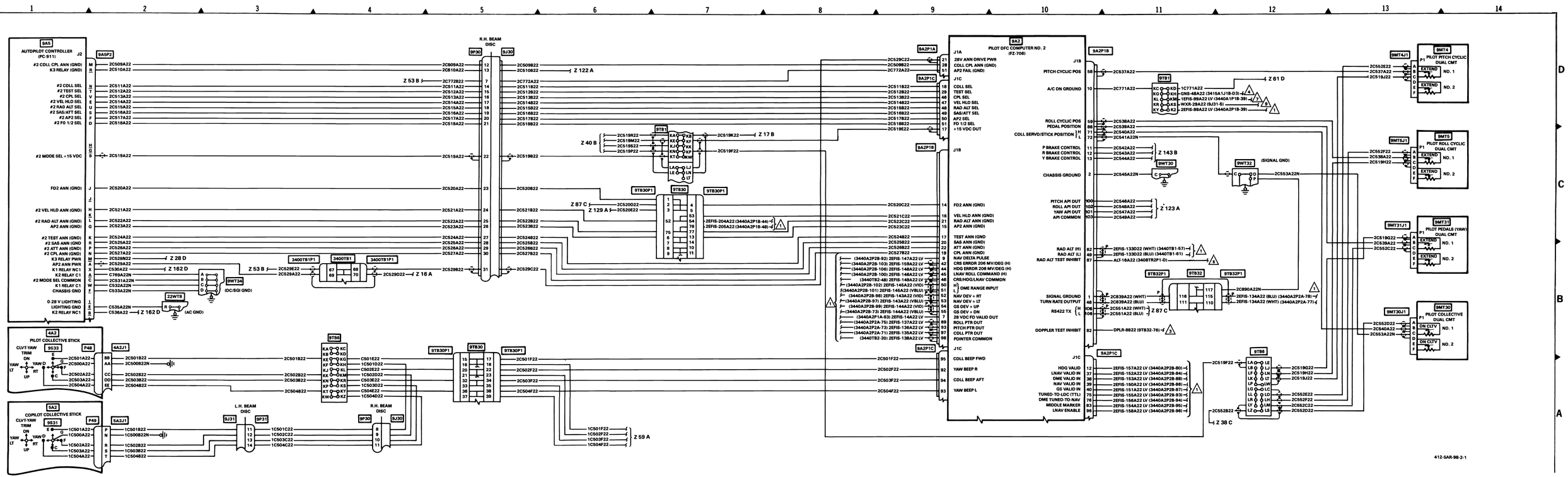
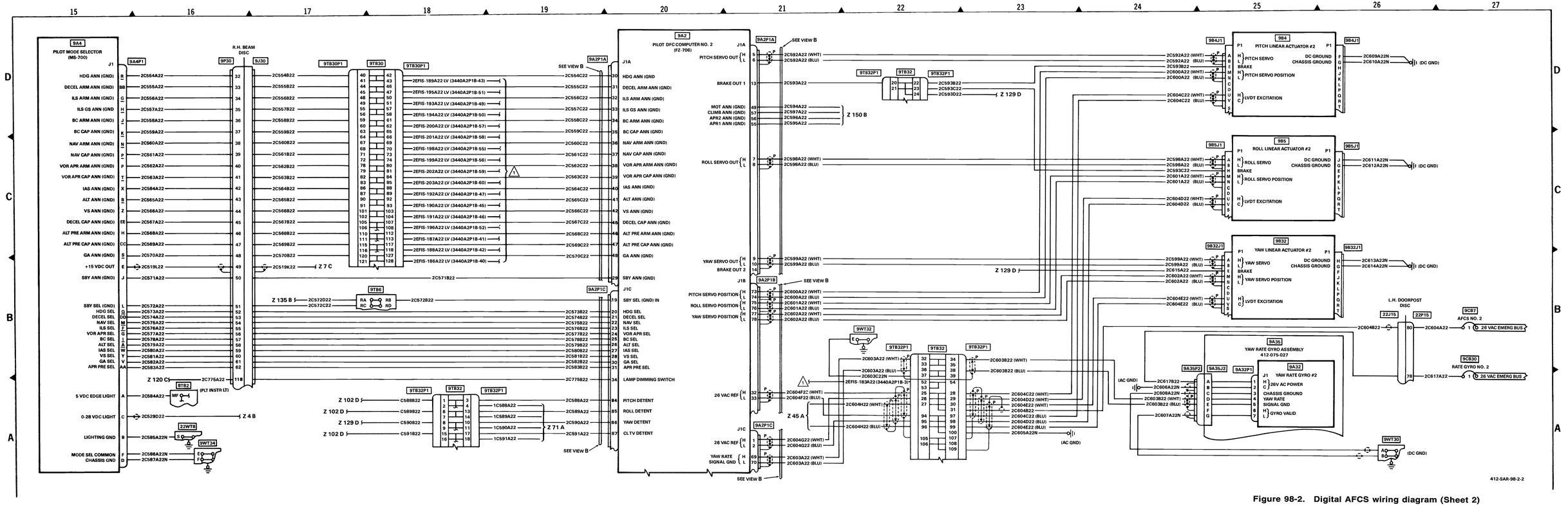
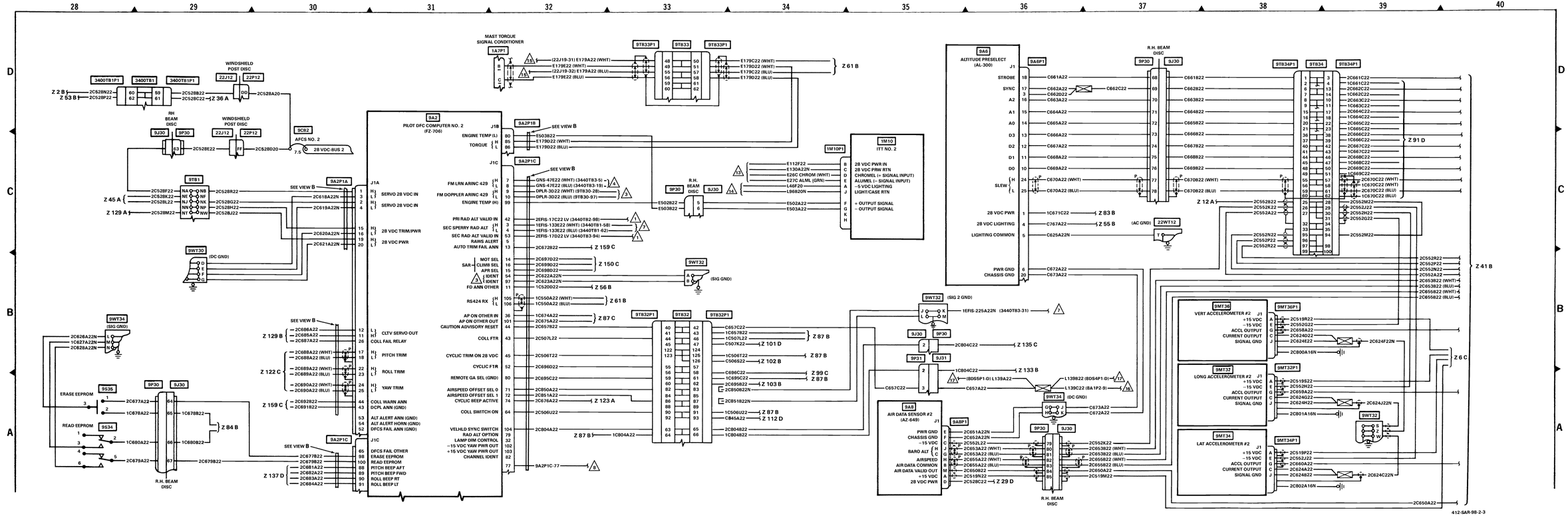



Figure 98-2. Digital AFCS wiring diagram (Sheet 1 of 13)


9A4
PILOT MODE SELECTOR (MS-700)
9A2
PILOT DFC COMPUTER NO. 2 (FZ-706)
R.H. BEAM DISC
9P30
9J30
9T30
9T32
9T6
9T8
9T34
22WT8
9A2P1A
9A2P1B
9A2P1C
9T32P1
9T30P1
9T8P2
9T32P2
9T8P1
9T8P2
9T32P2
9T32P1
9T8P1
9T8P2
9T32P2
9T32P1
9T8P1
9T8P2
9T32P2
9T32P1


28
29
30
31
32
33
34
35
36
37
38
39
40
D
C
B
A
9A2 PILOT DFC COMPUTER NO. 2 (FZ-706)
9A6 ALTITUDE PRESELECT (AL-300)
9A8 AIR DATA SENSOR #2 (AZ-649)
1M10 ITT NO. 2
MAST TORQUE SIGNAL CONDITIONER
WINDSHIELD POST DISC
RH BEAM DISC
R.H. BEAM DISC
AFCS NO. 2
28 VDC-BUS 2
ENGINE TEMP (L)
TORQUE
ENGINE TEMP (R)
FM LRN ARINC 429
FM DOPPLER ARINC 429
PRI RAD ALT VALID IN
SEC SPERRY RAD ALT
SEC RAD ALT VALID IN
RAWS ALERT
AUTO TRIM FAIL ANN
MOT SEL
SAR
CLIMB SEL
APR SEL
IDENT
FD ANN OTHER
RS424 RX
AP ON OTHER IN
AP ON OTHER OUT
CAUTION ADVISORY RESET
COLL FTR
CYCLIC TRIM ON 28 VDC
CYCLIC FTR
REMOTE GA SEL (GND)
AIRSPEED OFFSET SEL 0
AIRSPEED OFFSET SEL 1
CYCLIC BEEP ACTIVE
COLL SWITCH ON
VELHLD SYNC SWITCH
RAD ALT OPTION
LAMP DIM CONTROL
−15 VDC YAW PWR OUT
+15 VDC YAW PWR OUT
CHANNEL IDENT
SERVO 28 VDC IN
SERVO 28 VDC IN
28 VDC TRIM/PWR
28 VDC PWR
CLTV SERVO OUT
COLL FAIL RELAY
PITCH TRIM
ROLL TRIM
YAW TRIM
COLL WARN ANN
DCPL ANN (GND)
ALT ALERT ANN (GND)
ALT ALERT HORN (GND)
DFCS FAIL ANN (GND)
DFCS FAIL OTHER
ERASE EEPROM
READ EEPROM
PITCH BEEP AFT
PITCH BEEP FWD
ROLL BEEP RT
ROLL BEEP LT
ERASE EEPROM
READ EEPROM
28 VDC PWR IN
28 VDC PRW RTN
CHROMEL (+ SIGNAL INPUT)
ALUMEL (− SIGNAL INPUT)
− 5 VDC LIGHTING
LIGHT/CASE RTN
+ OUTPUT SIGNAL
− OUTPUT SIGNAL
STROBE
SYNC
A2
A1
A0
D3
D2
D1
D0
SLEW
28 VDC PWR
28 VDC LIGHTING
LIGHTING COMMON
PWR GND
CHASSIS GND
PWR GND
CHASSIS GND
BARO ALT
AIRSPEED
AIR DATA COMMON
AIR DATA VALID OUT
+15 VDC
28 VDC PWR
VERT ACCELEROMETER #2
LONG ACCELEROMETER #2
LAT ACCELEROMETER #2
+15 VDC
−15 VDC
ACCL OUTPUT
CURRENT OUTPUT
SIGNAL GND
Z 28 B
Z 53 B
Z 36 A
Z 45 A
Z 129 A
Z 129 B
Z 122 C
Z 159 C
Z 137 D
Z 84 B
Z 61 B
Z 87 C
Z 150 C
Z 56 B
Z 159 C
Z 123 A
Z 87 B
Z 101 D
Z 102 B
Z 103 B
Z 87 B
Z 99 C
Z 87 B
Z 112 D
Z 29 D
Z 135 C
Z 133 B
Z 83 B
Z 55 B
Z 12 A
Z 91 D
Z 41 B
Z 6 C
(AC GND)
(DC GND)
(SIG GND)
(SIG 2 GND)
412-SAR-98-2-3

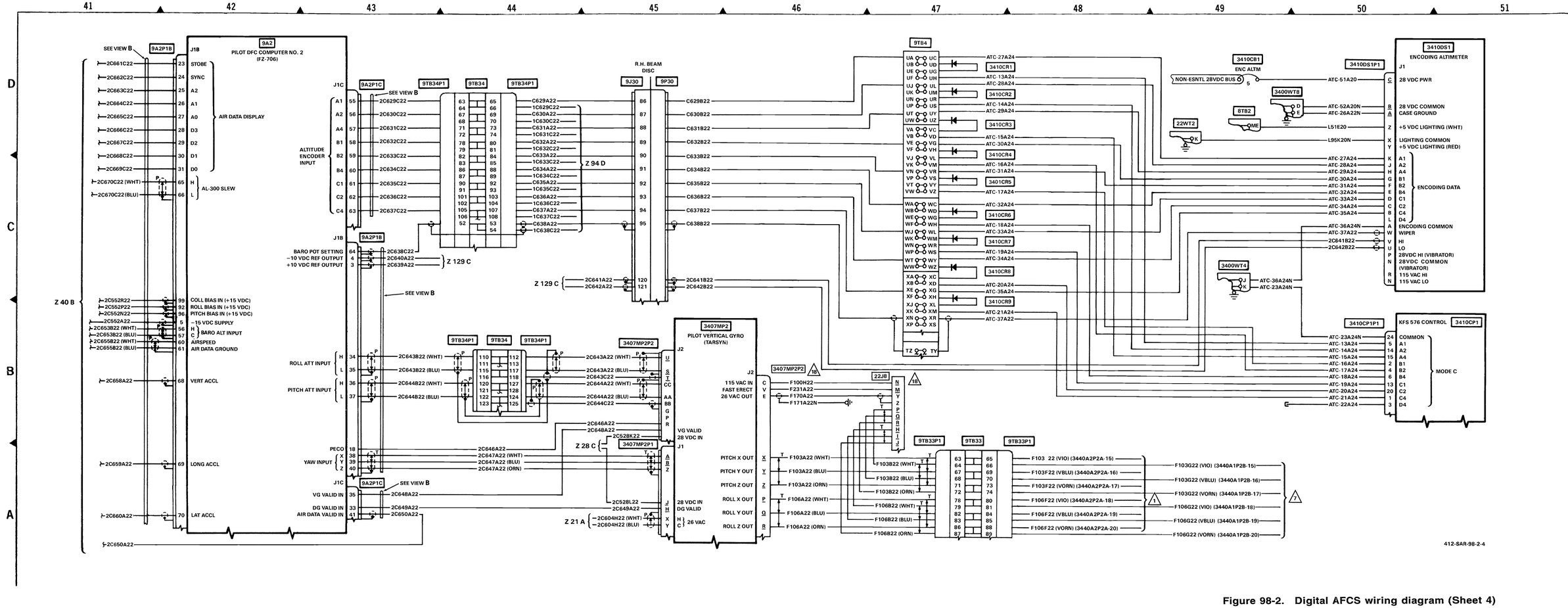



Figure 98-2. Digital AFCS wiring diagram (Sheet 4)

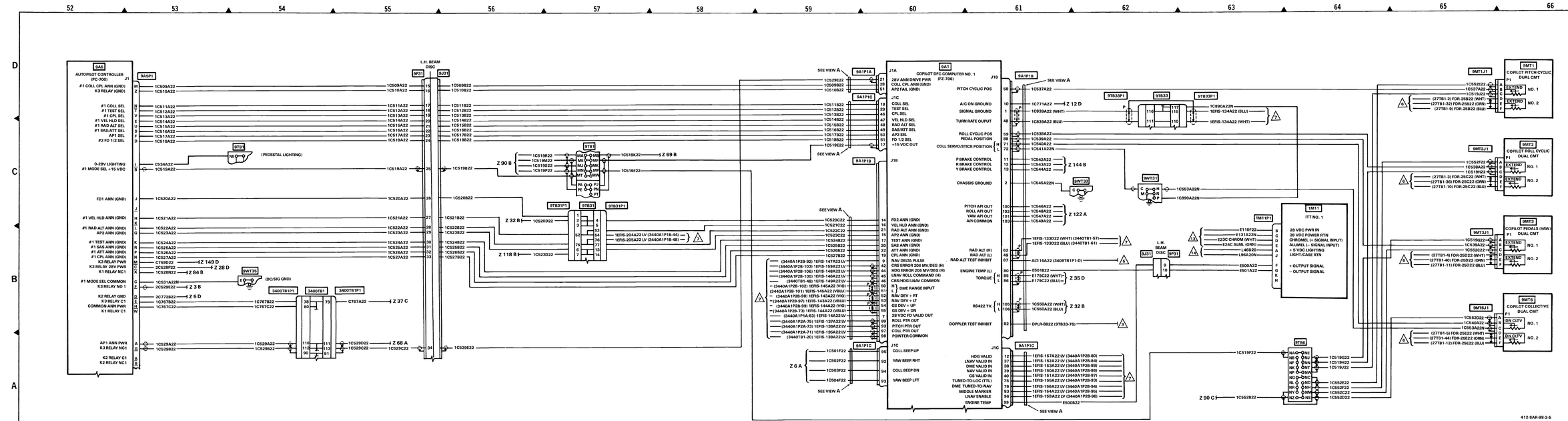



**Figure 98-2. Digital AFCS wiring diagram (Sheet 5)**

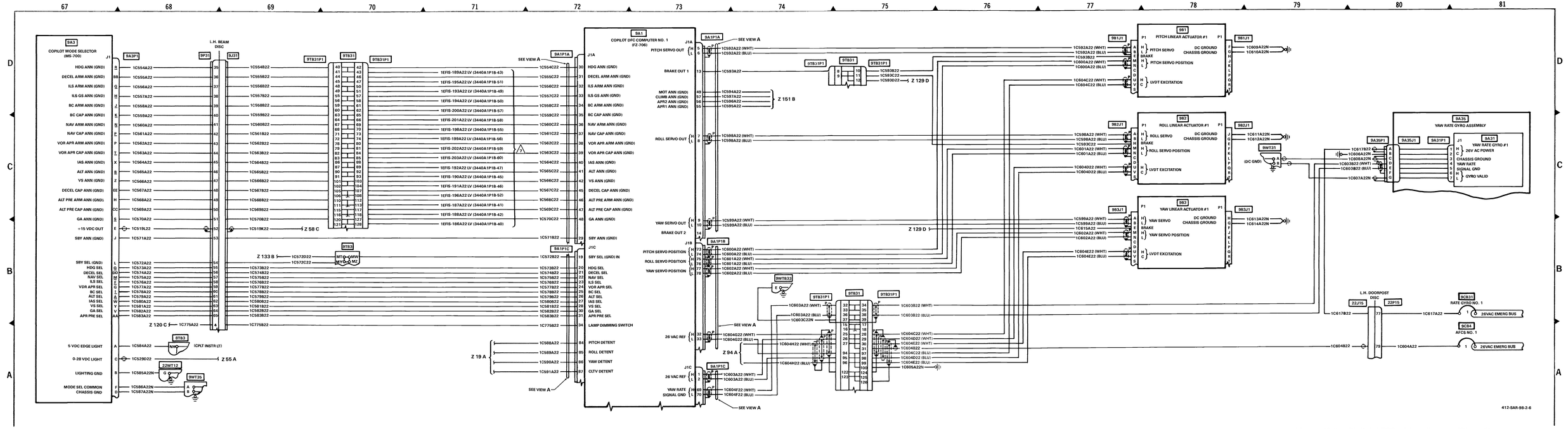
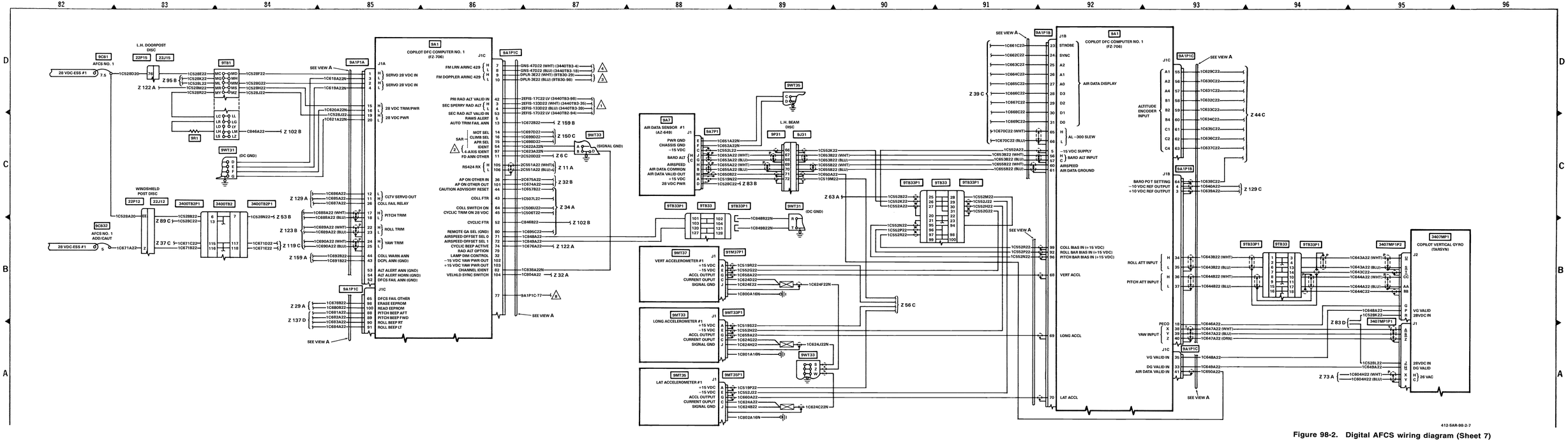



Figure 98-2. Digital AFCS wiring diagram (Sheet 6)




412-SAR-98-2-7

Figure 98-2. Digital AFCS wiring diagram (Sheet 7)

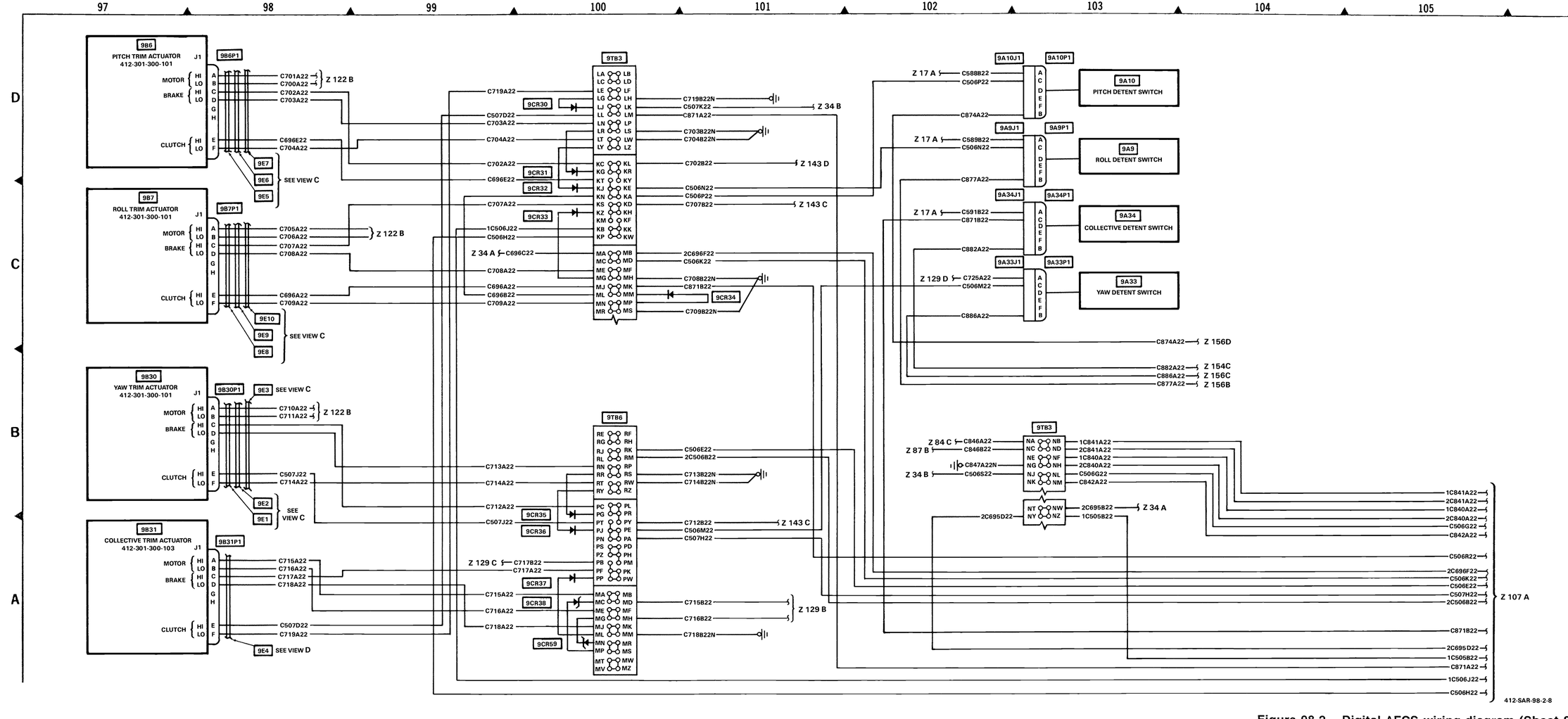



Figure 98-2. Digital AFCS wiring diagram (Sheet 8)

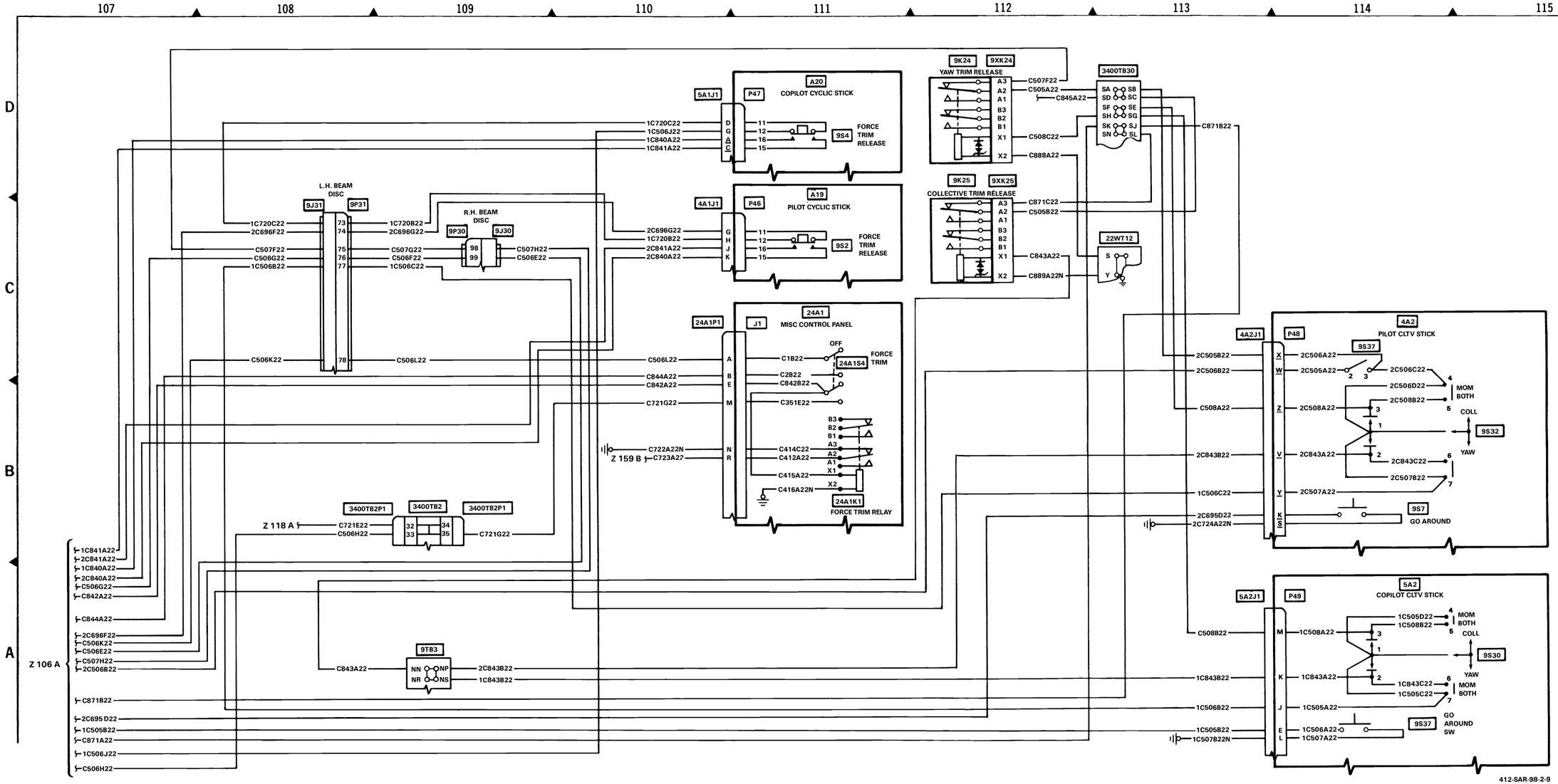



Figure 98-2. Digital AFCS wiring diagram (Sheet 9)

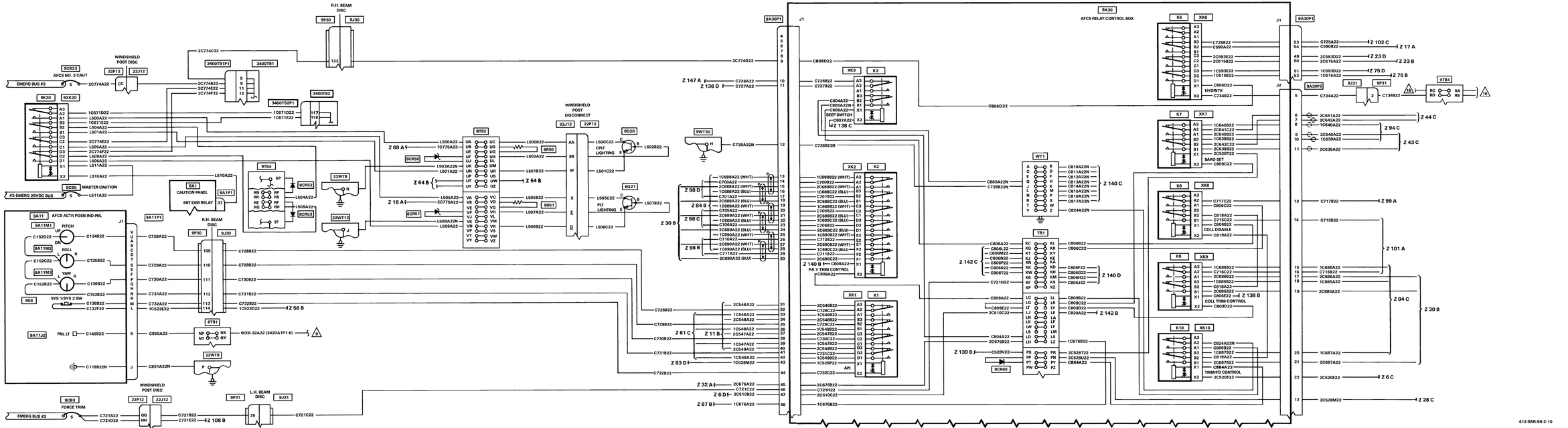



Figure 98-2. Digital AFCS wiring diagram (Sheet 10)

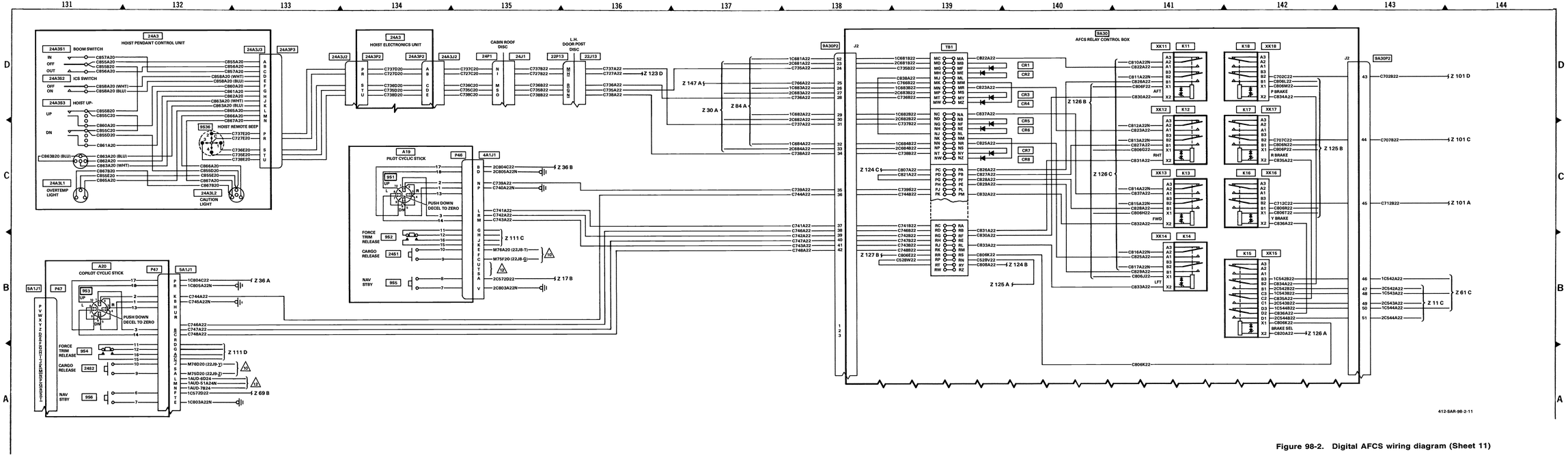



Figure 98-2. Digital AFCS wiring diagram (Sheet 11)

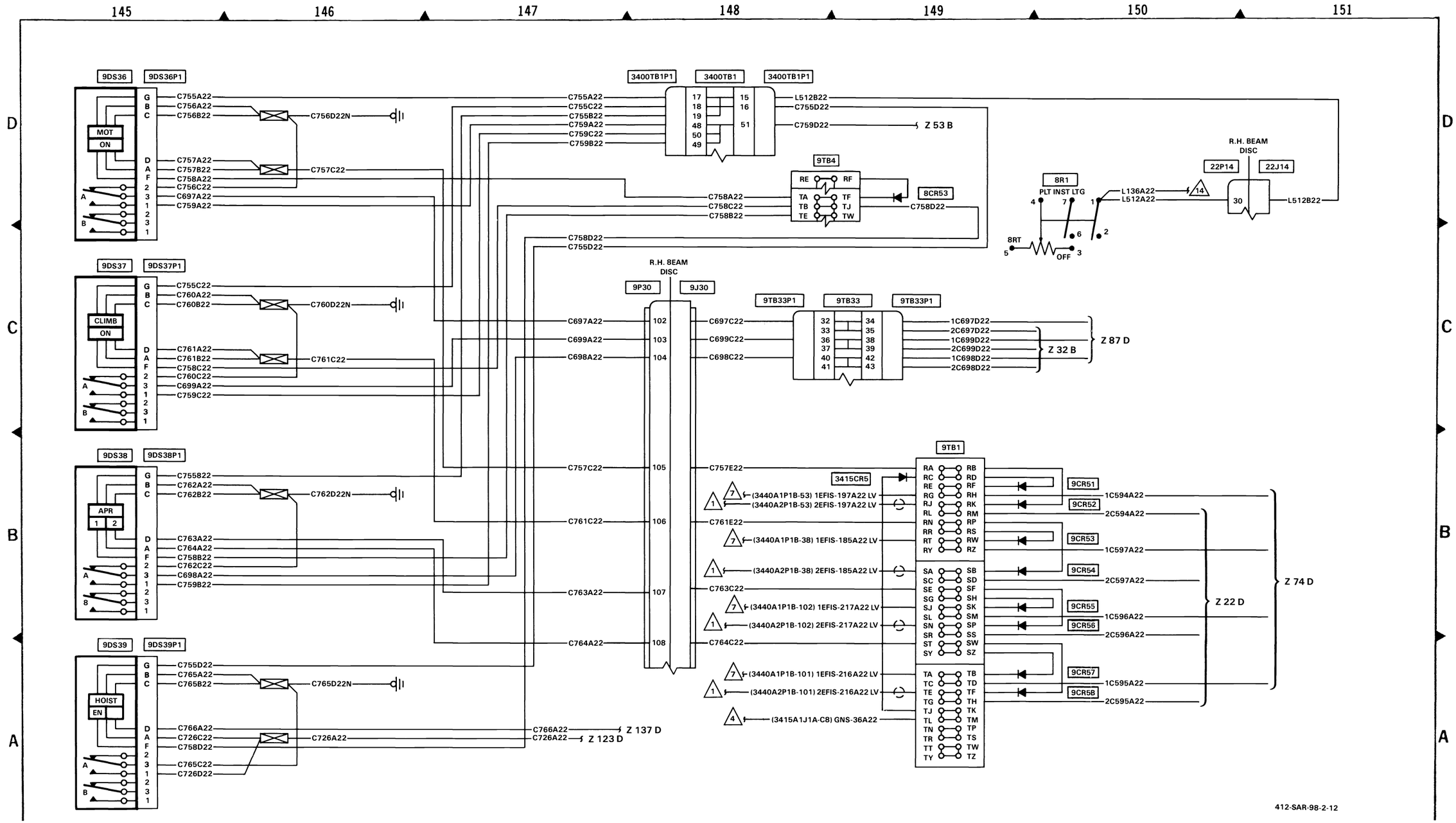
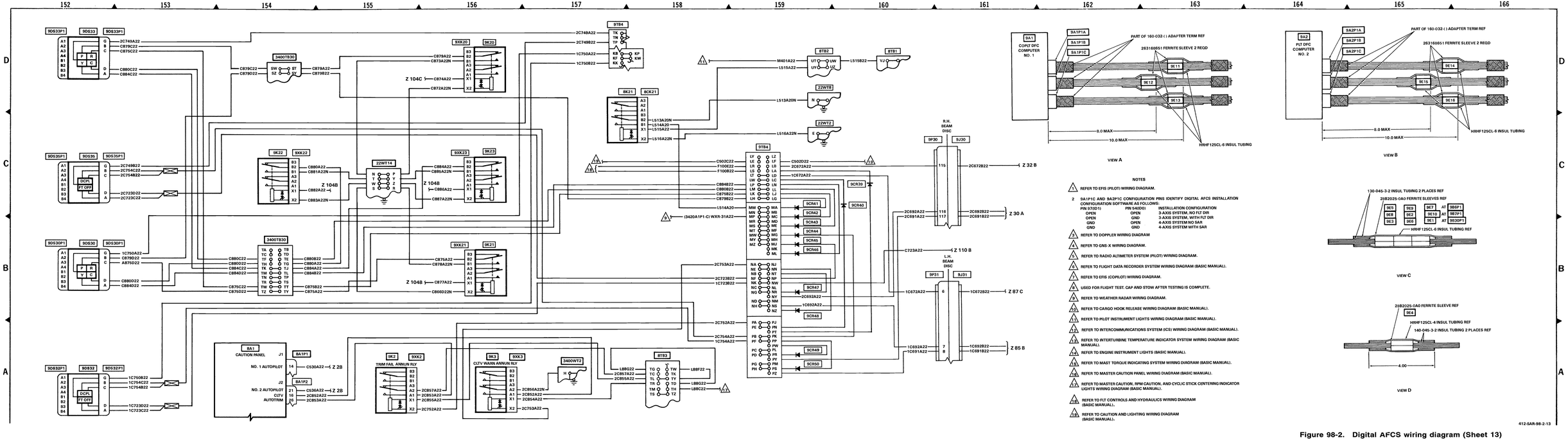



**Figure 98-2. Digital AFCS wiring diagram (Sheet 12)**




NOTES

1. REFER TO EFIS (PILOT) WIRING DIAGRAM.
2. 9A1P1C AND 9A2P1C CONFIGURATION PINS IDENTIFY DIGITAL AFCS INSTALLATION CONFIGURATION SOFTWARE AS FOLLOWS:

| PIN 57(D1) | PIN 54(D0) | INSTALLATION CONFIGURATION |
|---|---|---|
| OPEN | OPEN | 3-AXIS SYSTEM, NO FLT DIR |
| OPEN | GND | 3-AXIS SYSTEM, WITH FLT DIR |
| GND | OPEN | 4-AXIS SYSTEM NO SAR |
| GND | GND | 4-AXIS SYSTEM WITH SAR |

3. REFER TO DOPPLER WIRING DIAGRAM
4. REFER TO GNS-X WIRING DIAGRAM.
5. REFER TO RADIO ALTIMETER SYSTEM (PILOT) WIRING DIAGRAM.
6. REFER TO FLIGHT DATA RECORDER SYSTEM WIRING DIAGRAM (BASIC MANUAL).
7. REFER TO EFIS (COPILOT) WIRING DIAGRAM.
8. USED FOR FLIGHT TEST. CAP AND STOW AFTER TESTING IS COMPLETE.
9. REFER TO WEATHER RADAR WIRING DIAGRAM.
10. REFER TO CARGO HOOK RELEASE WIRING DIAGRAM (BASIC MANUAL).
11. REFER TO PILOT INSTRUMENT LIGHTS WIRING DIAGRAM (BASIC MANUAL).
12. REFER TO INTERCOMMUNICATIONS SYSTEM (ICS) WIRING DIAGRAM (BASIC MANUAL).
13. REFER TO INTERTURBINE TEMPERATURE INDICATOR SYSTEM WIRING DIAGRAM (BASIC MANUAL).
14. REFER TO ENGINE INSTRUMENT LIGHTS (BASIC MANUAL).
15. REFER TO MAST TORQUE INDICATING SYSTEM WIRING DIAGRAM (BASIC MANUAL).
16. REFER TO MASTER CAUTION PANEL WIRING DIAGRAM (BASIC MANUAL).
17. REFER TO MASTER CAUTION, RPM CAUTION, AND CYCLIC STICK CENTERING INDICATOR LIGHTS WIRING DIAGRAM (BASIC MANUAL).
18. REFER TO FLT CONTROLS AND HYDRAULICS WIRING DIAGRAM (BASIC MANUAL).
19. REFER TO CAUTION AND LIGHTING WIRING DIAGRAM (BASIC MANUAL).

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**Figure 98-2. Digital AFCS wiring diagram (Sheet 13)**

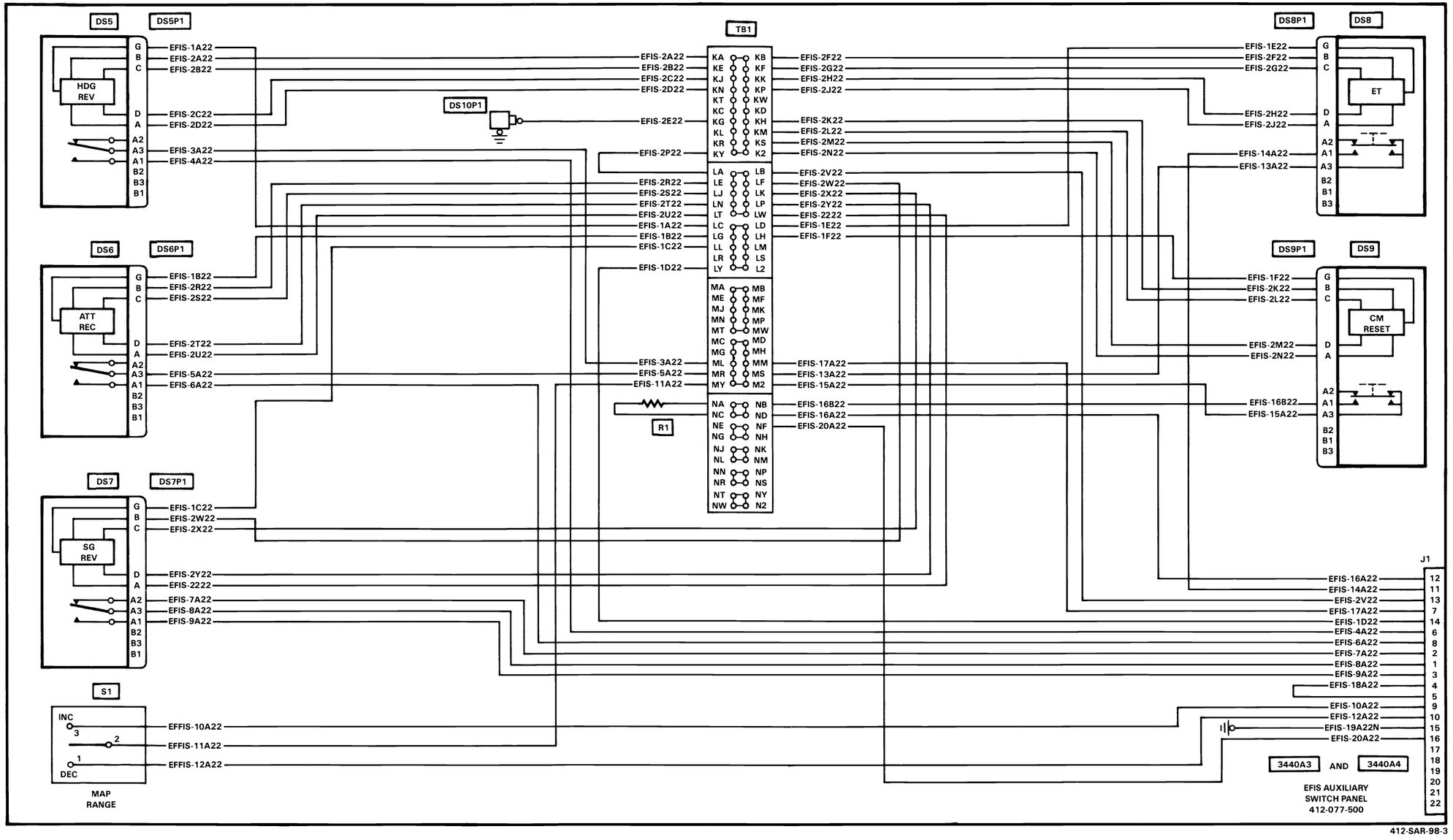



**Figure 98-3. EFIS AUX switch panel wiring diagram**

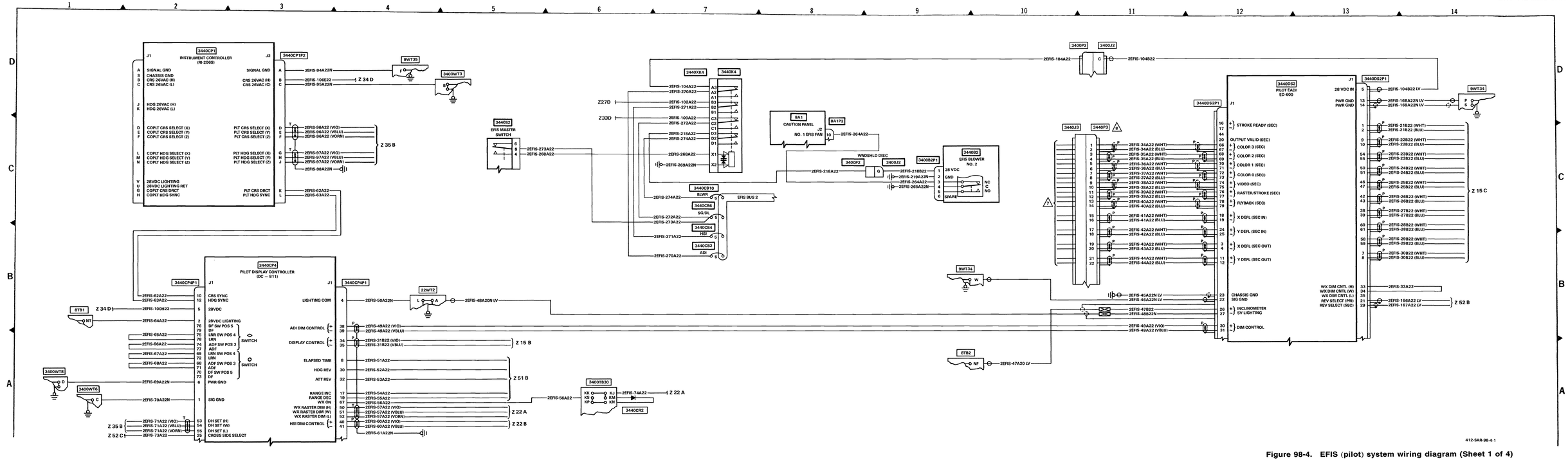



Figure 98-4. EFIS (pilot) system wiring diagram (Sheet 1 of 4)

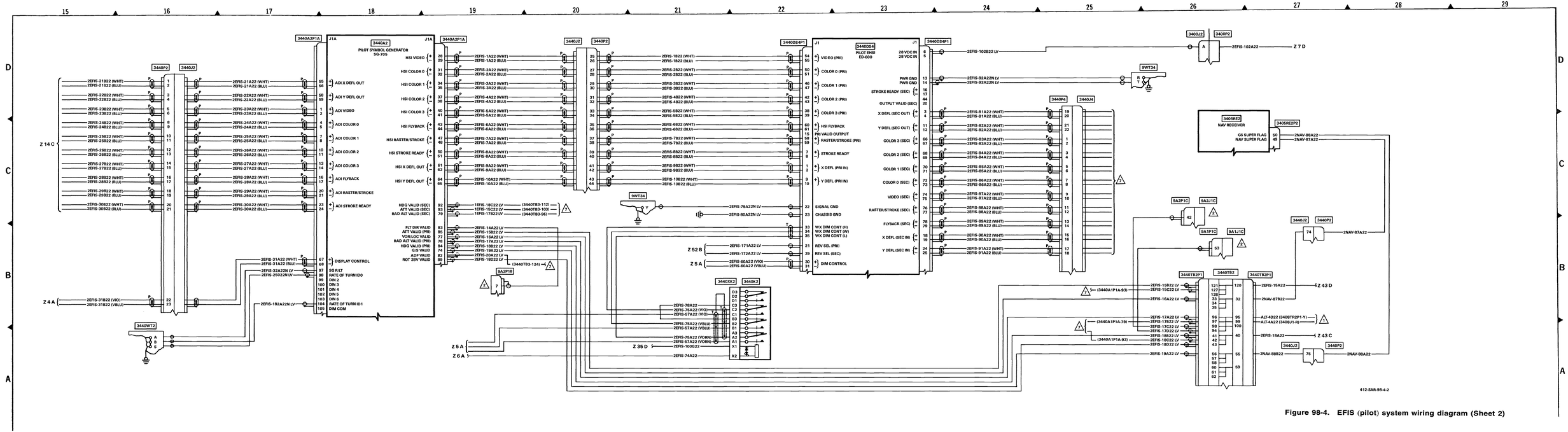



**Figure 98-4. EFIS (pilot) system wiring diagram (Sheet 2)**

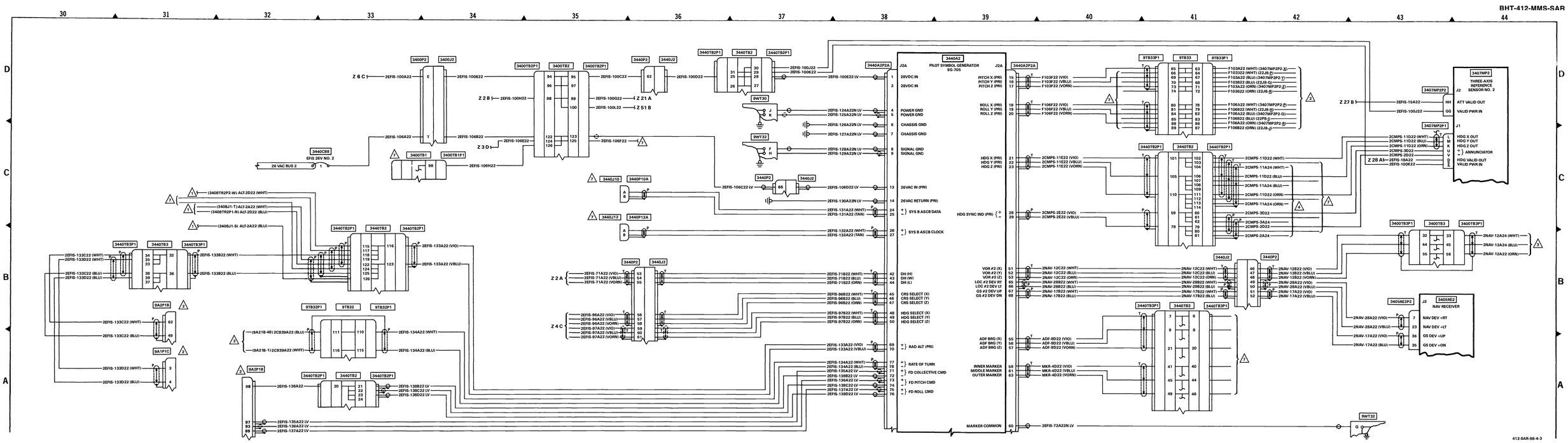



Figure 98-4. EFIS (pilot) system wiring diagram (Sheet 3)

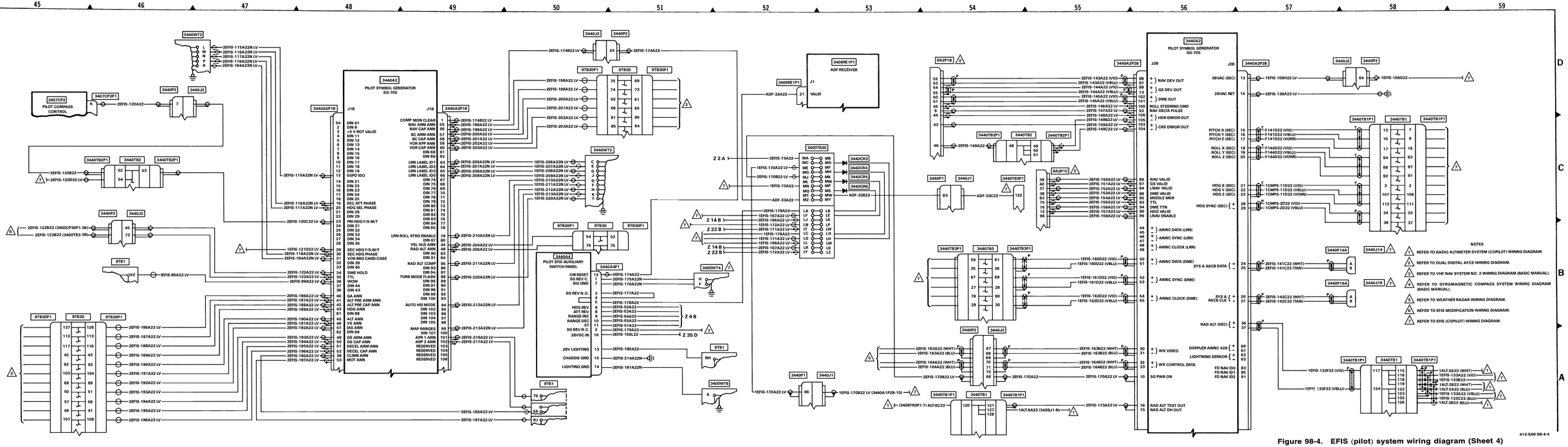



Figure 98-4. EFIS (pilot) system wiring diagram (Sheet 4)

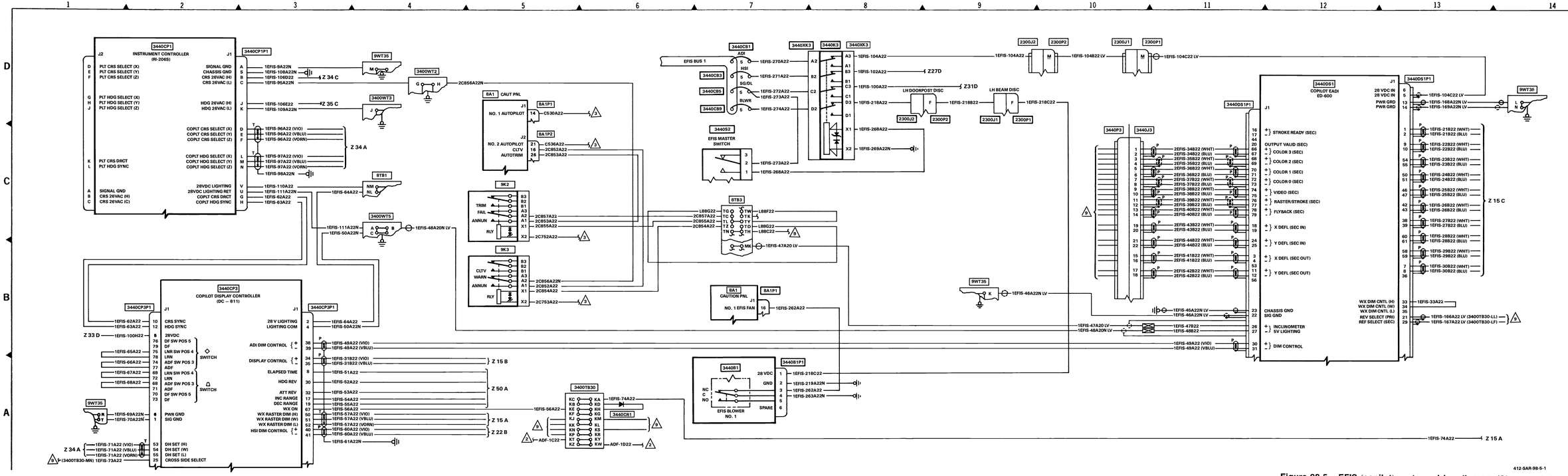



Figure 98-5. EFIS (copilot) system wiring diagram (Sheet 1 of 4)

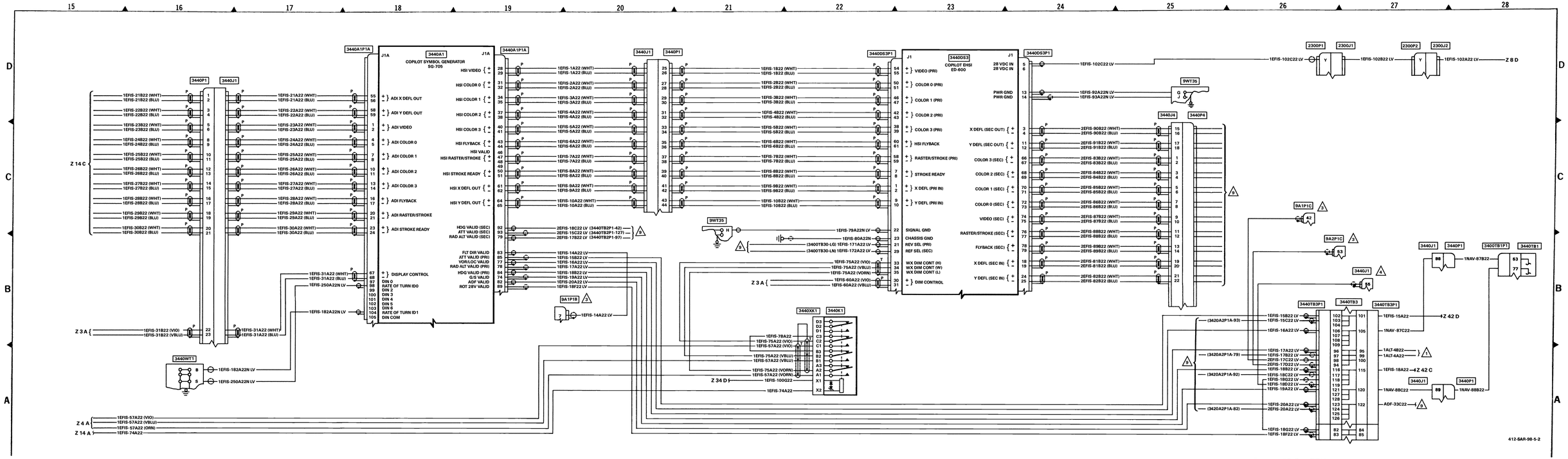



Figure 98-5. EFIS (copilot) system wiring diagram (Sheet 2)

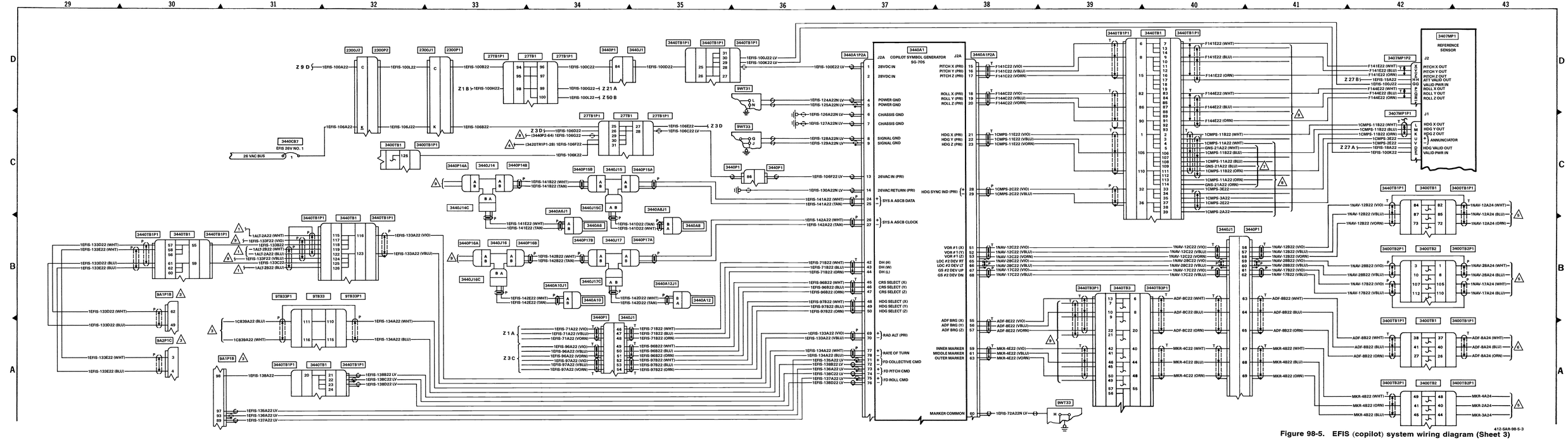



Figure 98-5. EFIS (copilot) system wiring diagram (Sheet 3)

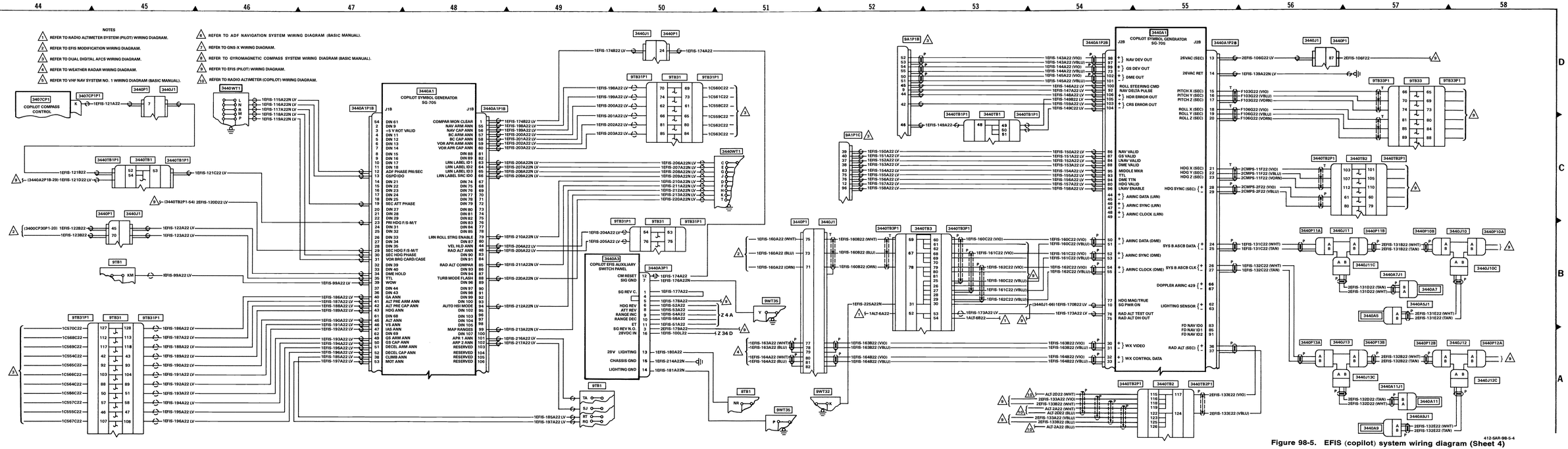



Figure 98-5. EFIS (copilot) system wiring diagram (Sheet 4)

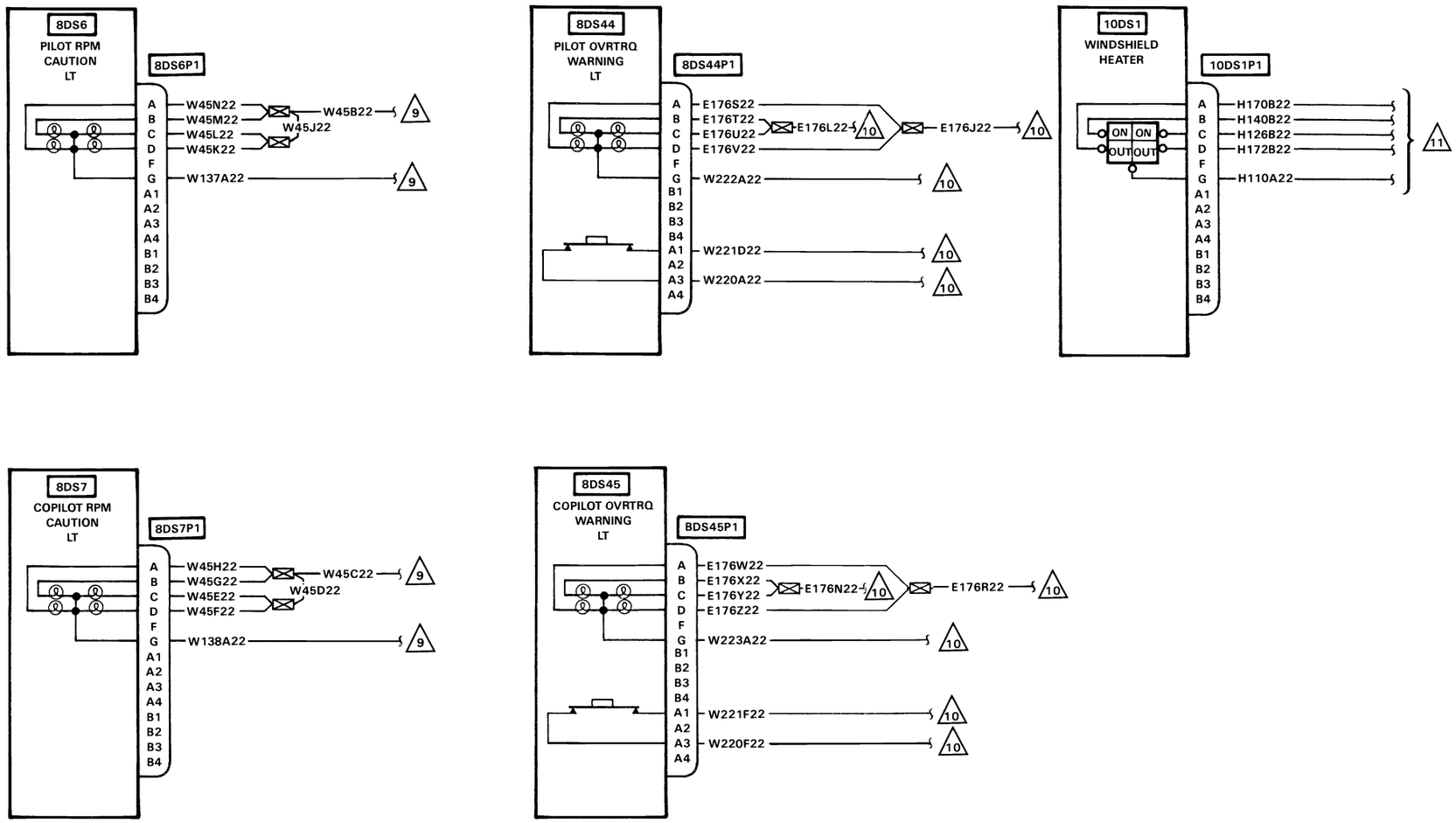
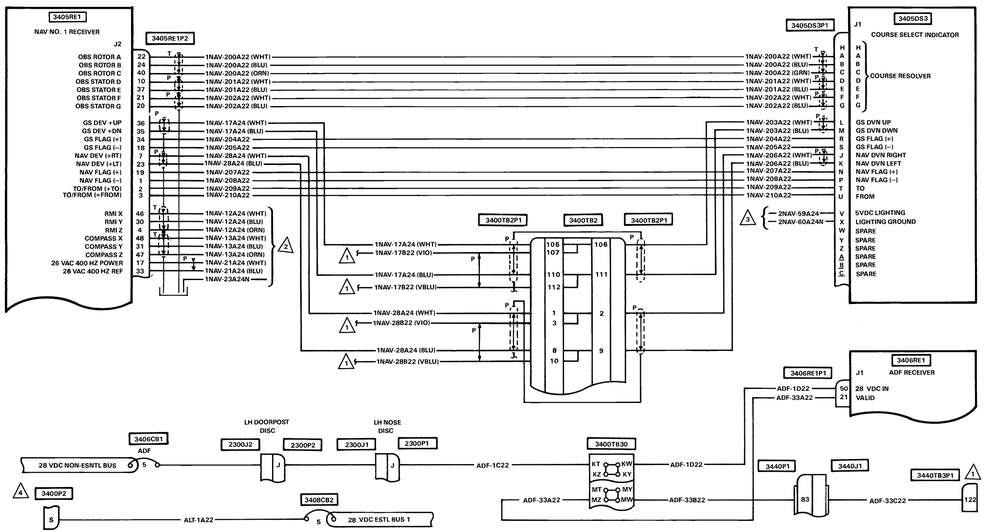
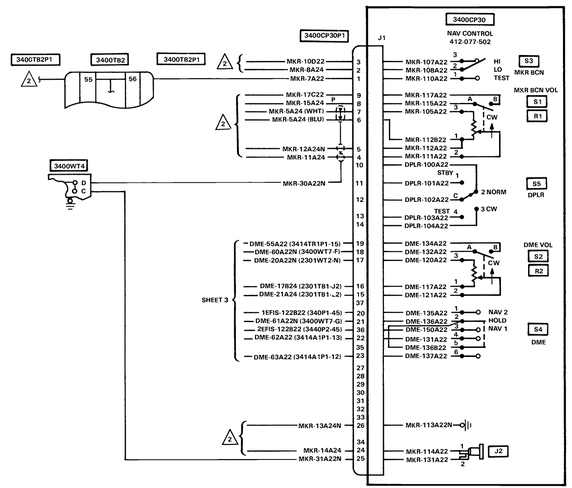
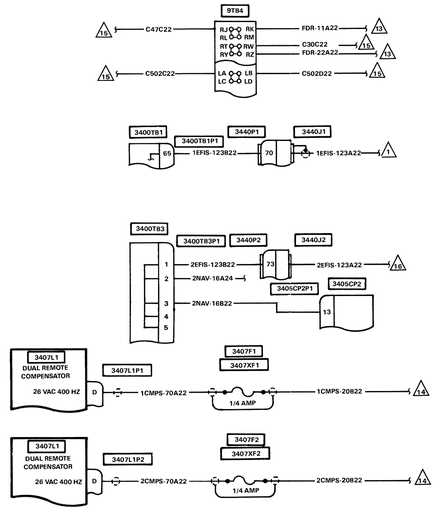
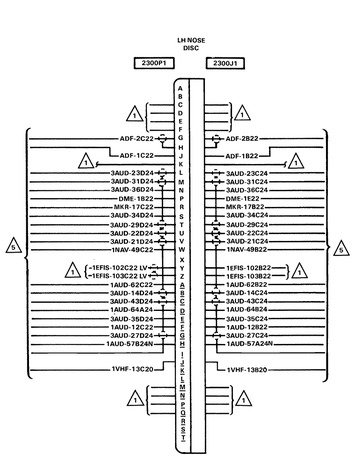



**Figure 98-6. EFIS (modification) system wiring diagram (Sheet 1 of 5)**




412-SAR-98-6-2

**Figure 98-6. EFIS (modification) system wiring diagram (Sheet 2)**

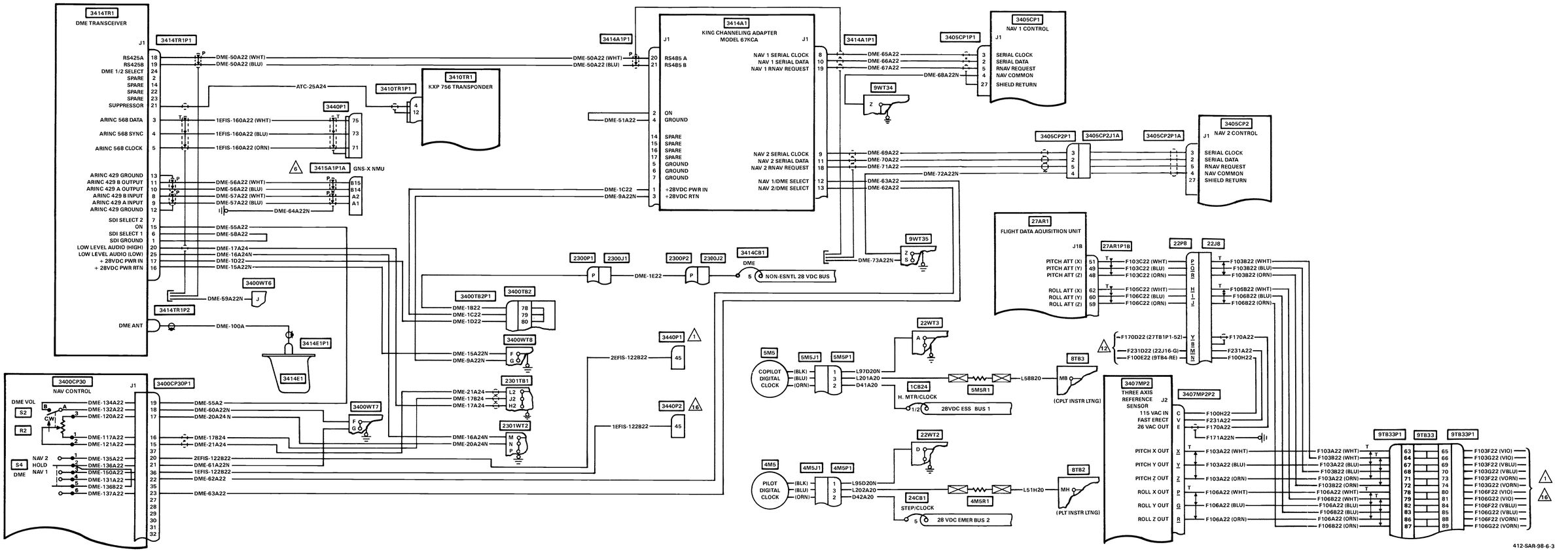



**Figure 98-6. EFIS (modification) system wiring diagram (Sheet 3)**

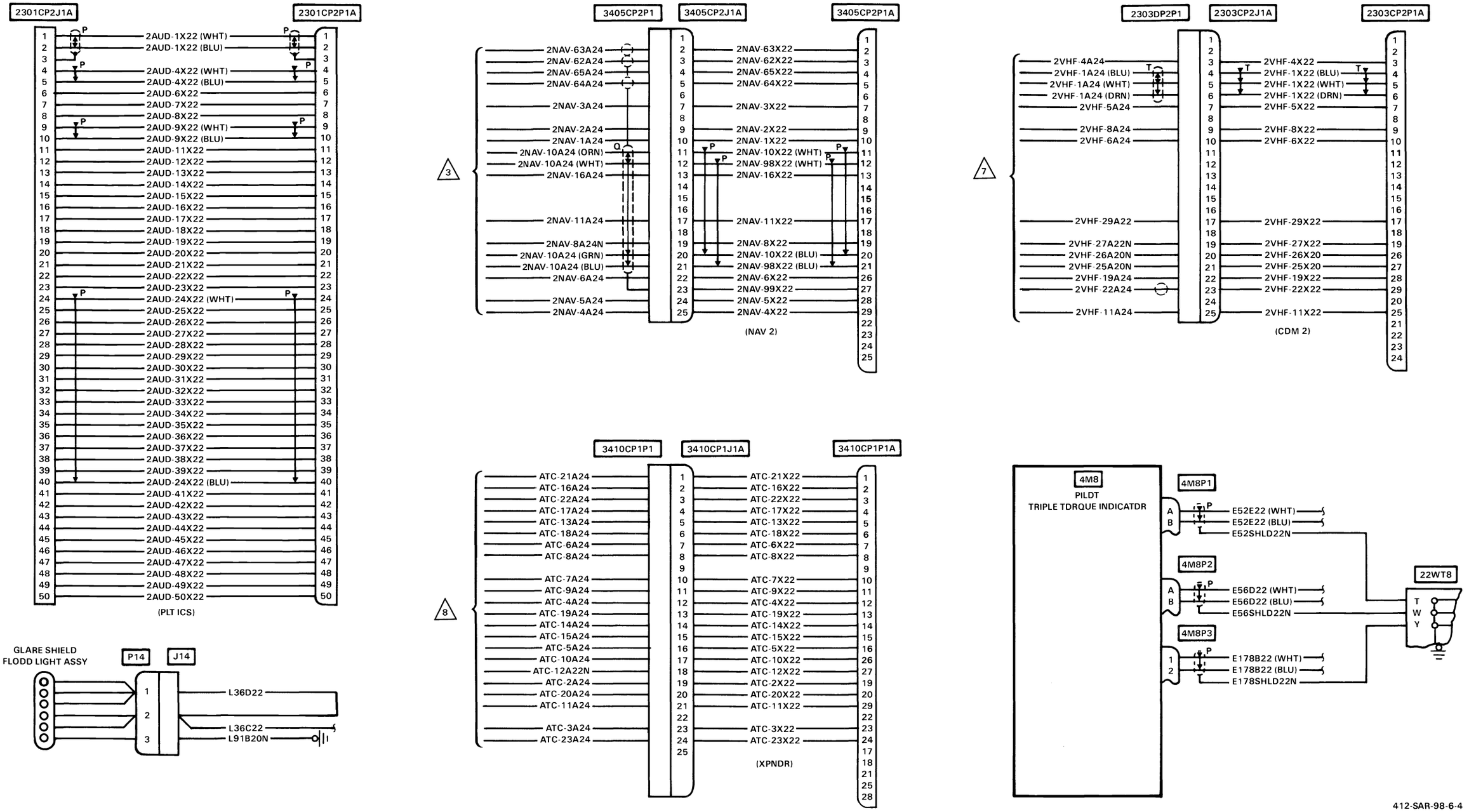
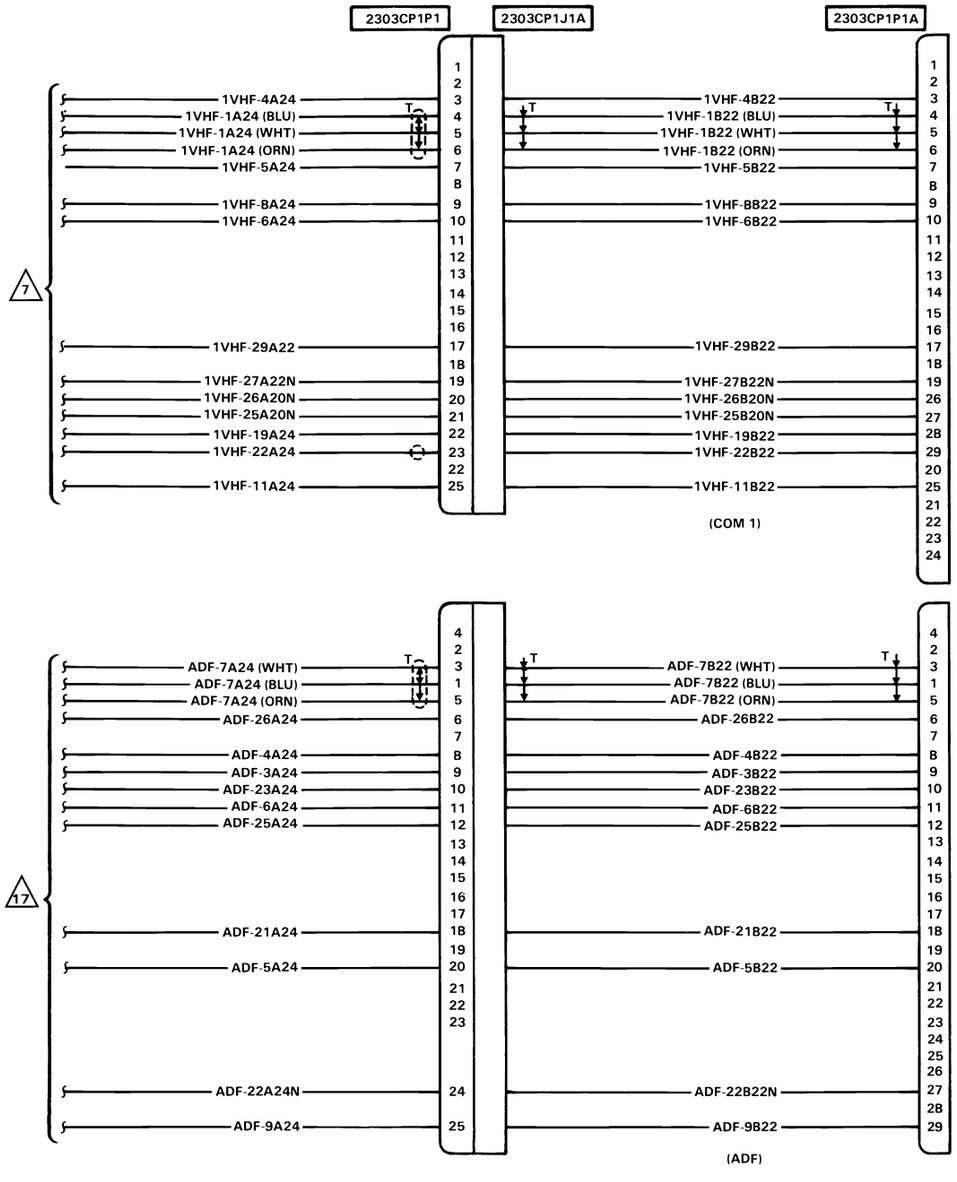
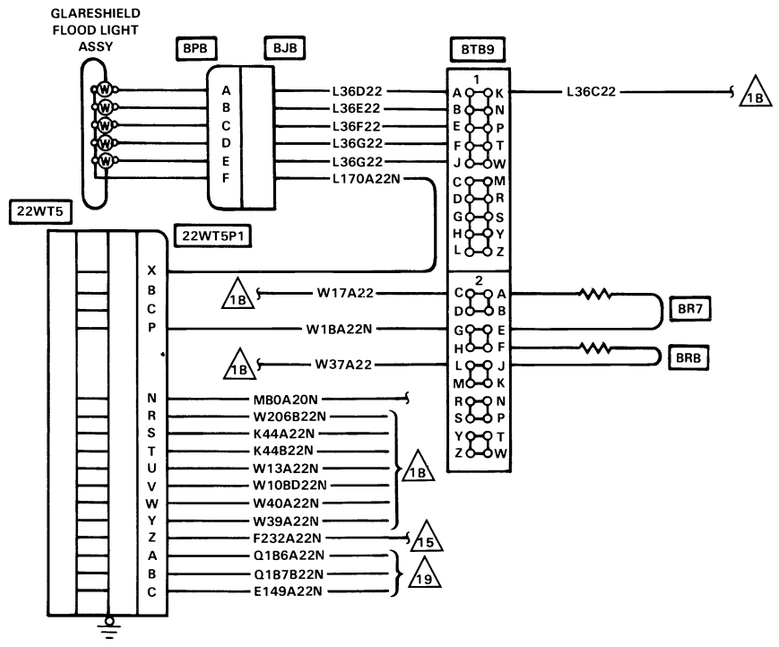
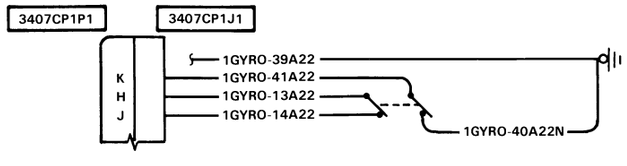



**Figure 98-6. EFIS (modification) system wiring diagram (Sheet 4)**




(FOR FOR SHIPS 36052 AND 36053)



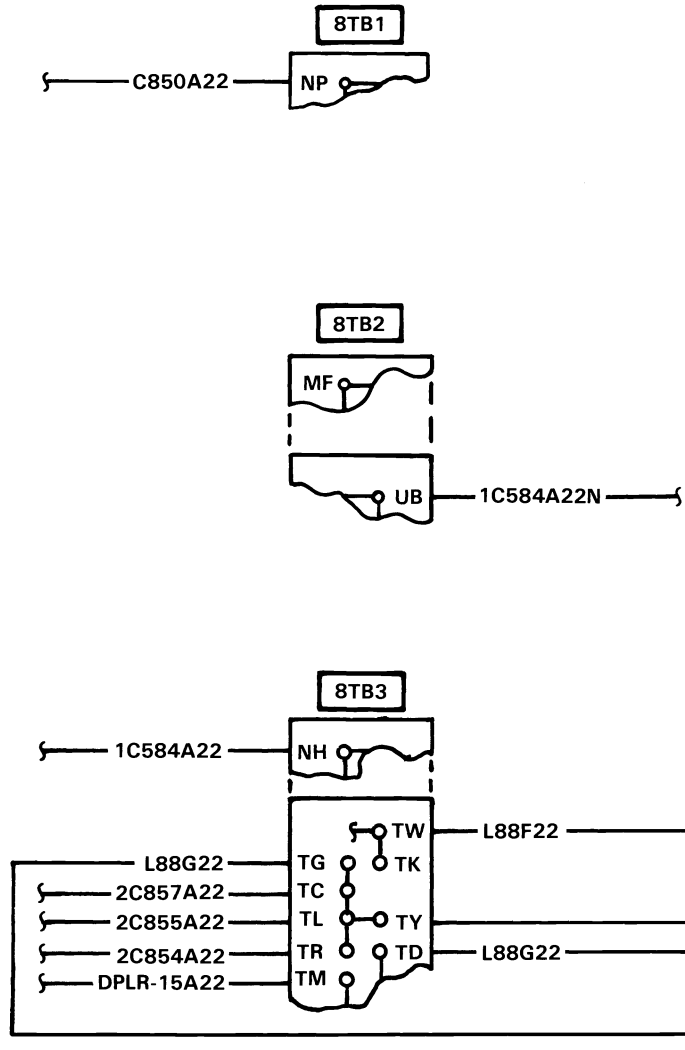

(FOR FOR SHIPS 36052 AND 36053)

NOTES

1. REFER TO EFIS (COPILOT) WIRING DIAGRAM.
2. REFER TO VHF NAV SYSTEM NO. 1 WIRING DIAGRAM (BASIC MANUAL).
3. REFER TO VHF NAV SYSTEM NO. 2 WIRING DIAGRAM (BASIC MANUAL).
4. REFER TO RADAR ALTIMETER WIRING DIAGRAM (BASIC MANUAL).
5. REFER TO INTERCOMMUNICATION SYSTEM (ICS) WIRING DIAGRAM (BASIC MANUAL).
6. REFER TO GNS-X WIRING DIAGRAM.
7. REFER TO VHF COMMUNICATION SYSTEM WIRING DIAGRAM (BASIC MANUAL).
8. REFER TO ATC TRANSPONDER WIRING DIAGRAM (BASIC MANUAL).
9. SEE FIGURE 98-7B BASIC MANUAL.
10. SEE FIGURE 98-47 BASIC MANUAL.
11. REFER TO SI-412-6.
12. REFER TO PILOT ATTITUDE DIRECTOR SYSTEM WIRING DIAGRAM (BASIC MANUAL).
13. REFER TO FLIGHT DATA RECORDER SYSTEM WIRING DIAGRAM (BASIC MANUAL).
14. REFER TO GYROMAGNETIC COMPASS SYSTEM WIRING DIAGRAM (BASIC MANUAL).
15. REFER TO HYDRAULIC SYSTEM WIRING DIAGRAM (BASIC MANUAL).
16. REFER TO EFIS (PILOT) WIRING DIAGRAM.
17. REFER TO ADF WIRING DIAGRAM (BASIC MANUAL).
18. REFER TO CAUTION AND LIGHTING WIRING DIAGRAM (BASIC MANUAL).
19. REFER TO POWER PLANT AND TRANSMISSION WIRING DIAGRAM (BASIC MANUAL).

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**Figure 98-6. EFIS (modification) system wiring diagram (Sheet 5)**




NOTE

REFER TO CHAPTER 98 OF BASIC MANUAL FOR REMAINDER OF 8TB1, 8TB2, and 8TB3.

412-SAR-98-7-1

**Figure 98-7. Component replacement (Sheet 1 of 59)**

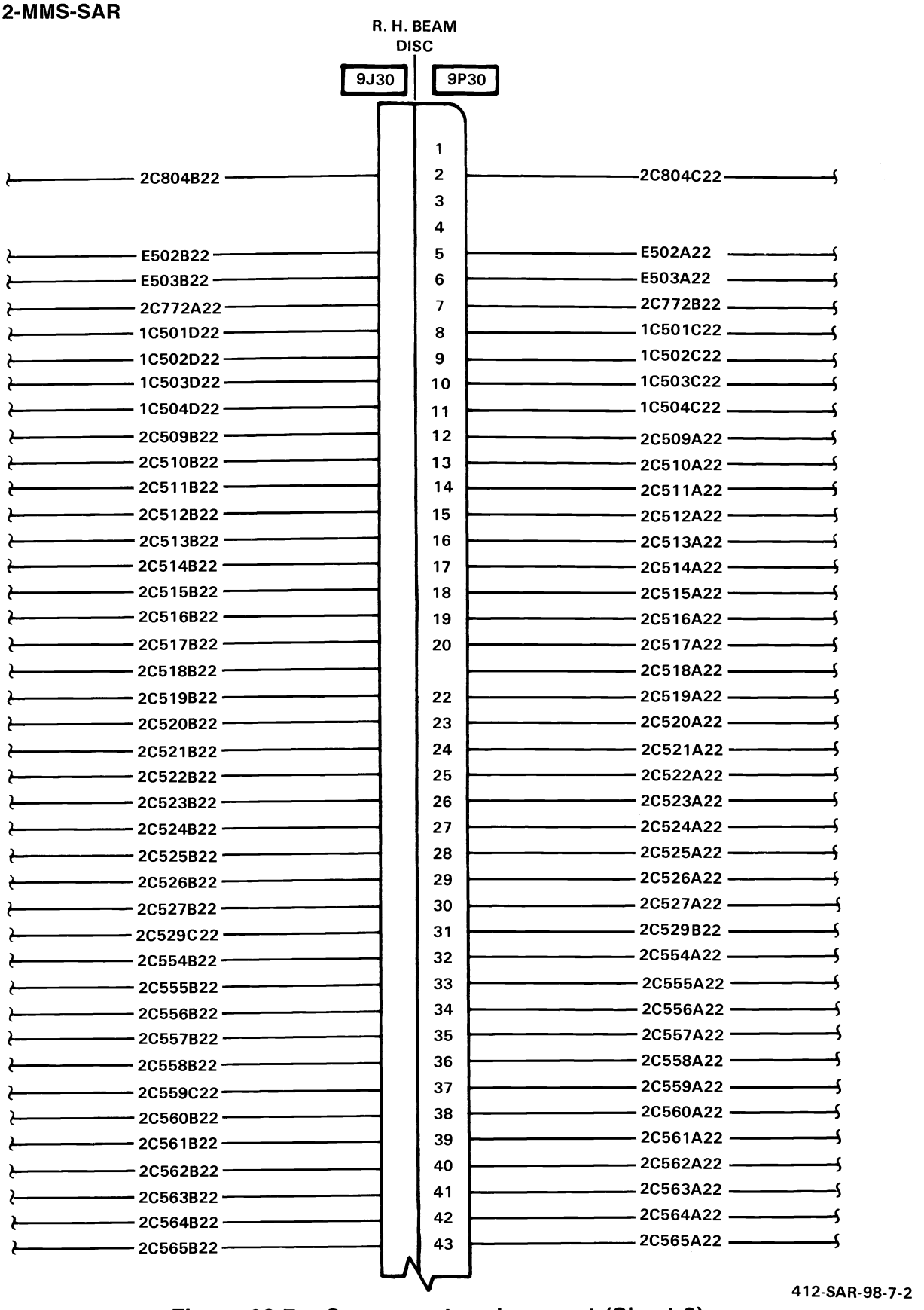
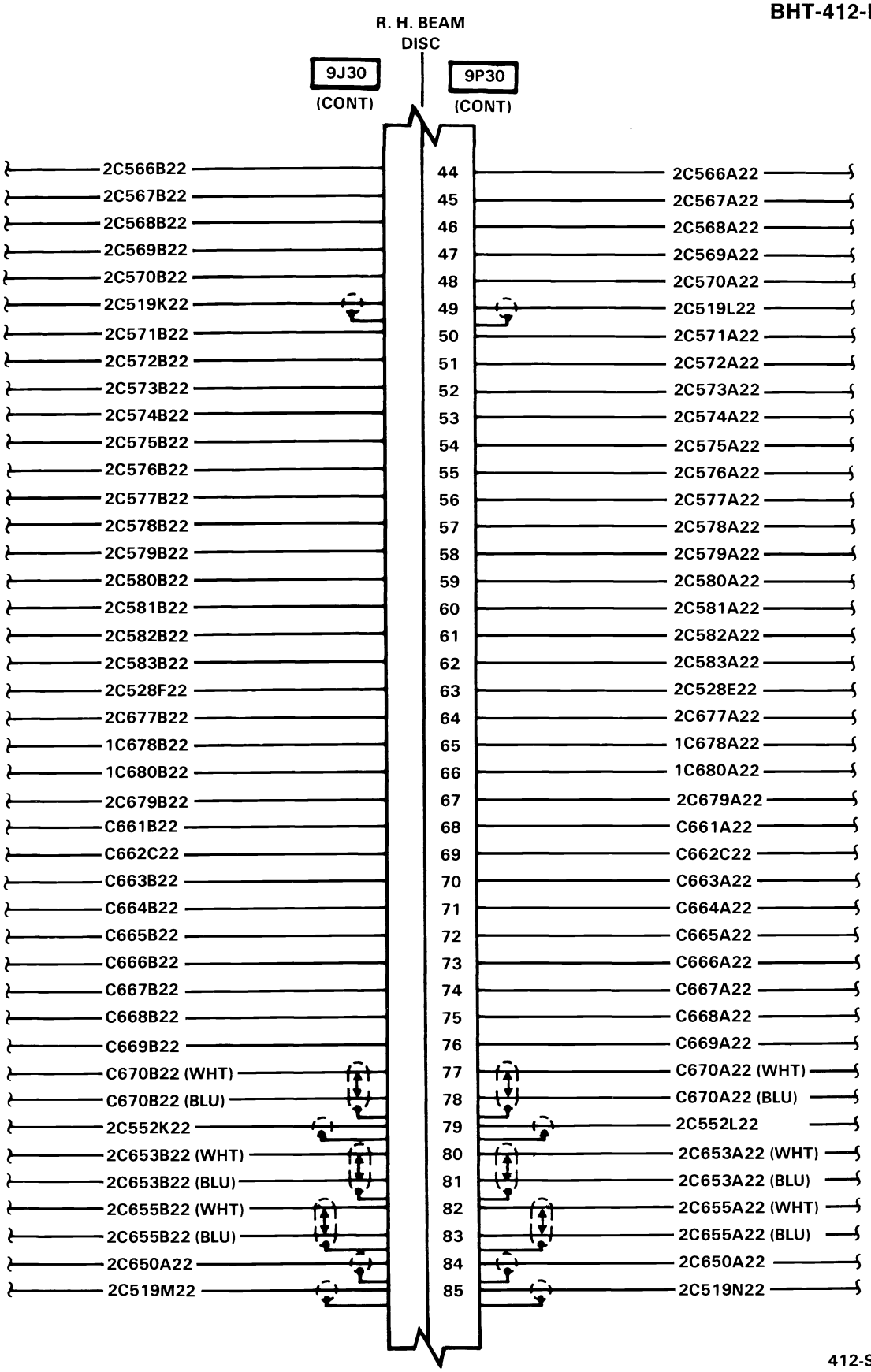
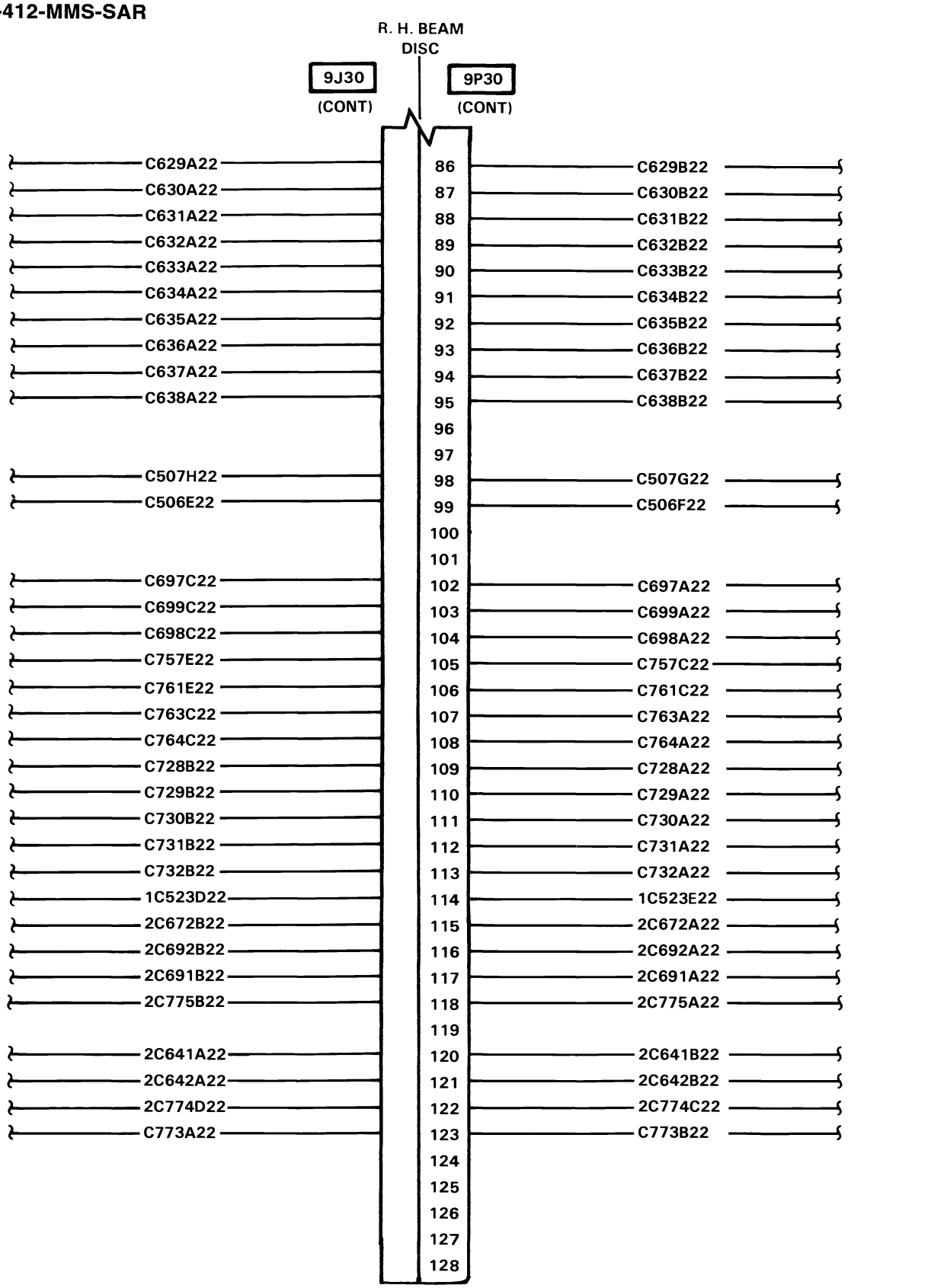



**Figure 98-7. Component replacement (Sheet 2)**



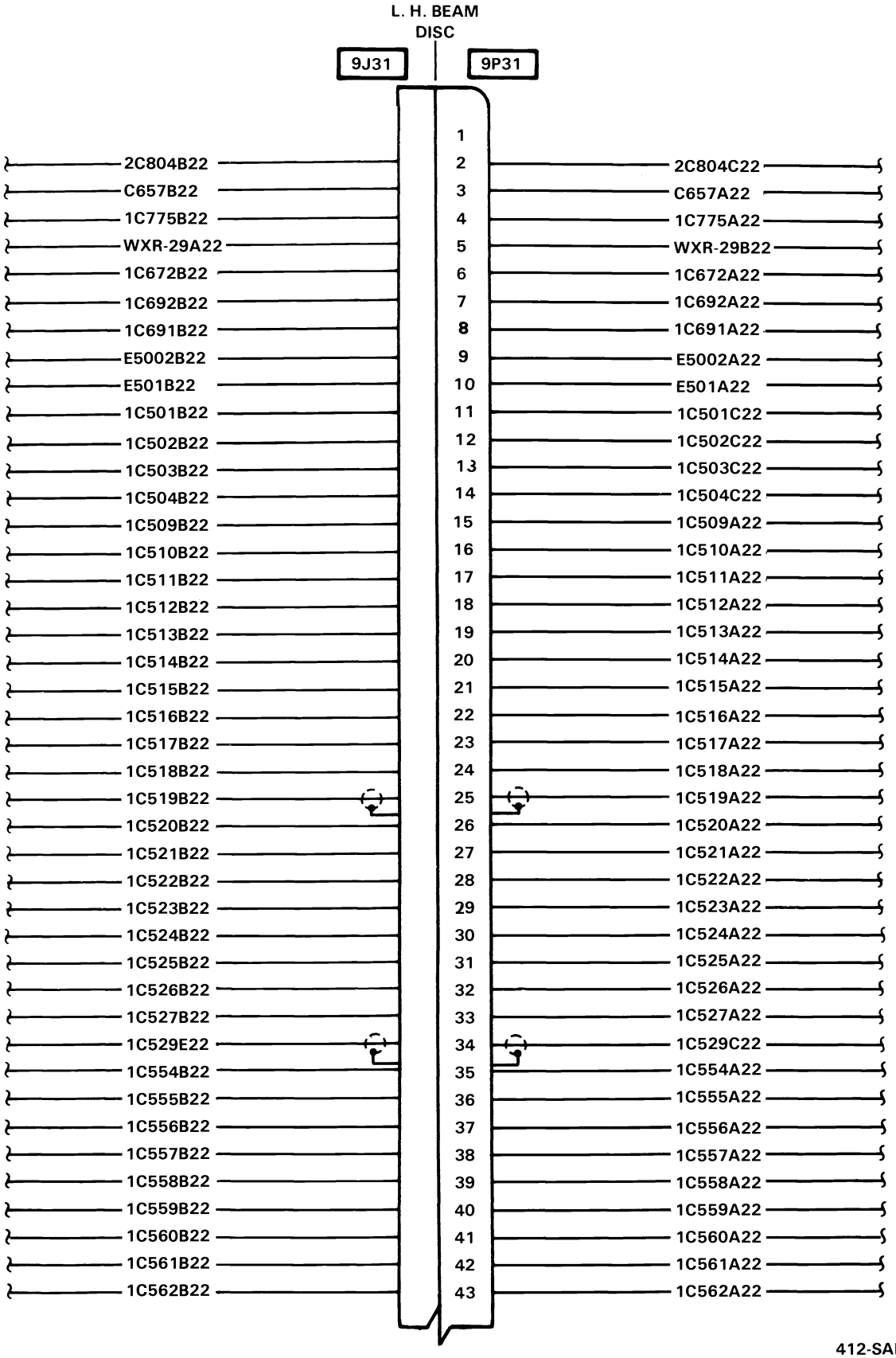

412-SAR-98-7-3

**Figure 98-7. Component replacement (Sheet 3)**



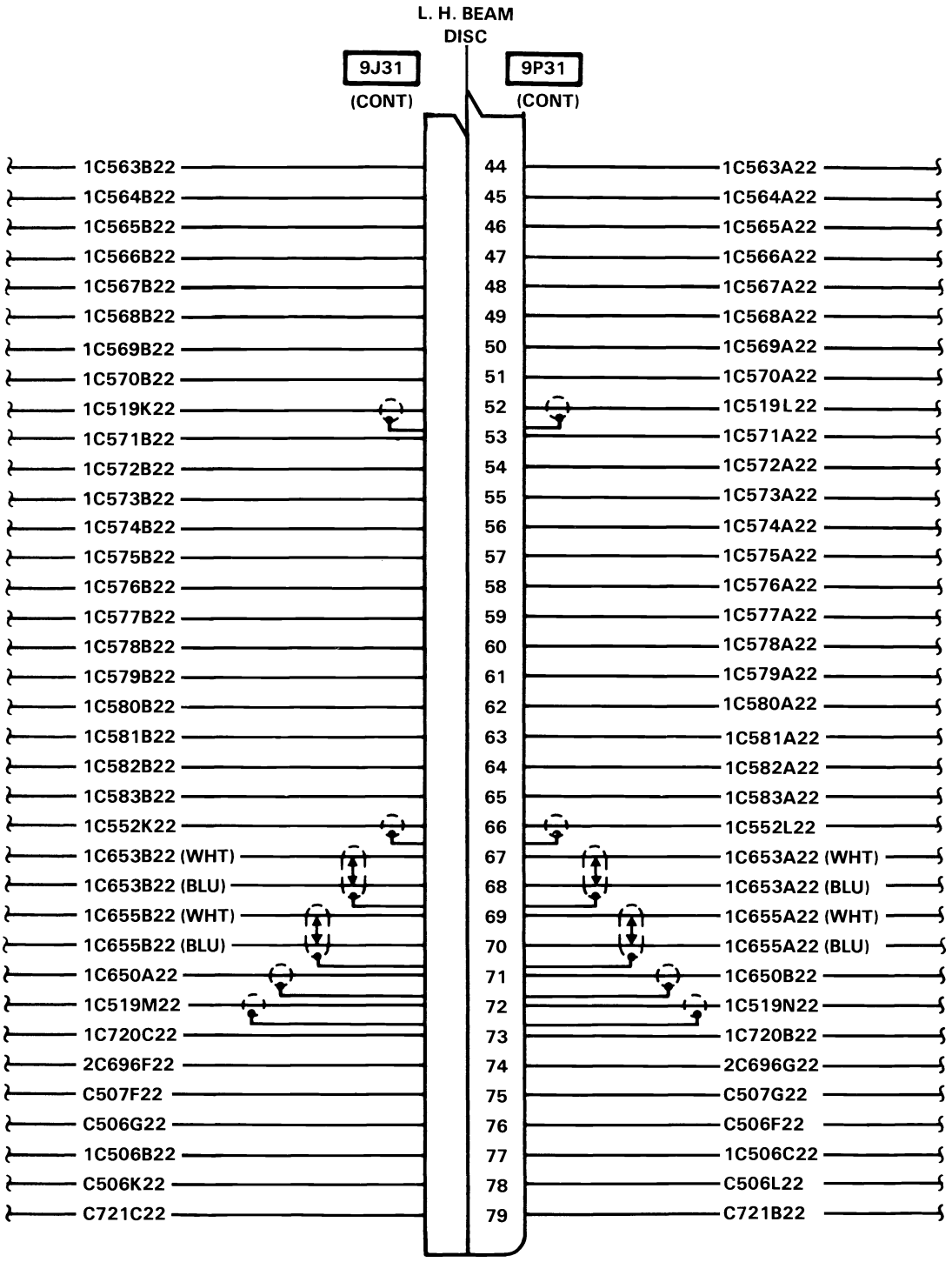

412-SAR-98-7-4

**Figure 98-7. Component replacement (Sheet 4)**



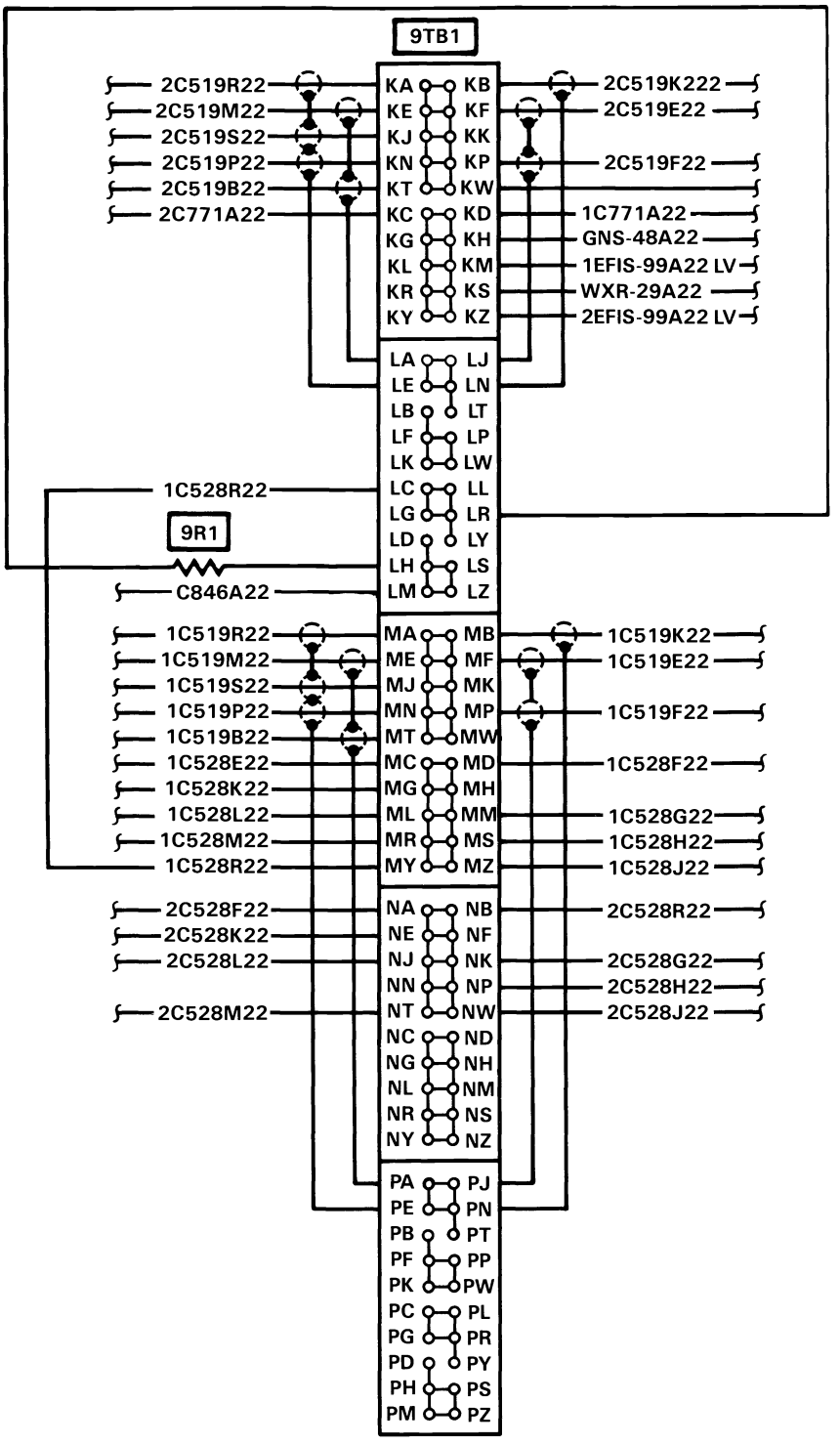

412-SAR-98-7-5

**Figure 98-7. Component replacement (Sheet 5)**



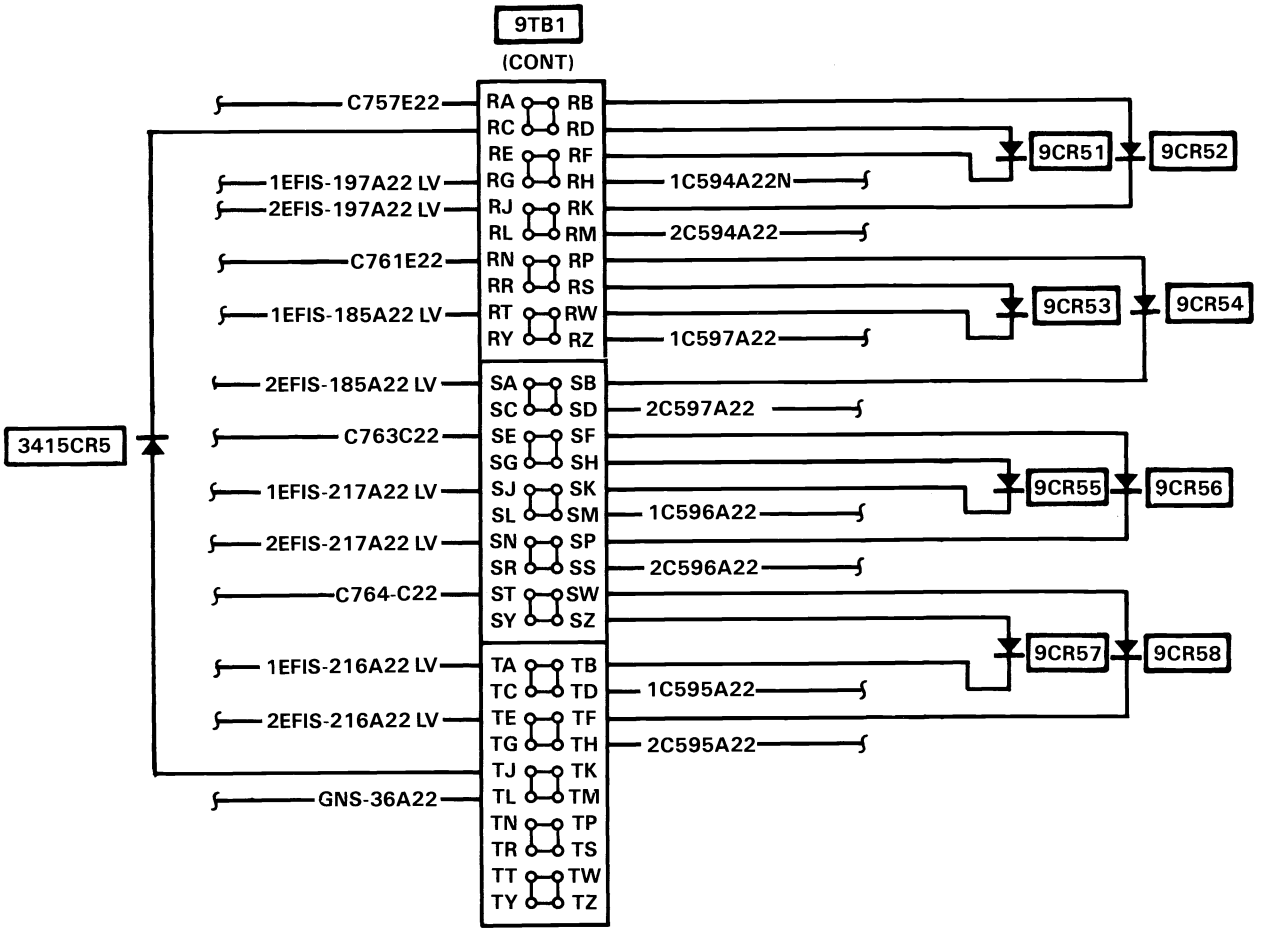

412-SAR-98-7-6

**Figure 98-7. Component replacement (Sheet 6)**



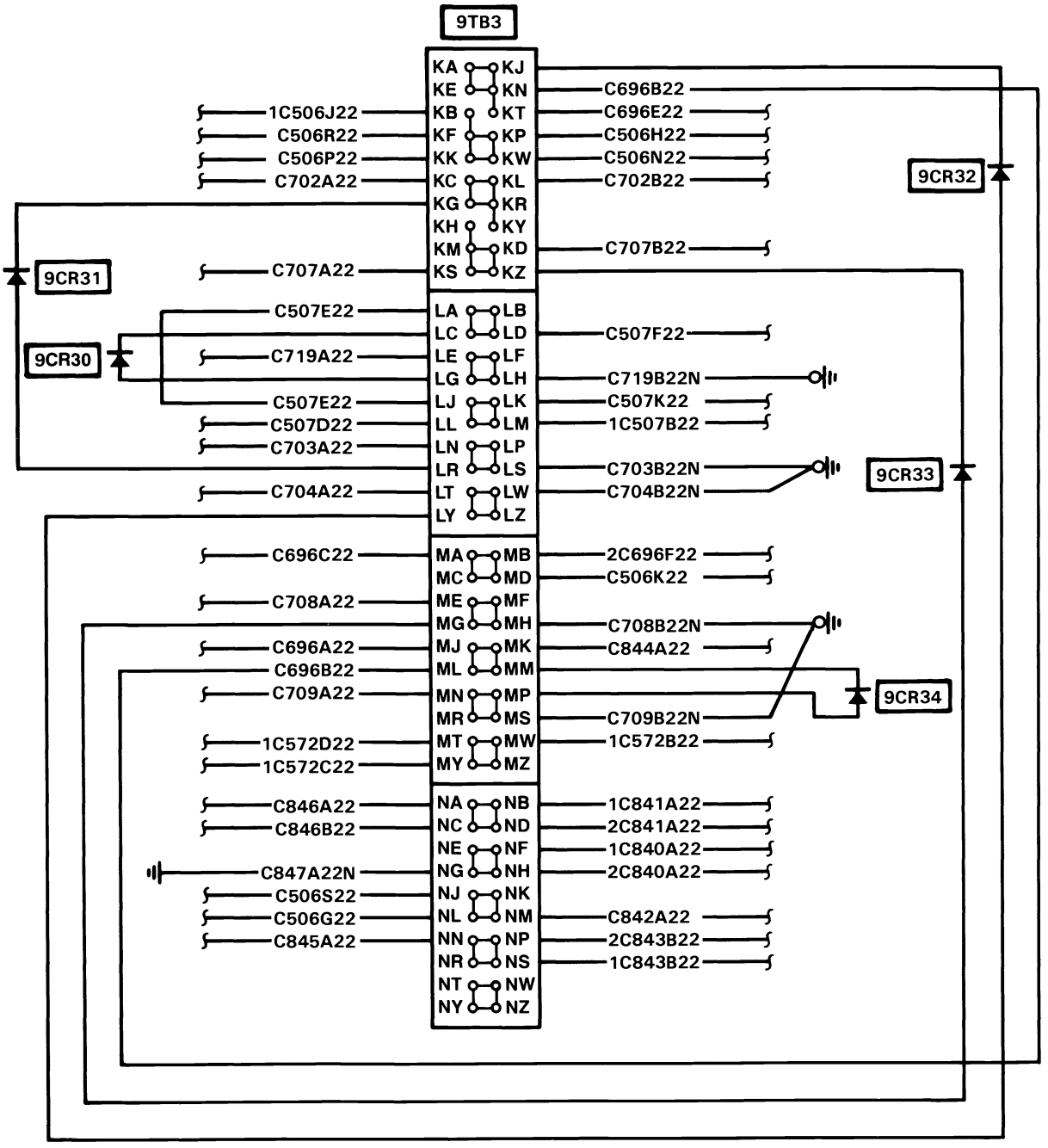

412-SAR-98-7-7

**Figure 98-7. Component replacement (Sheet 7)**




412-SAR-98-7-8

**Figure 98-7. Component replacement (Sheet 8)**




412-SAR-98-7-9

**Figure 98-7. Component replacement (Sheet 9)**

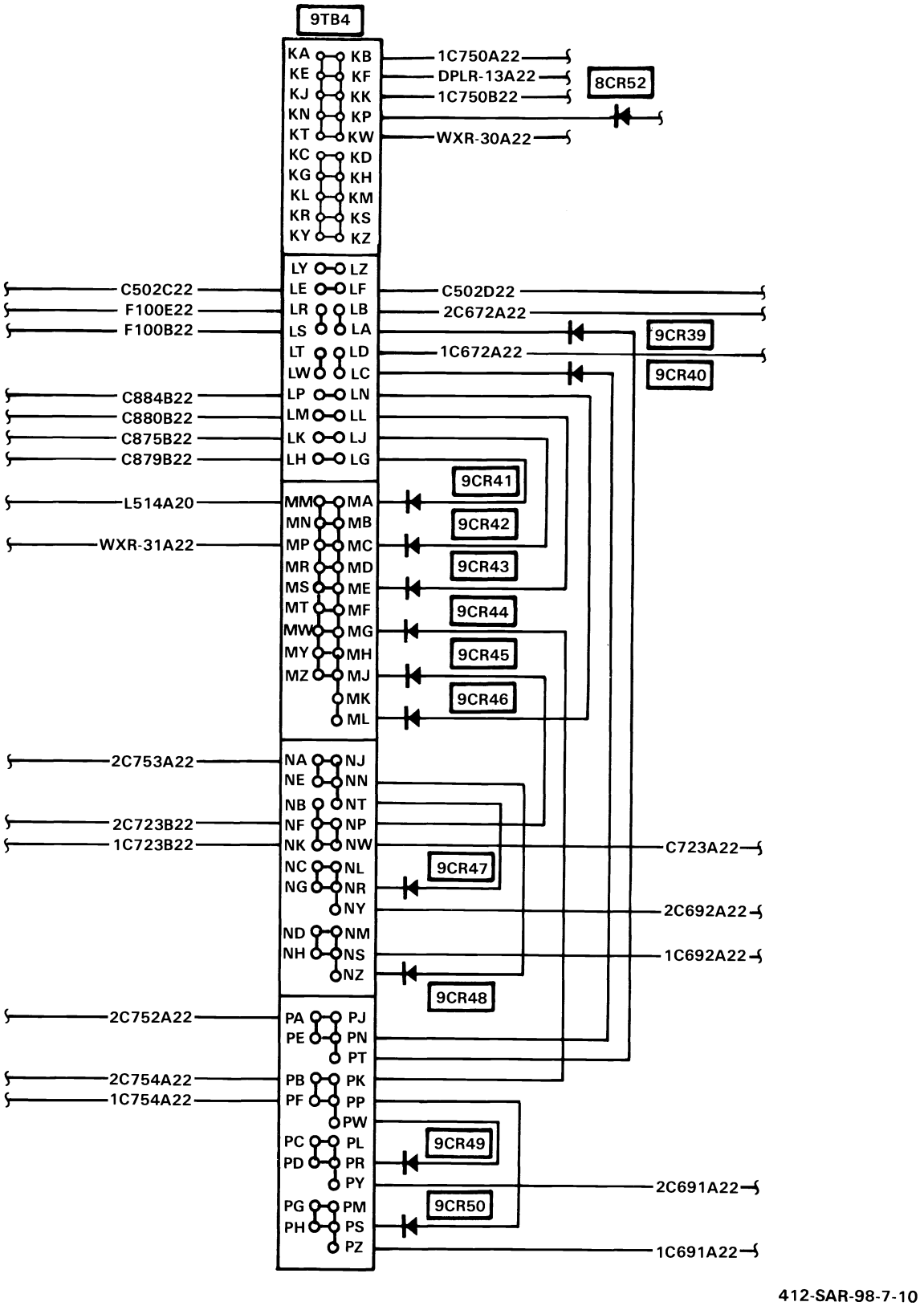
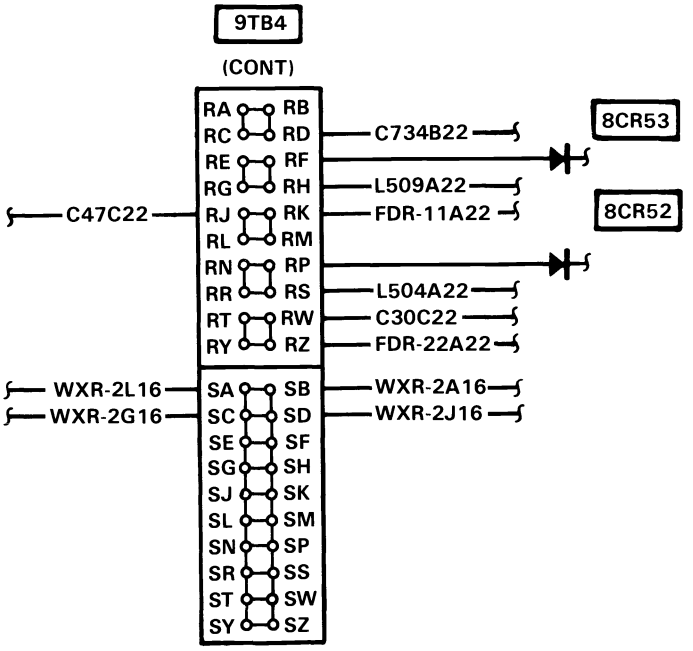
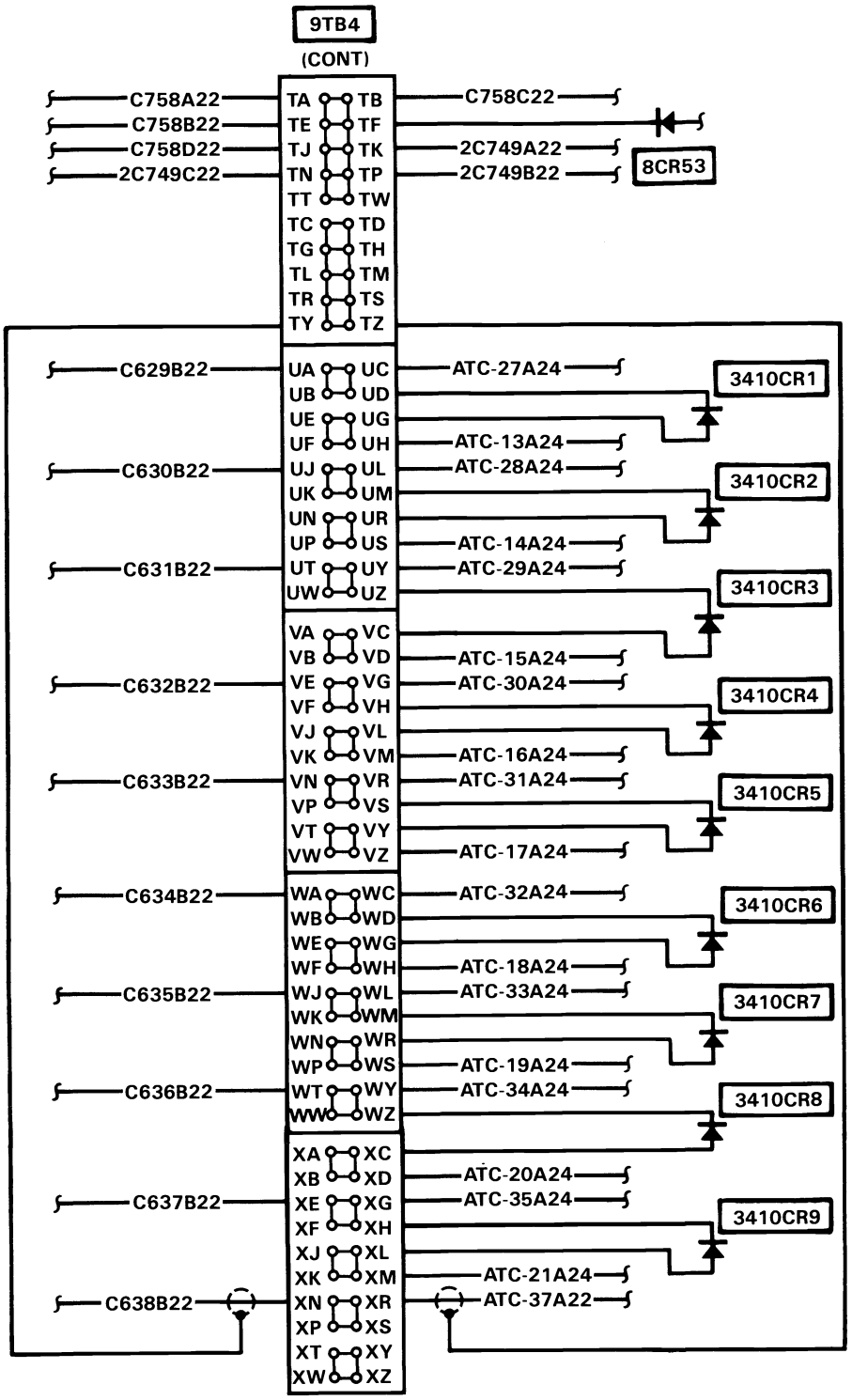



**Figure 98-7. Component replacement (Sheet 10)**



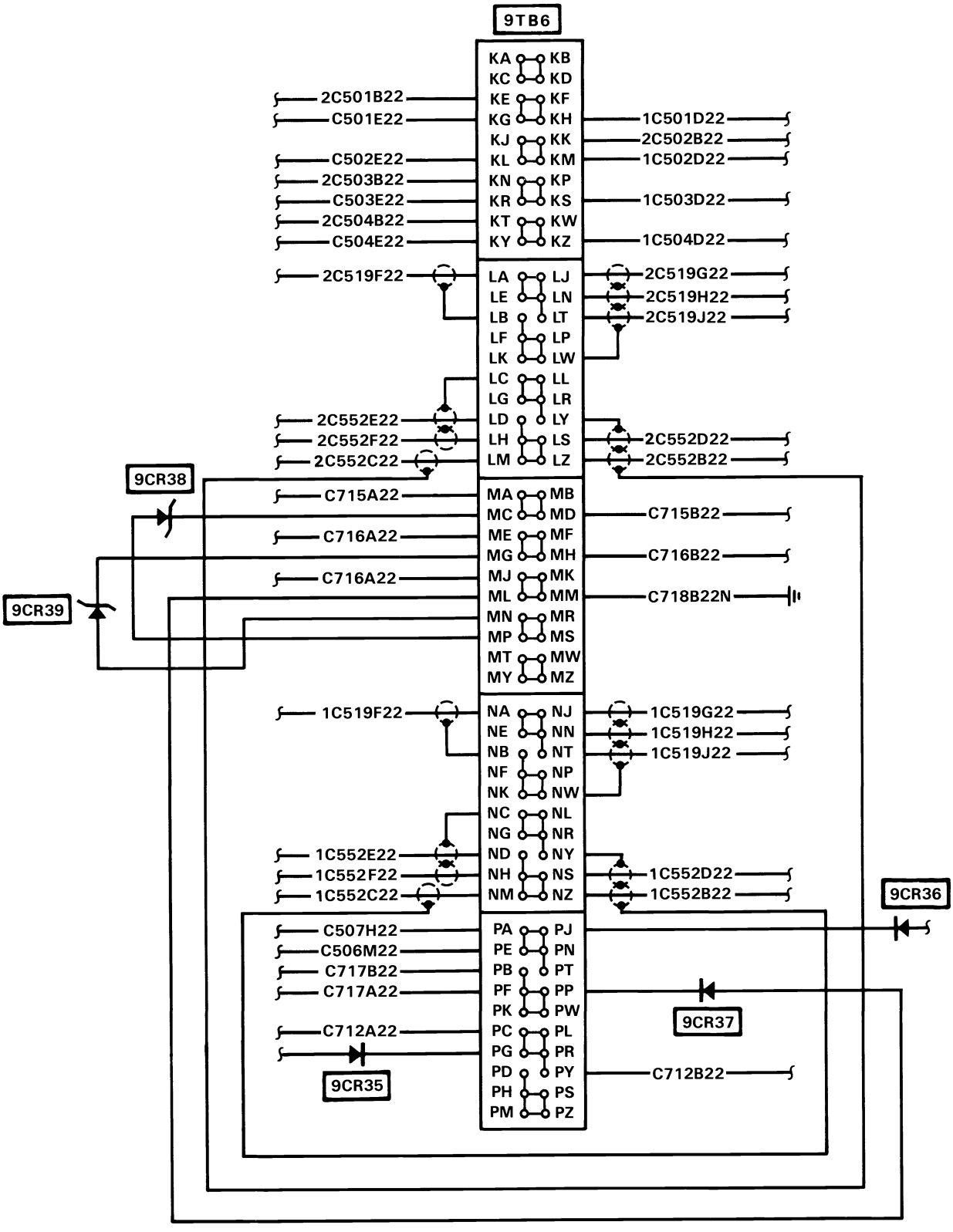

412-SAR-98-7-11

**Figure 98-7. Component replacement (Sheet 11)**



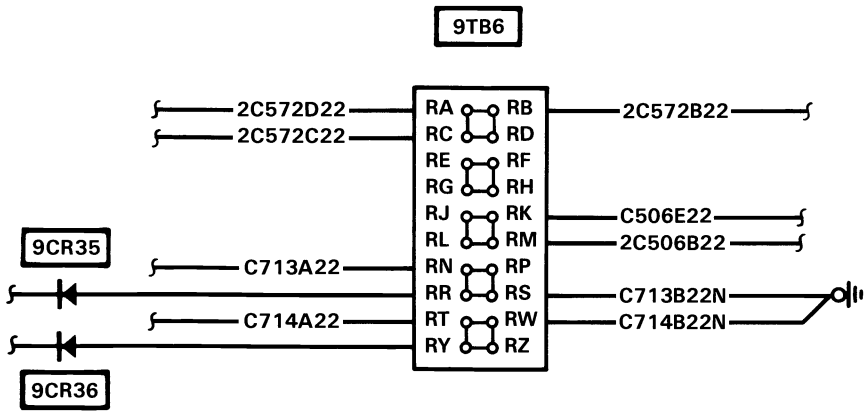

412-SAR-98-7-12

**Figure 98-7. Component replacement (Sheet 12)**



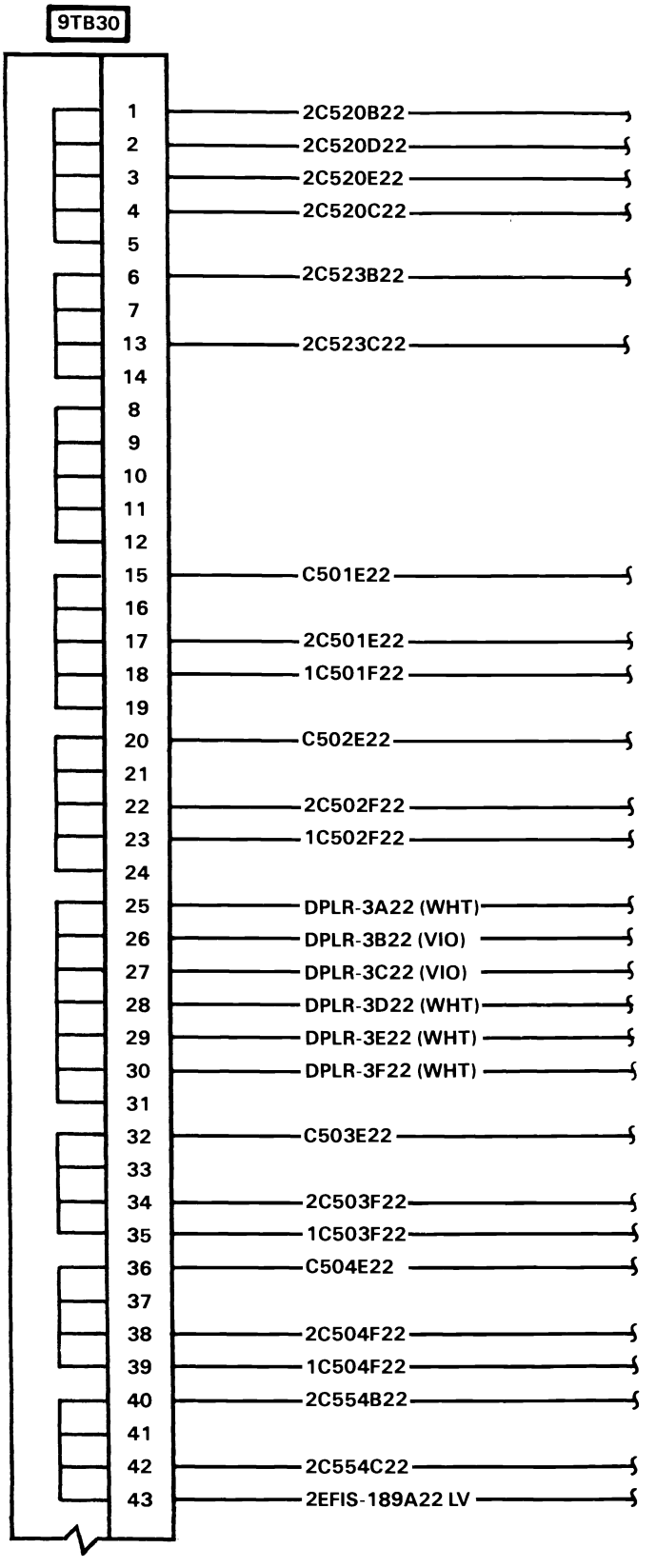

412-SAR-98-7-13

**Figure 98-7. Component replacement (Sheet 13)**




412-SAR-98-7-14

**Figure 98-7. Component replacement (Sheet 14)**




412-SAR-98-7-15

**Figure 98-7. Component replacement (Sheet 15)**

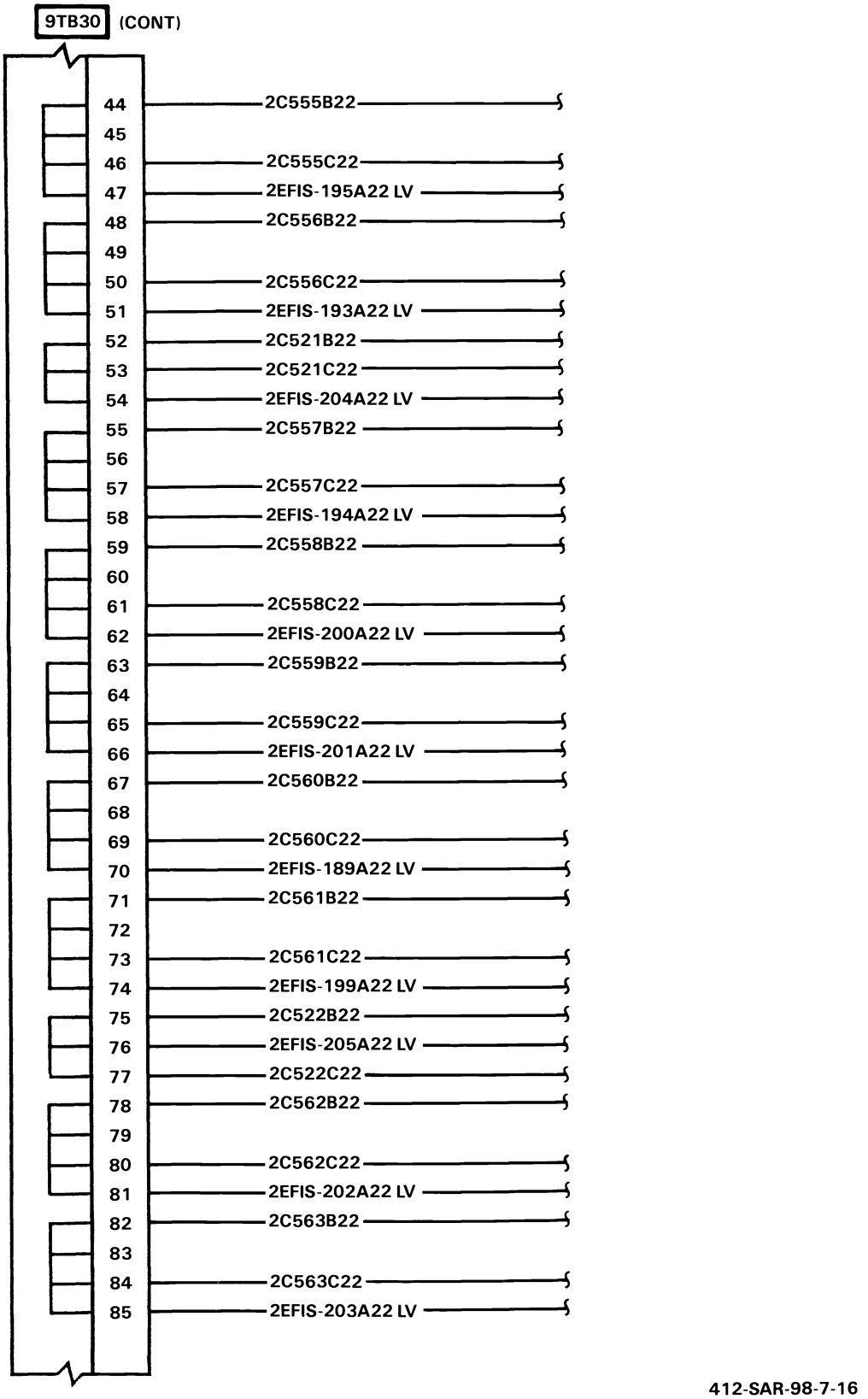
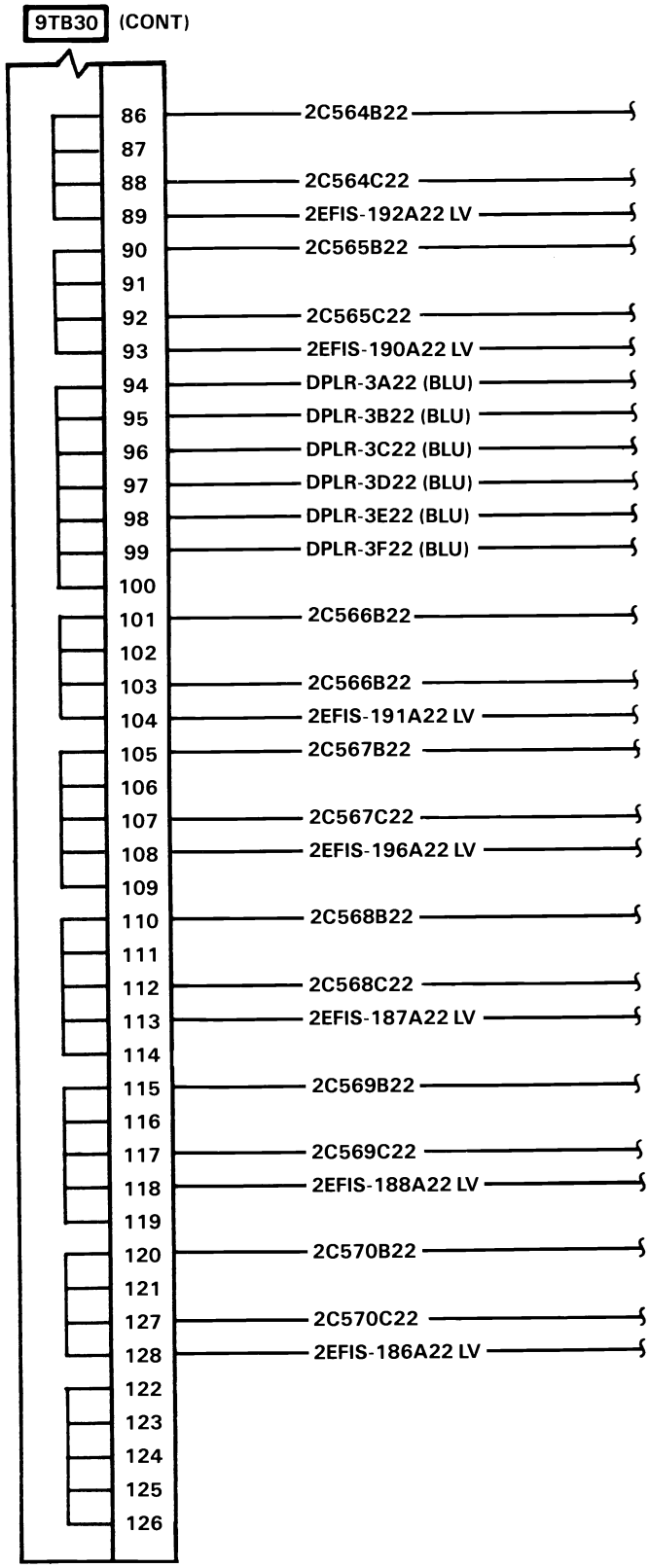



**Figure 98-7. Component replacement (Sheet 16)**




412-SAR-98-7-17

**Figure 98-7. Component replacement (Sheet 17)**

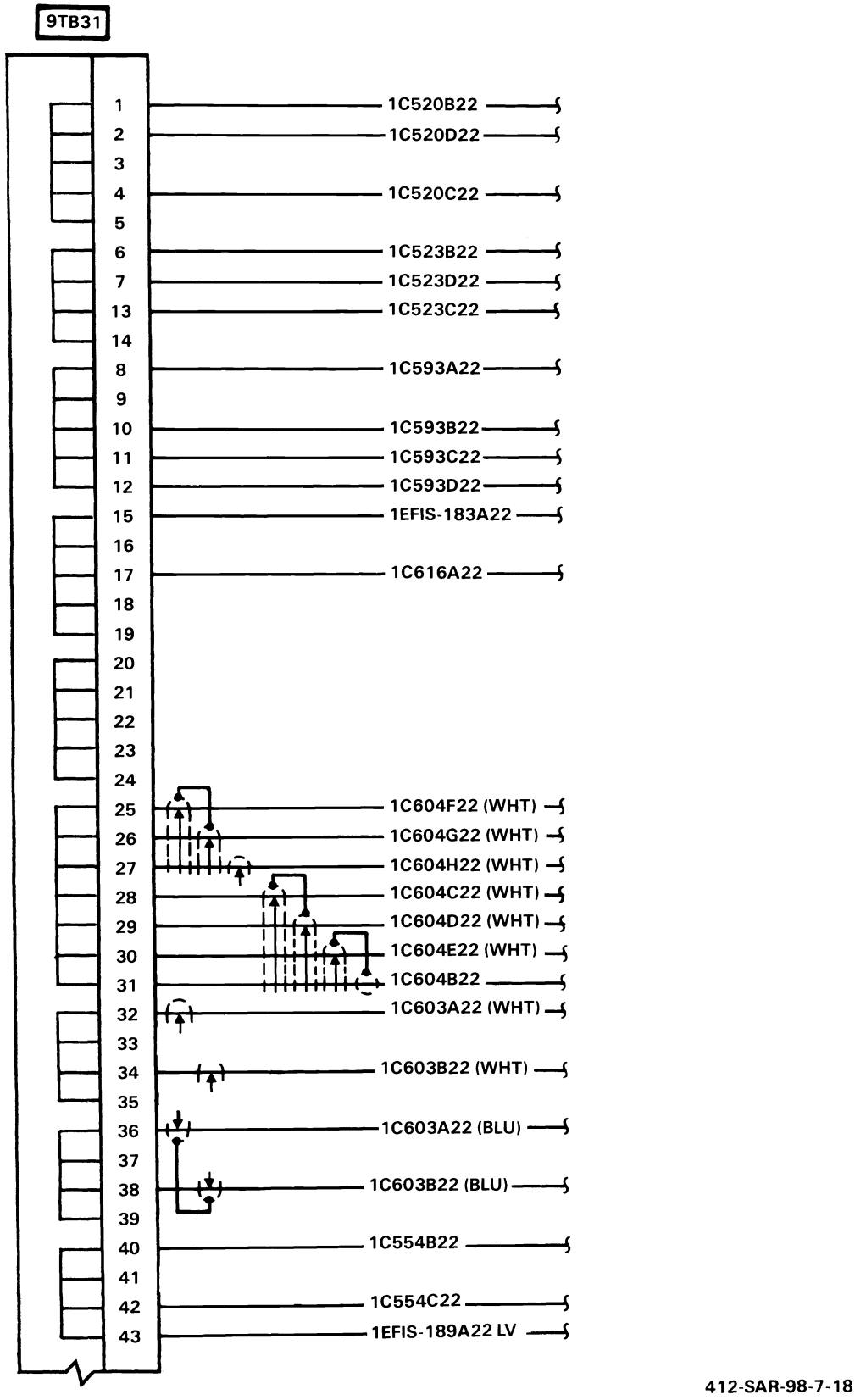



**Figure 98-7. Component replacement (Sheet 18)**

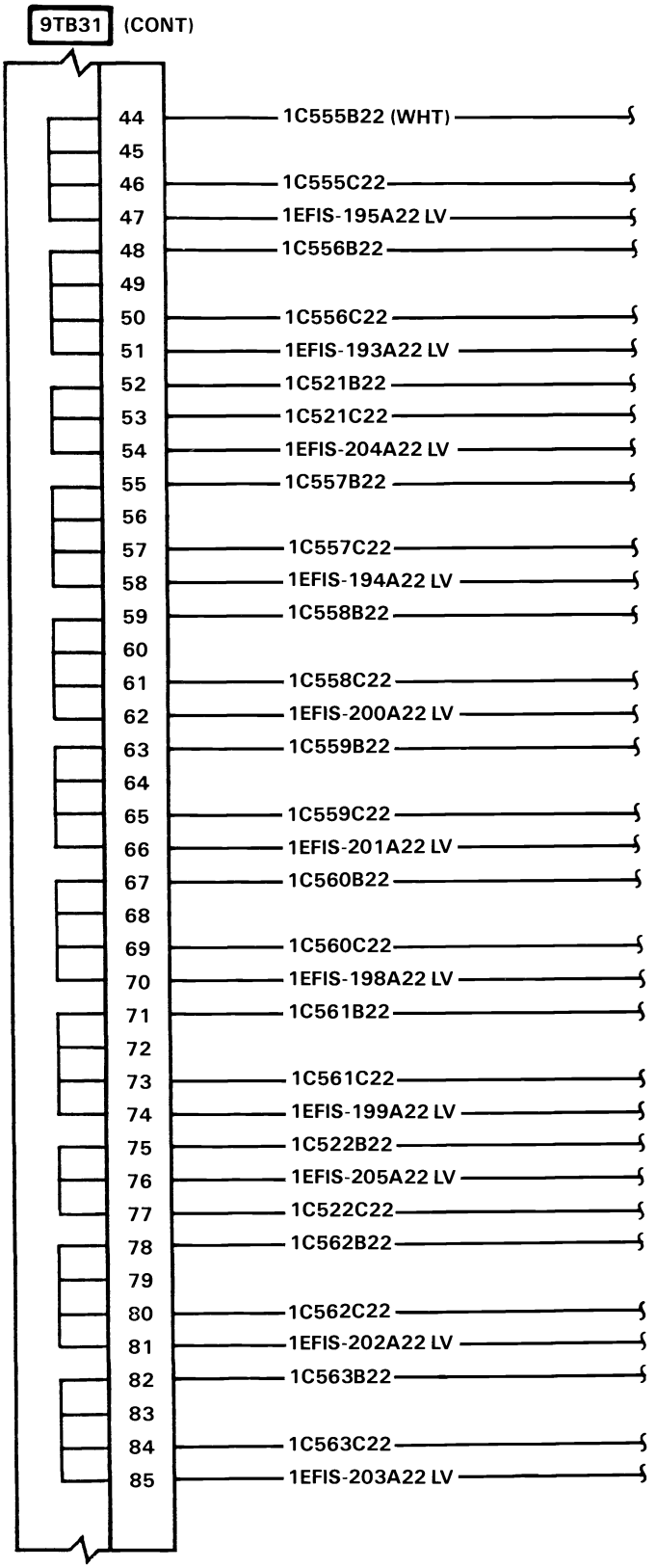



**Figure 98-7. Component replacement (Sheet 19)**

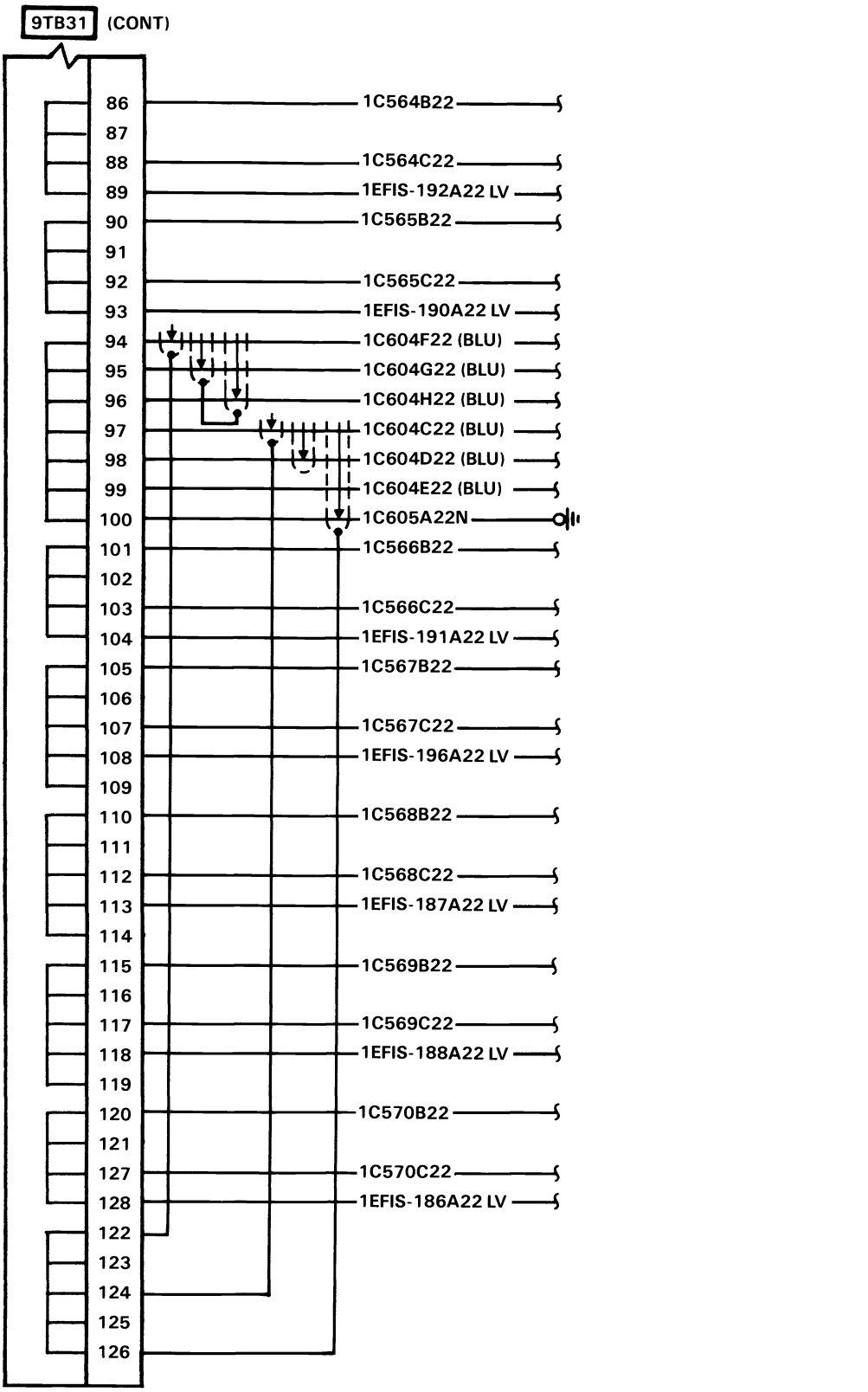



**Figure 98-7. Component replacement (Sheet 20)**

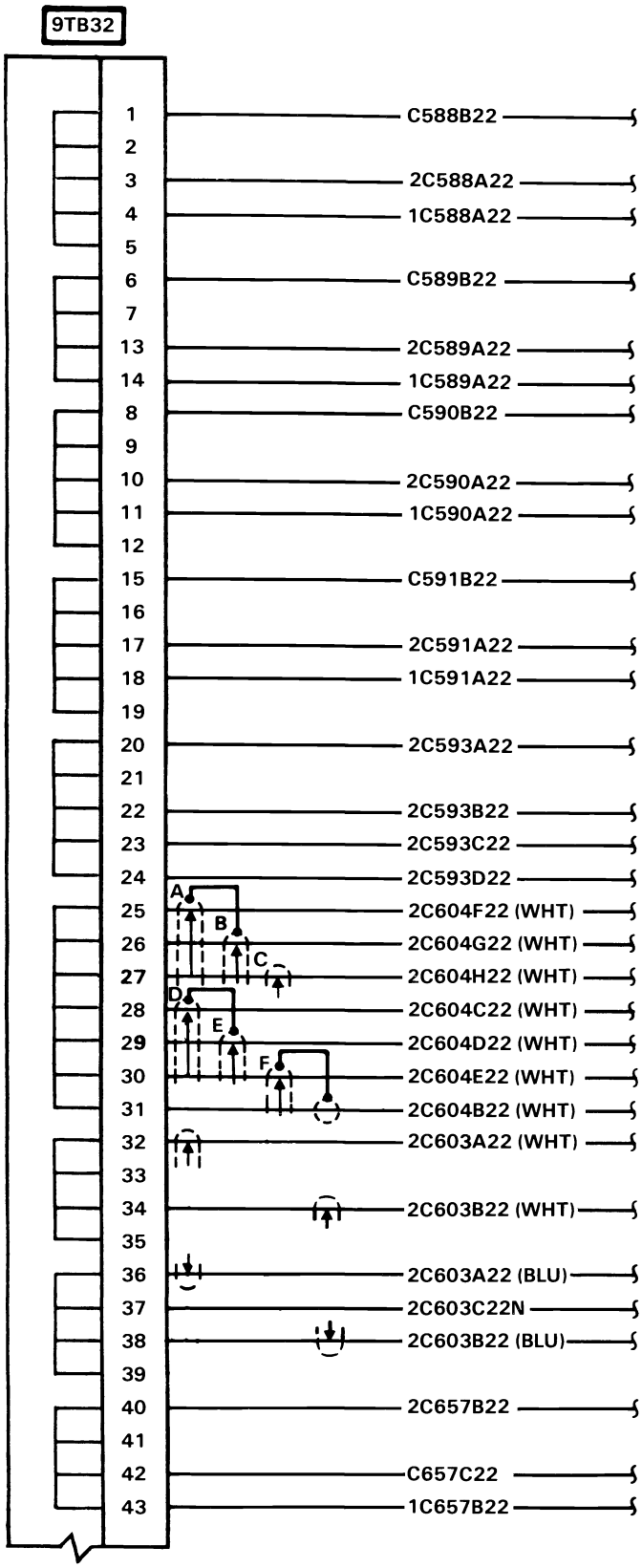
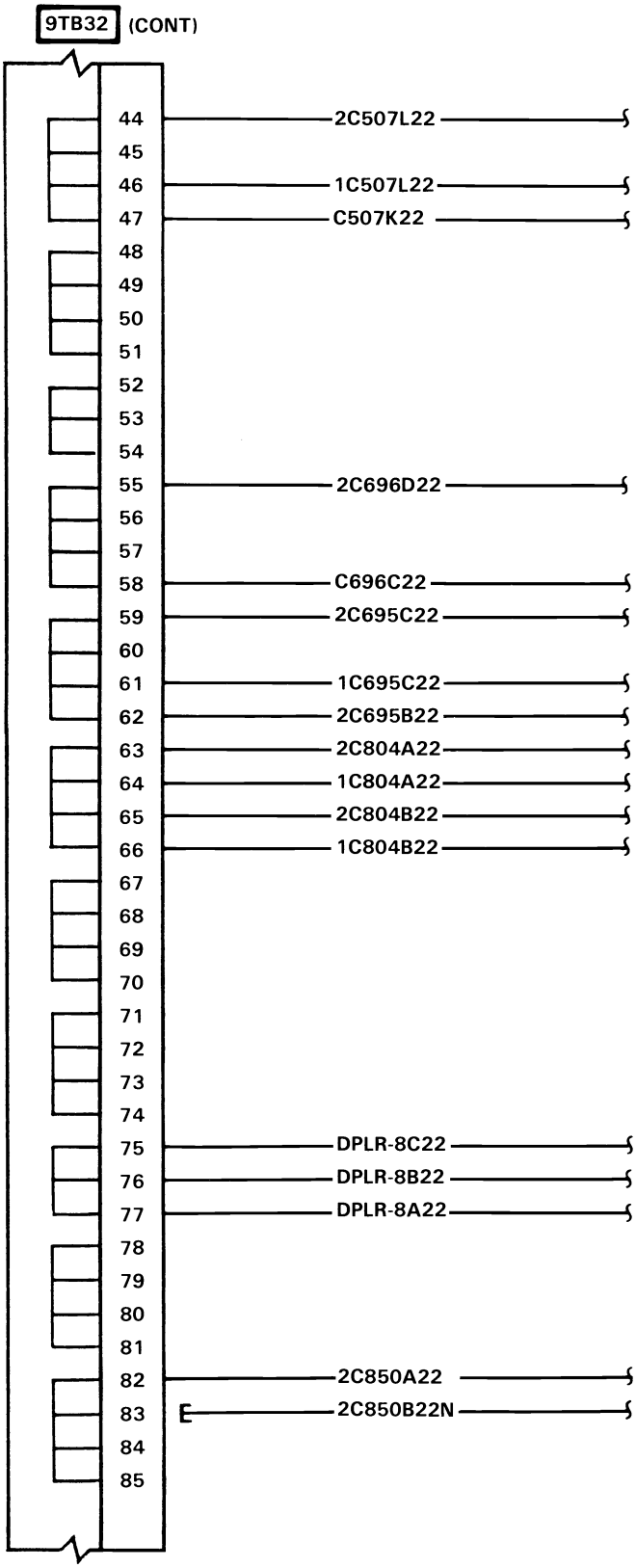
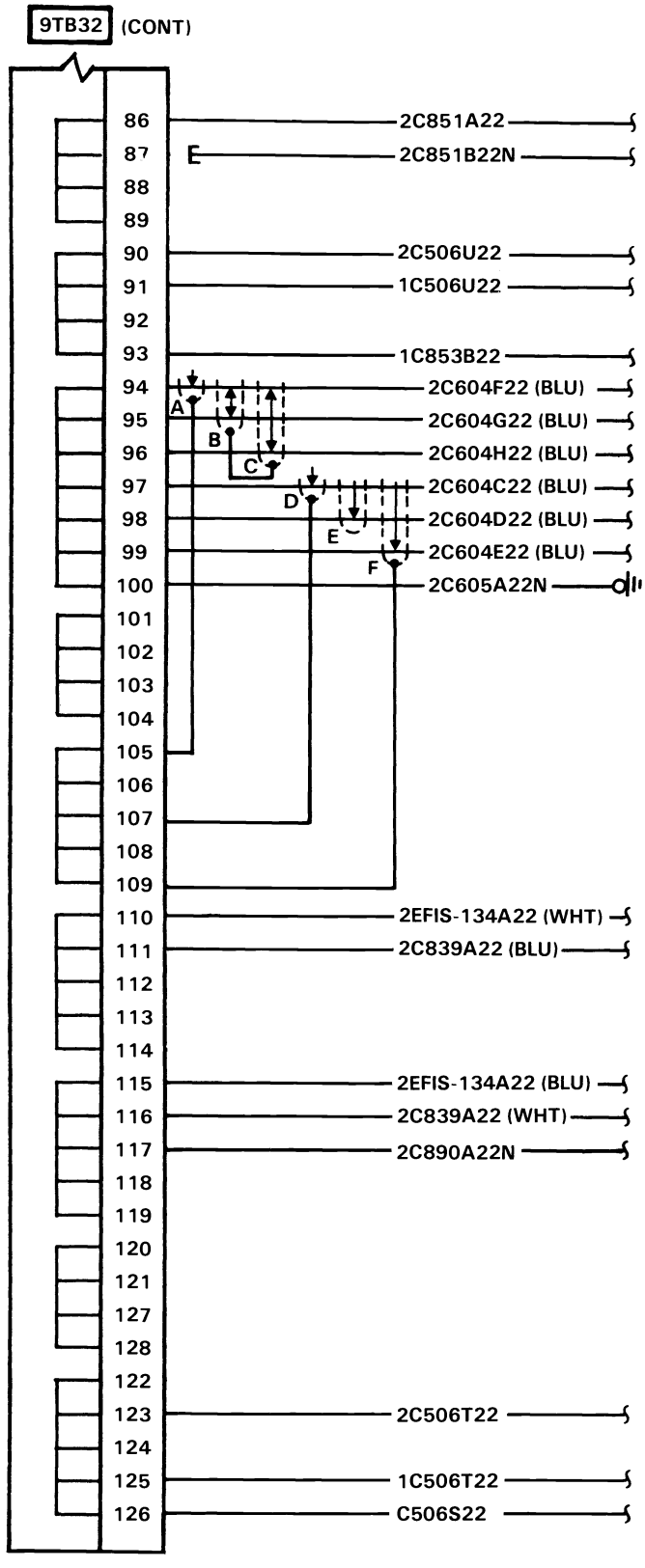



**Figure 98-7. Component replacement (Sheet 21)**



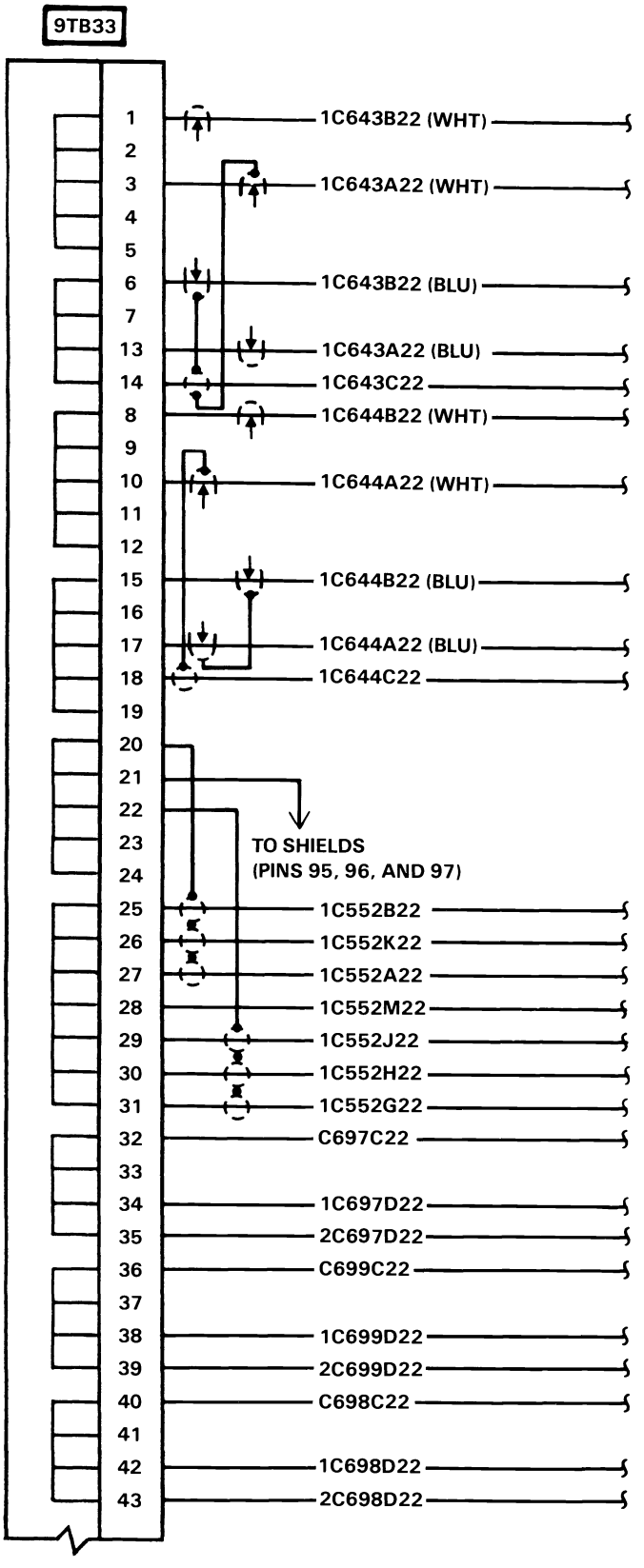

412-SAR-98-7-22

**Figure 98-7. Component replacement (Sheet 22)**



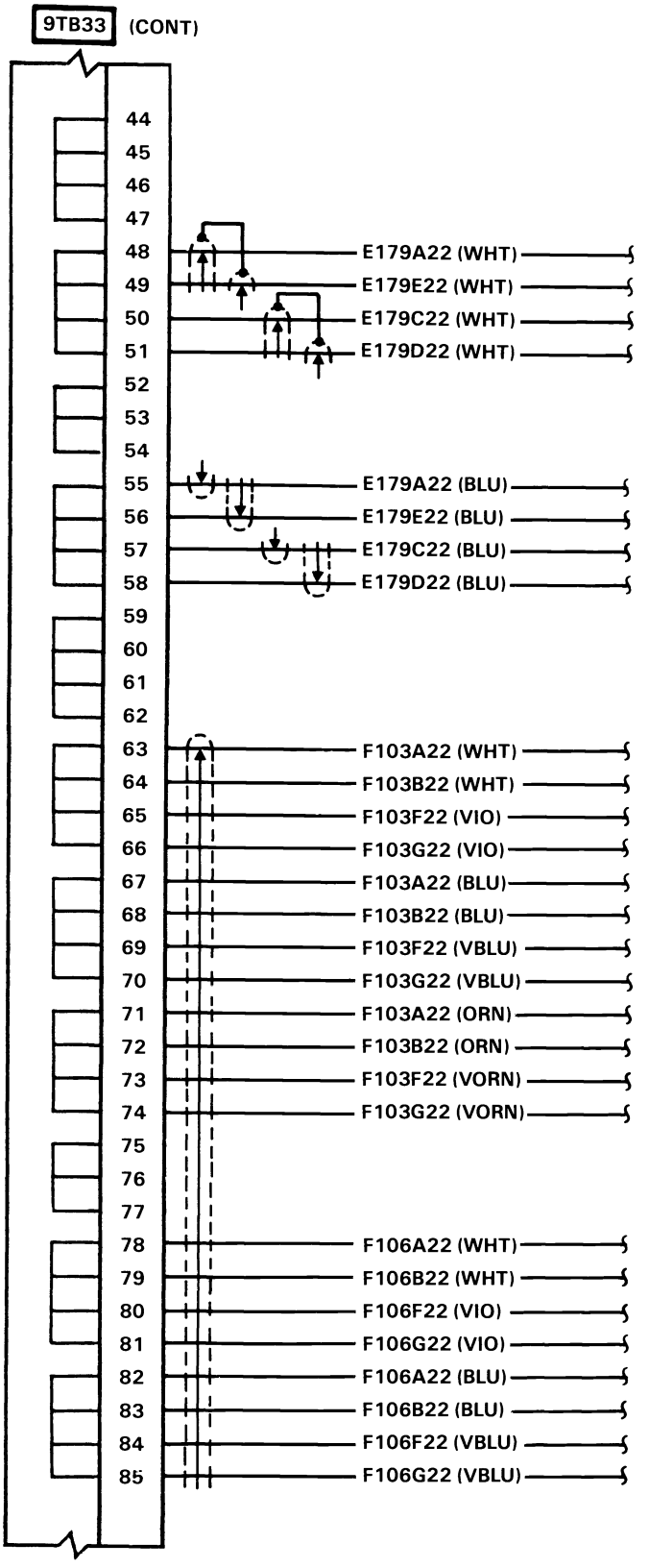

412-SAR-98-7-23

**Figure 98-7. Component replacement (Sheet 23)**



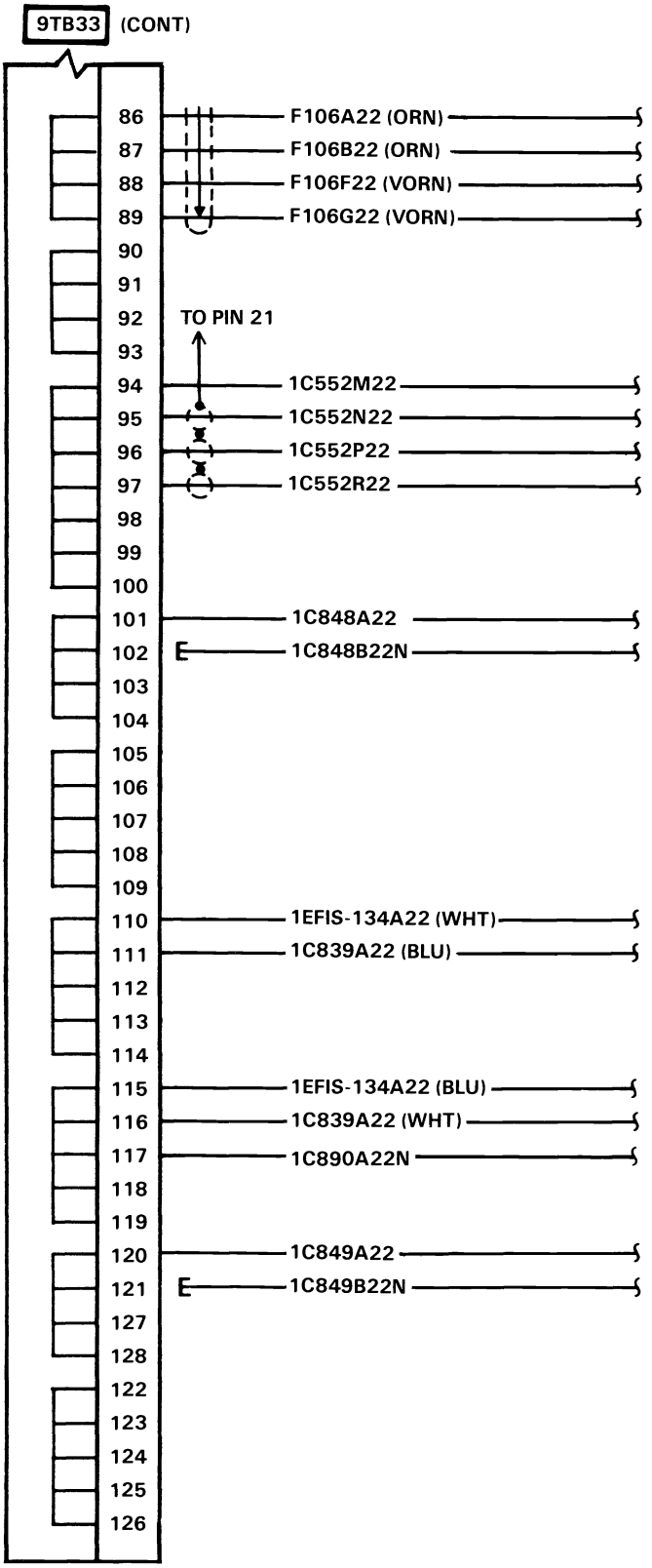

412-SAR-98-7-24

**Figure 98-7. Component replacement (Sheet 24)**




412-SAR-98-7-25

**Figure 98-7. Component replacement (Sheet 25)**




412-SAR-98-7-26

**Figure 98-7. Component replacement (Sheet 26)**

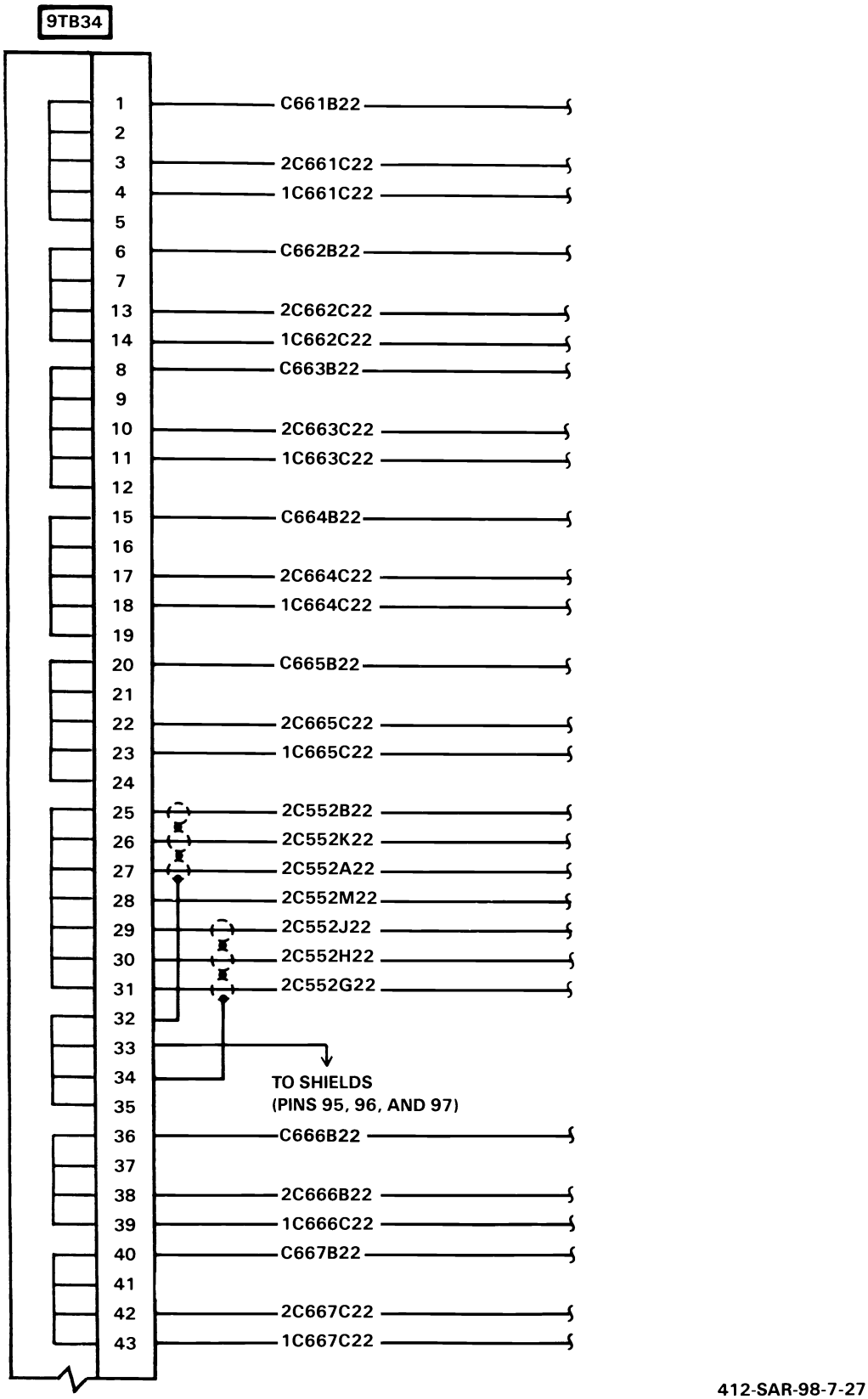



**Figure 98-7. Component replacement (Sheet 27)**

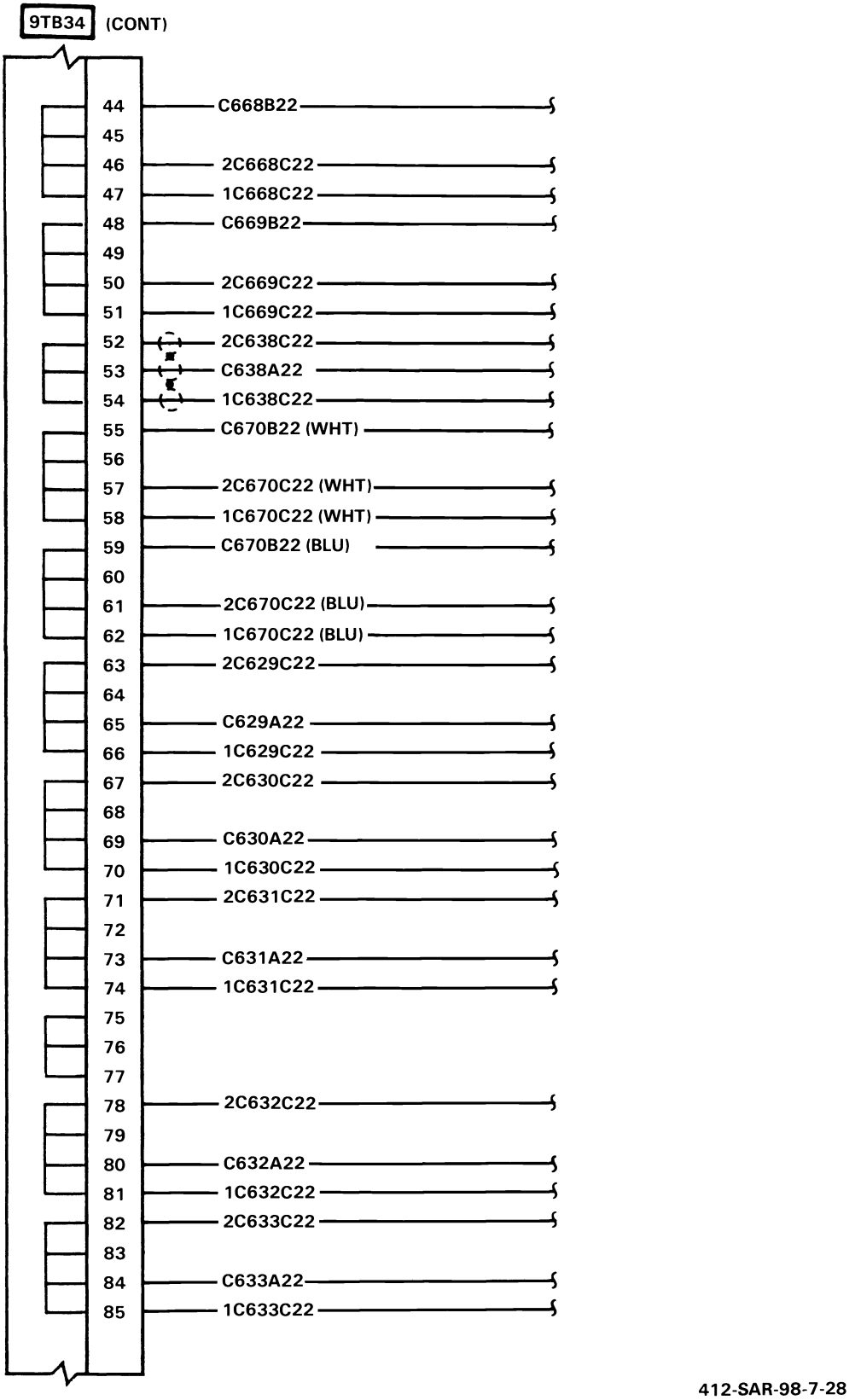
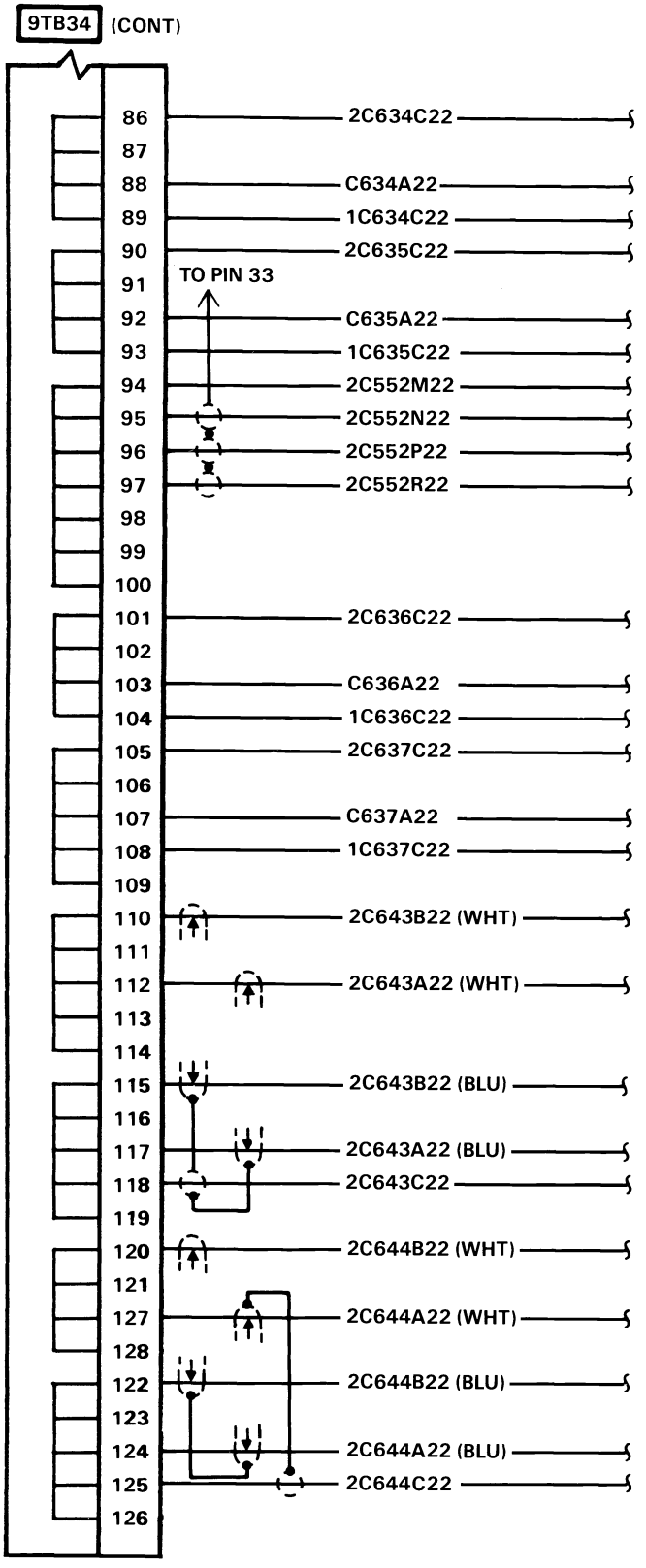
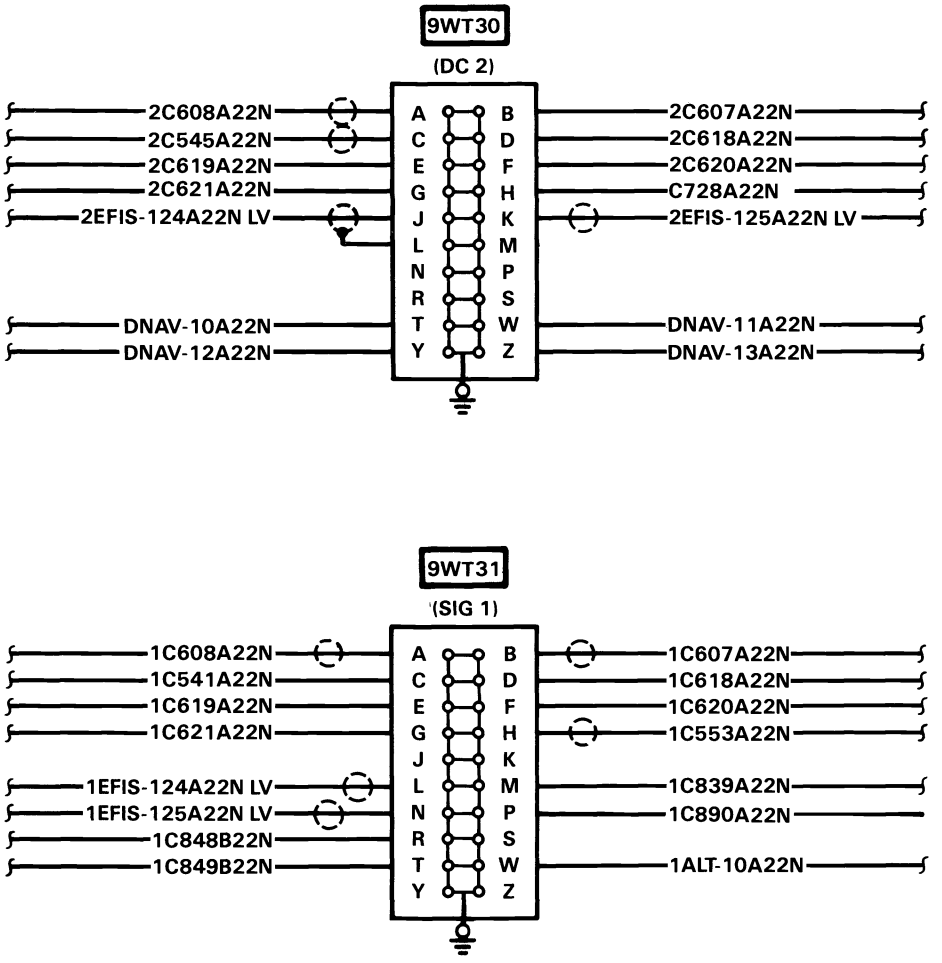



**Figure 98-7. Component replacement (Sheet 28)**



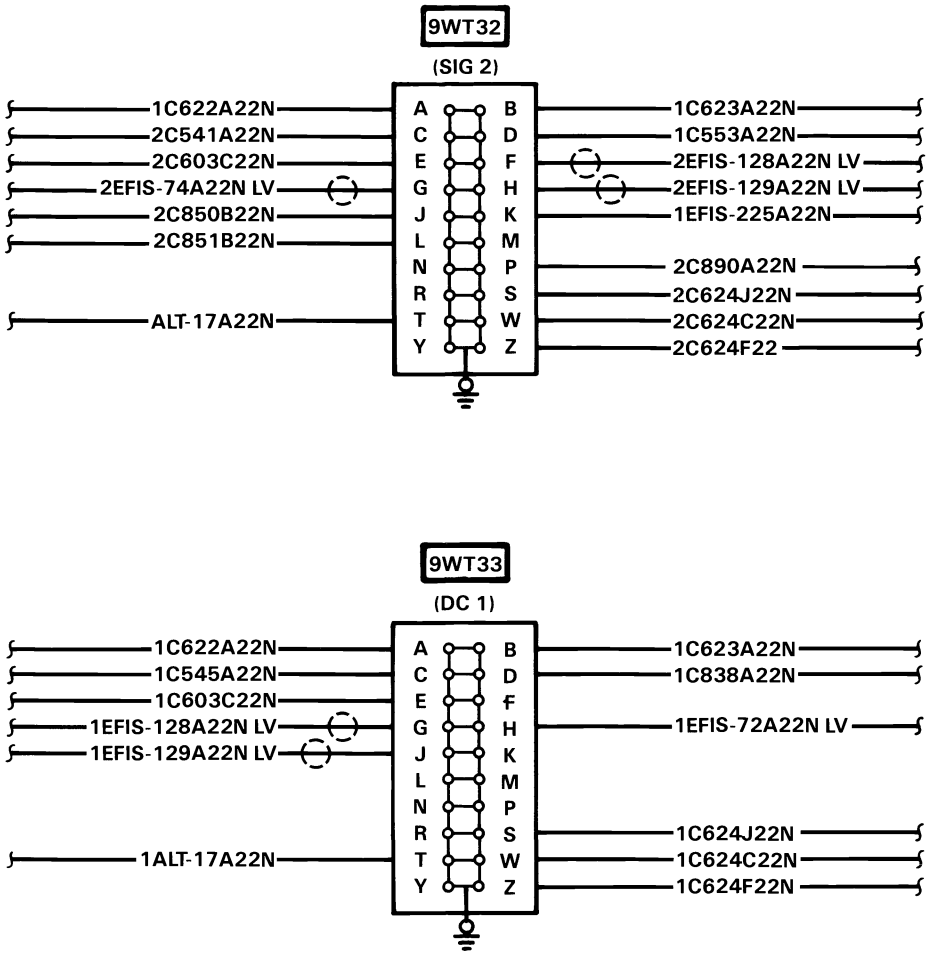

412-SAR-98-7-29

**Figure 98-7. Component replacement (Sheet 29)**



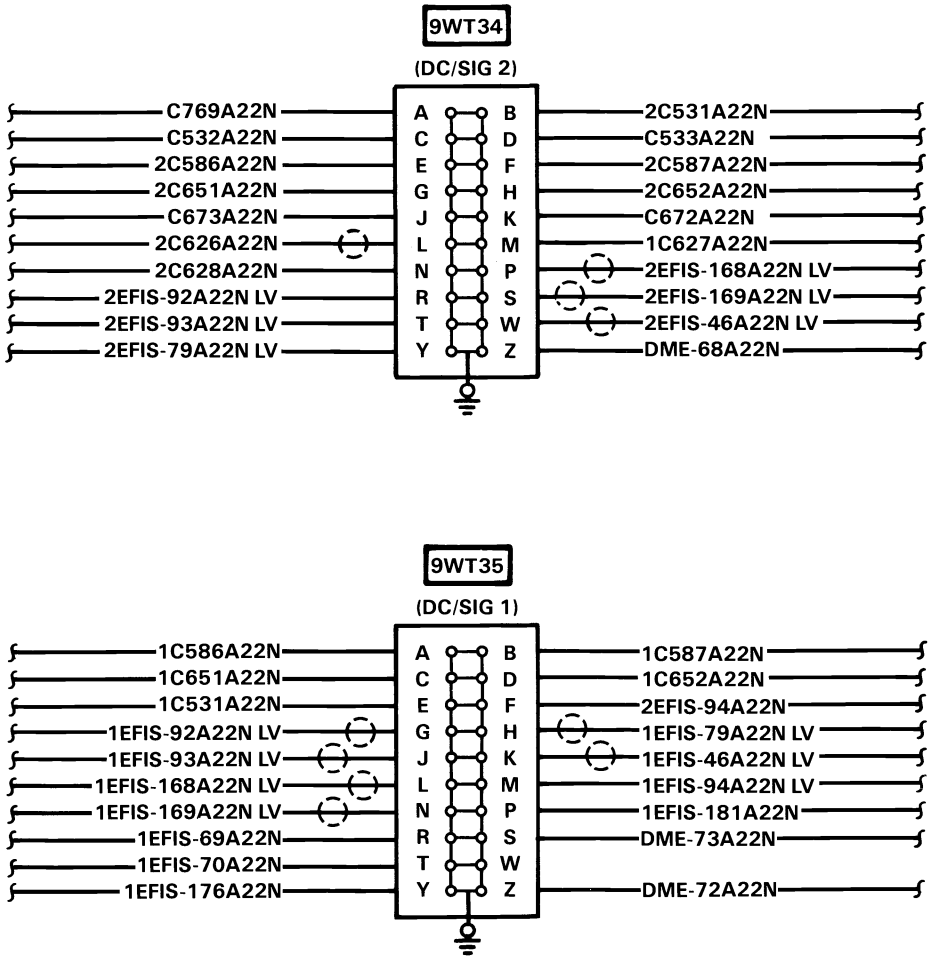

412-SAR-98-7-30

**Figure 98-7. Component replacement (Sheet 30)**



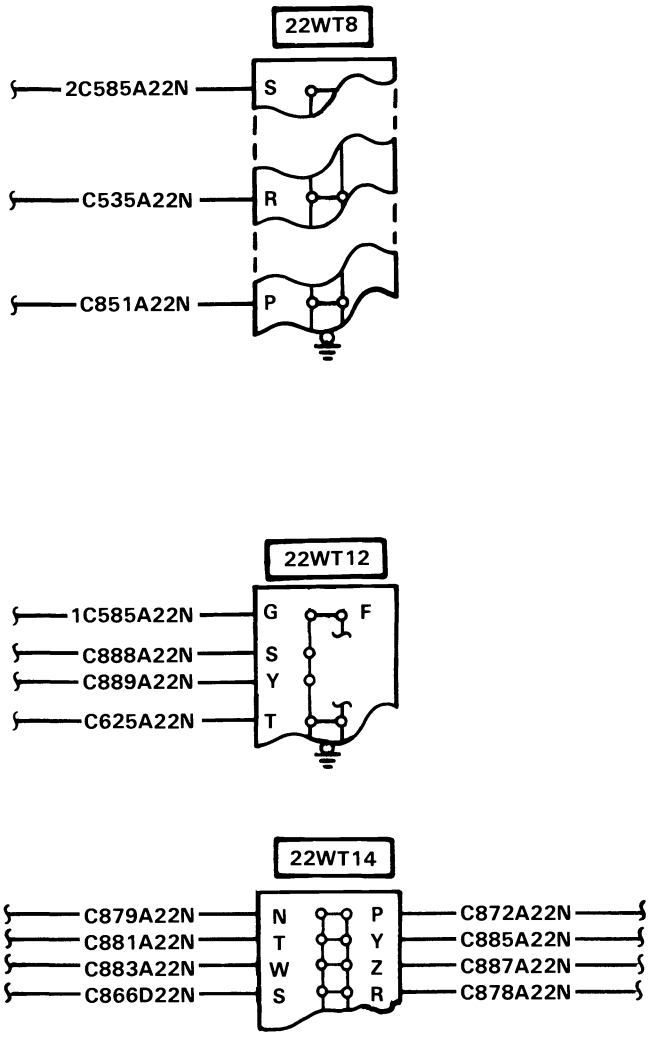

412-SAR-98-7-31

**Figure 98-7. Component replacement (Sheet 31)**



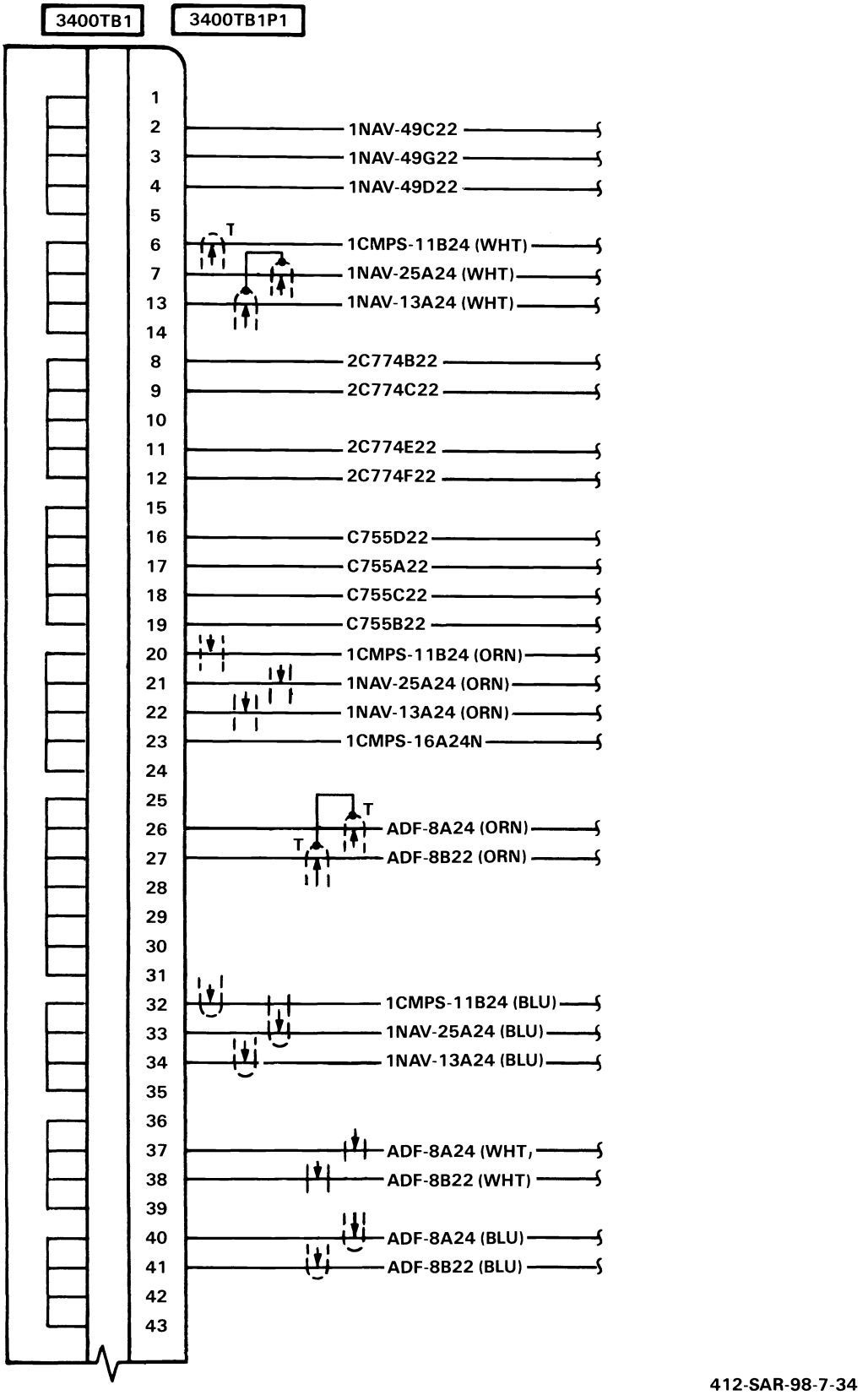

412-SAR-98-7-32

**Figure 98-7. Component replacement (Sheet 32)**




412-SAR-98-7-33

**Figure 98-7. Component replacement (Sheet 33)**




412-SAR-98-7-34

**Figure 98-7. Component replacement (Sheet 34)**

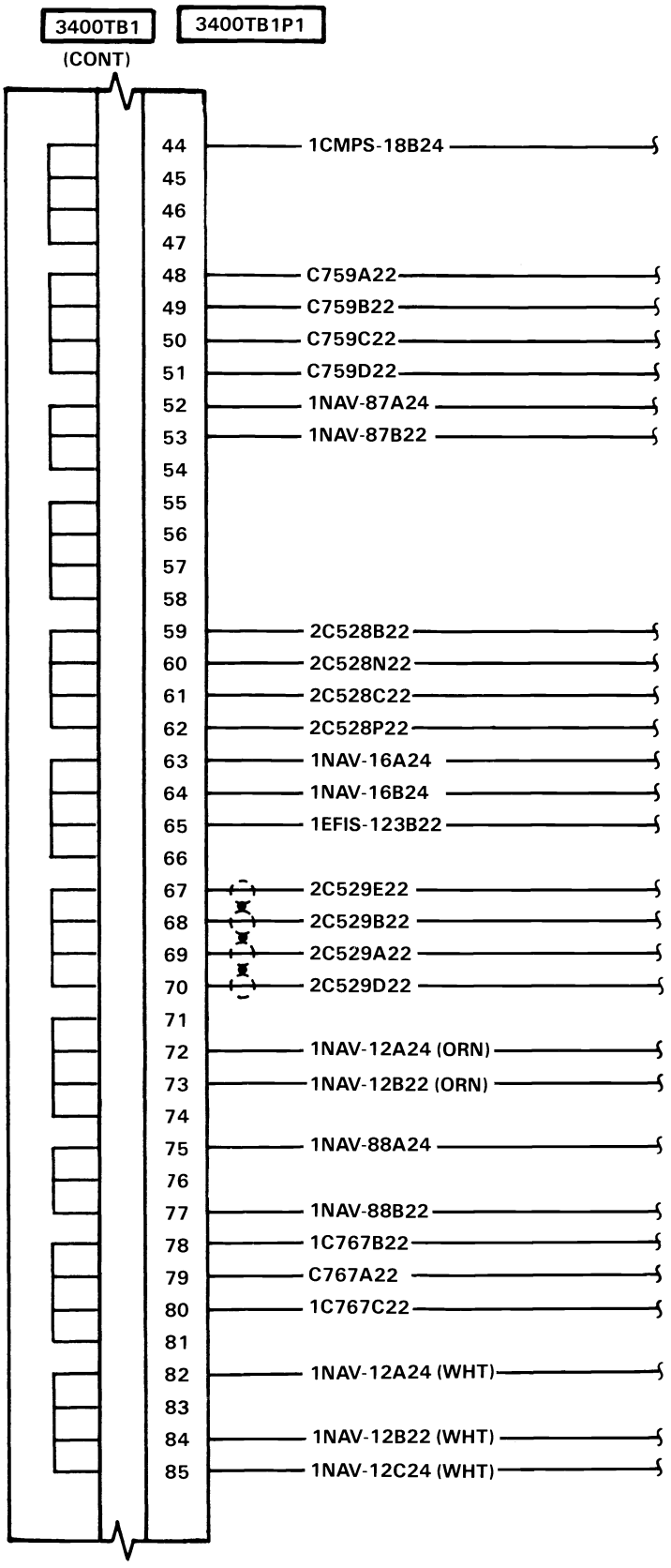



**Figure 98-7. Component replacement (Sheet 35)**

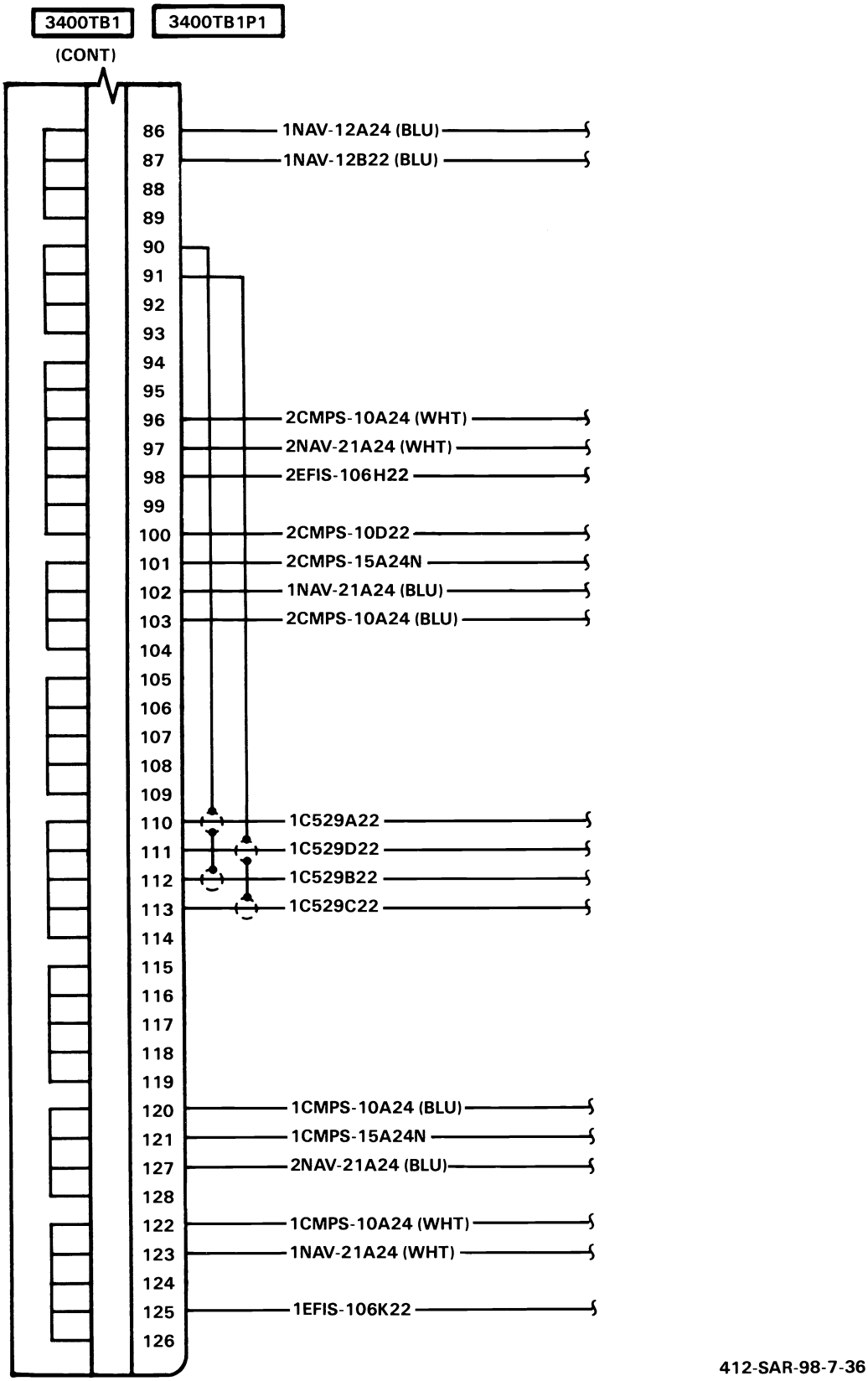



**Figure 98-7. Component replacement (Sheet 36)**

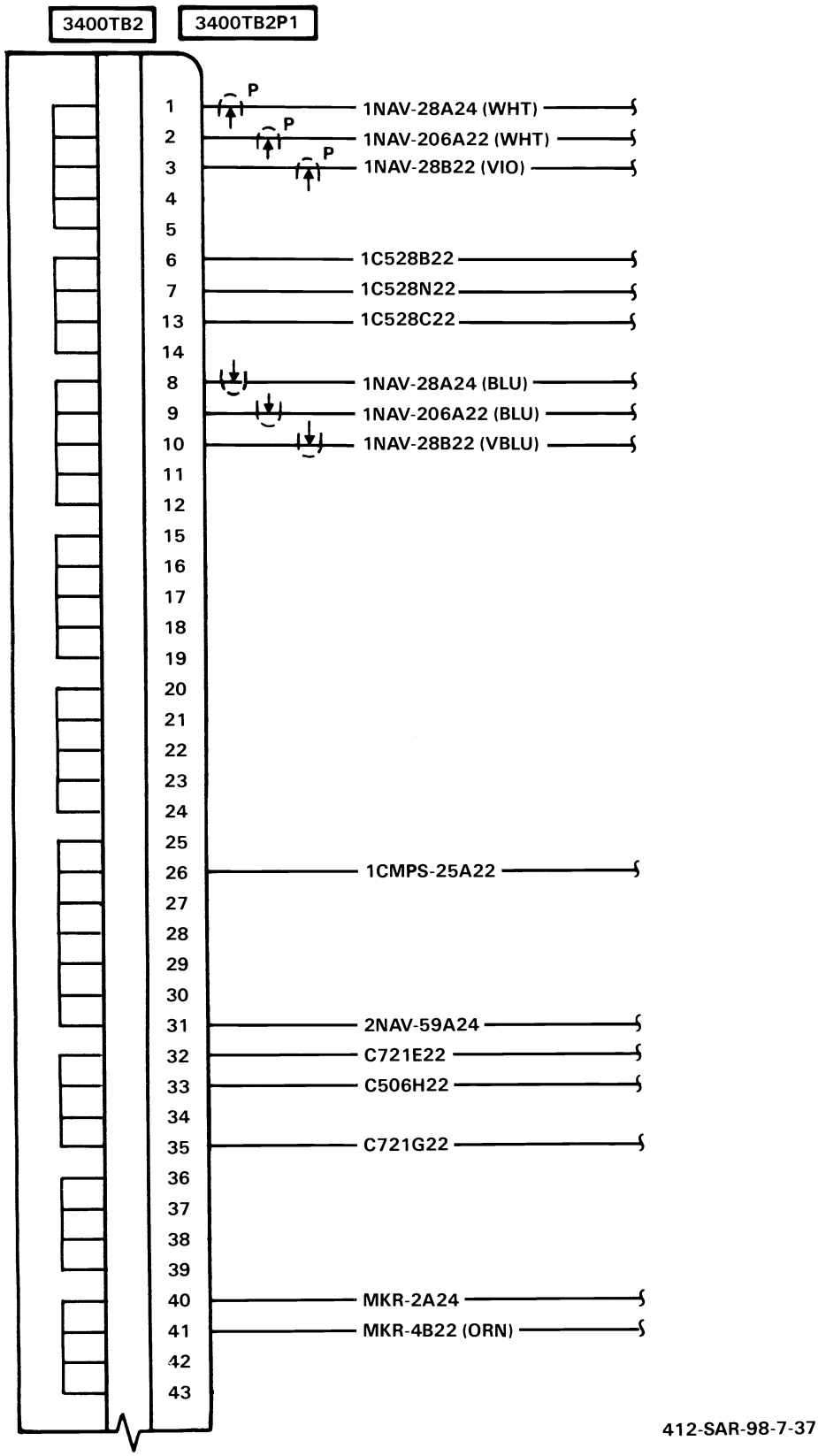



**Figure 98-7. Component replacement (Sheet 37)**

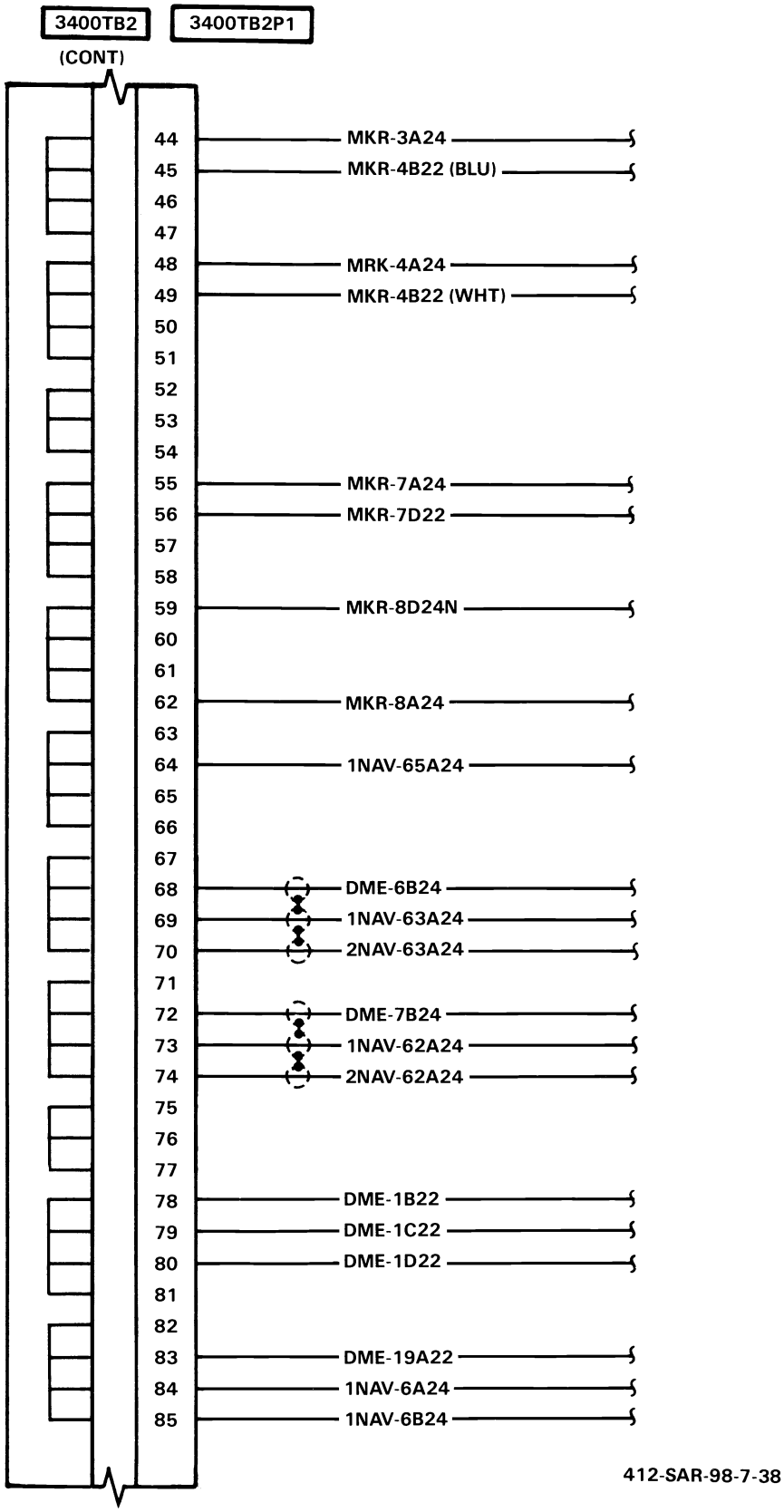



**Figure 98-7. Component replacement (Sheet 38)**

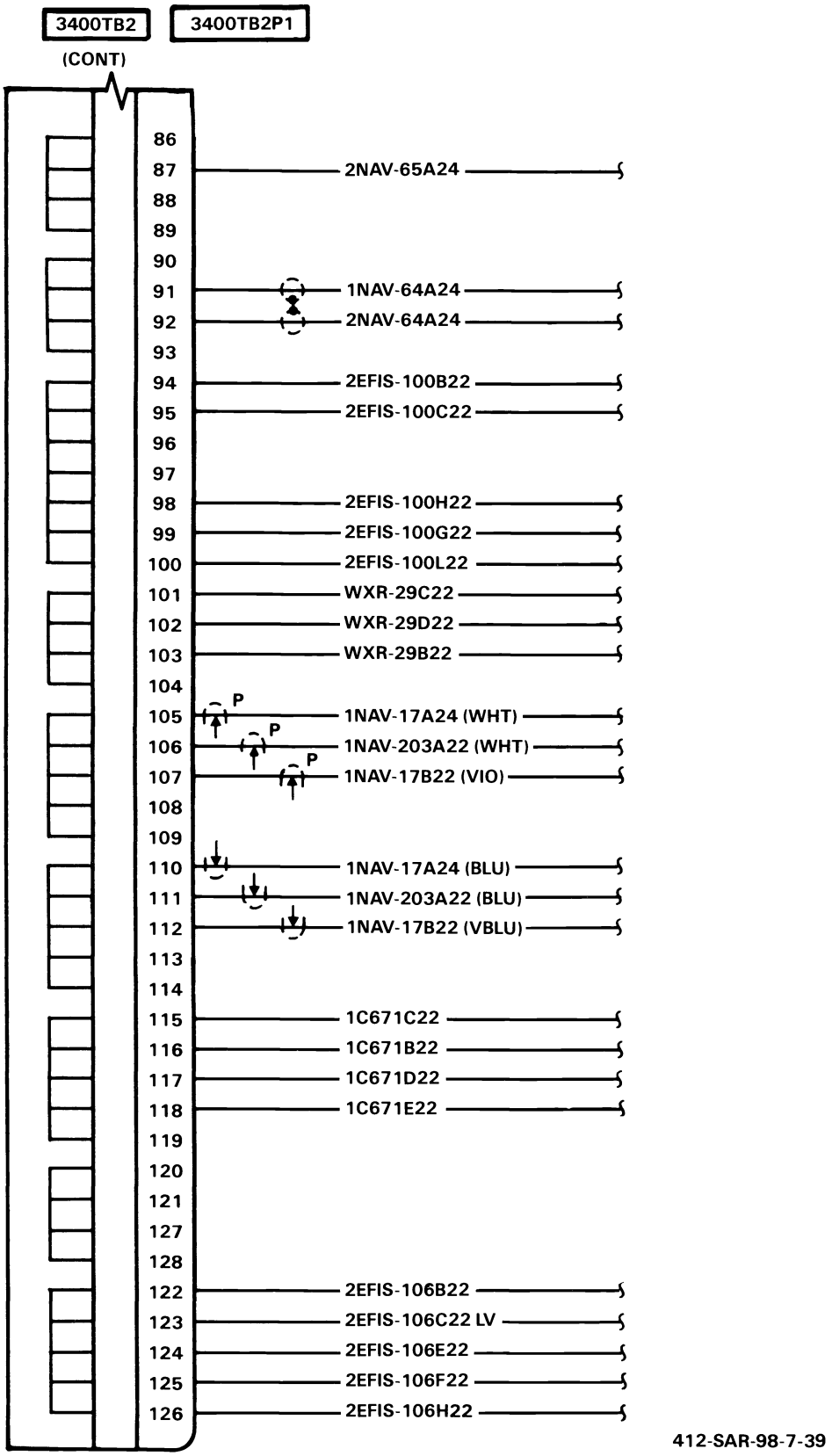
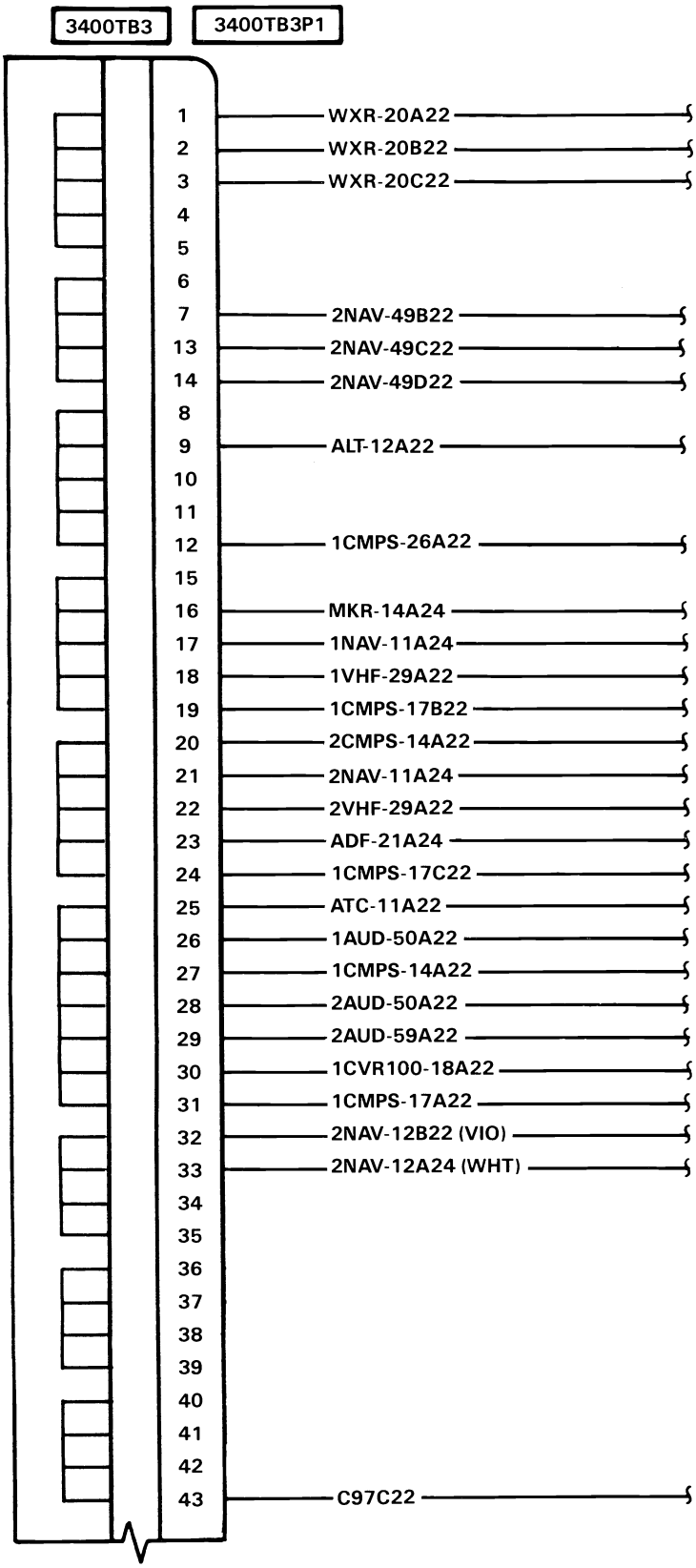
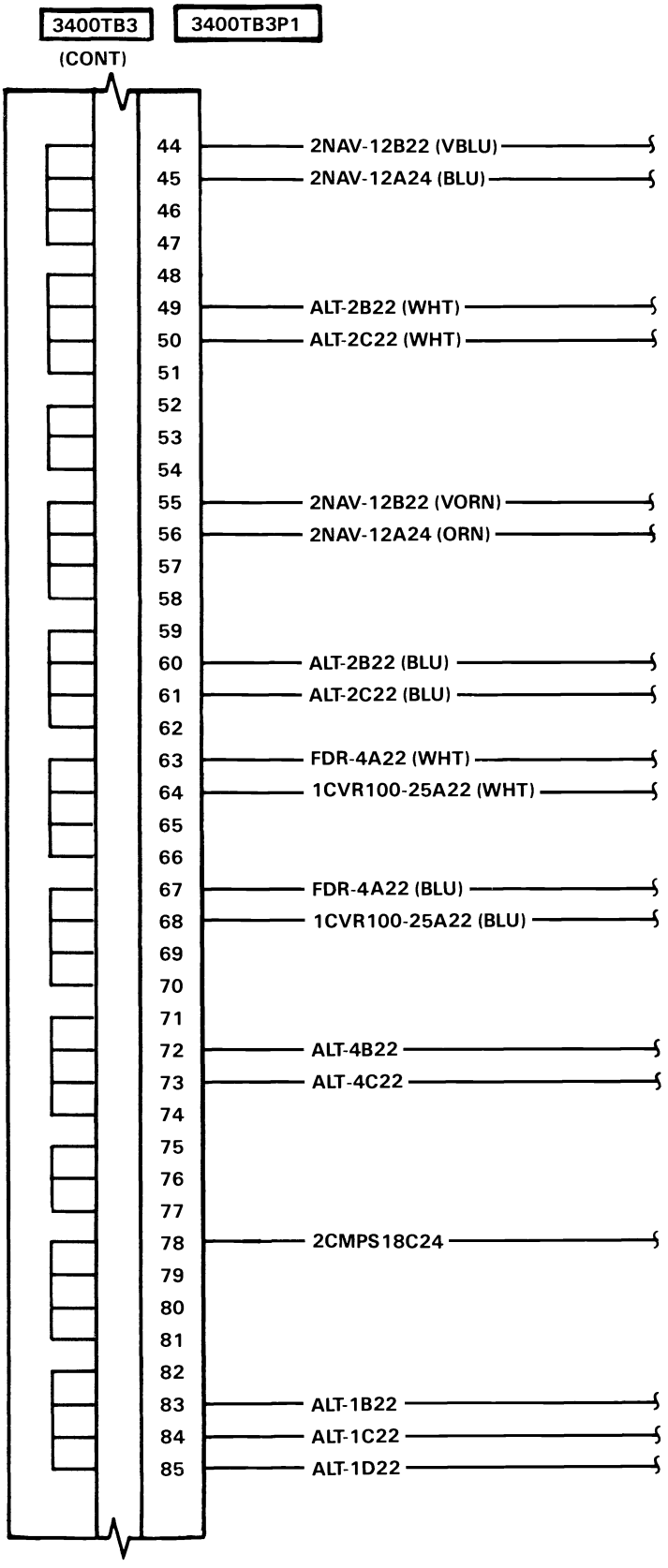



**Figure 98-7. Component replacement (Sheet 39)**



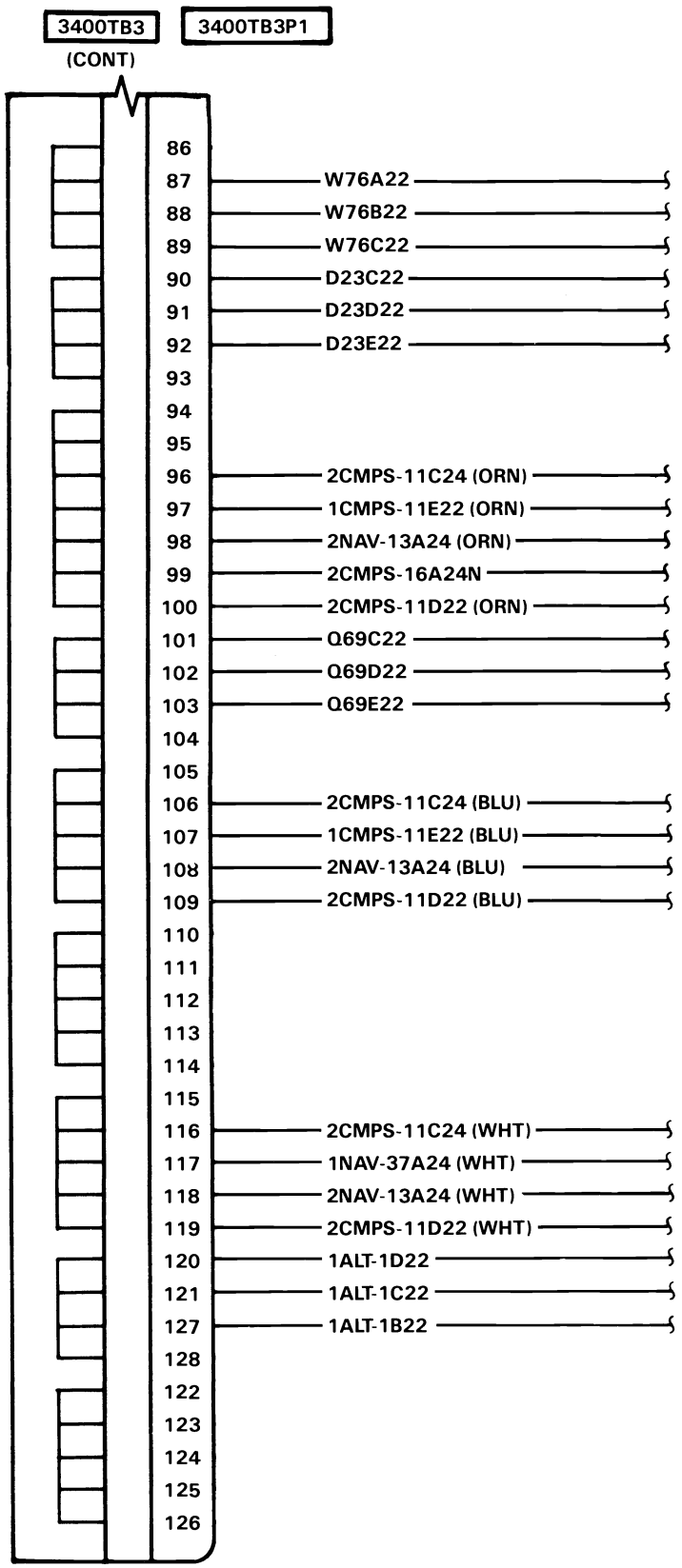

412-SAR-98-7-40

**Figure 98-7. Component replacement (Sheet 40)**



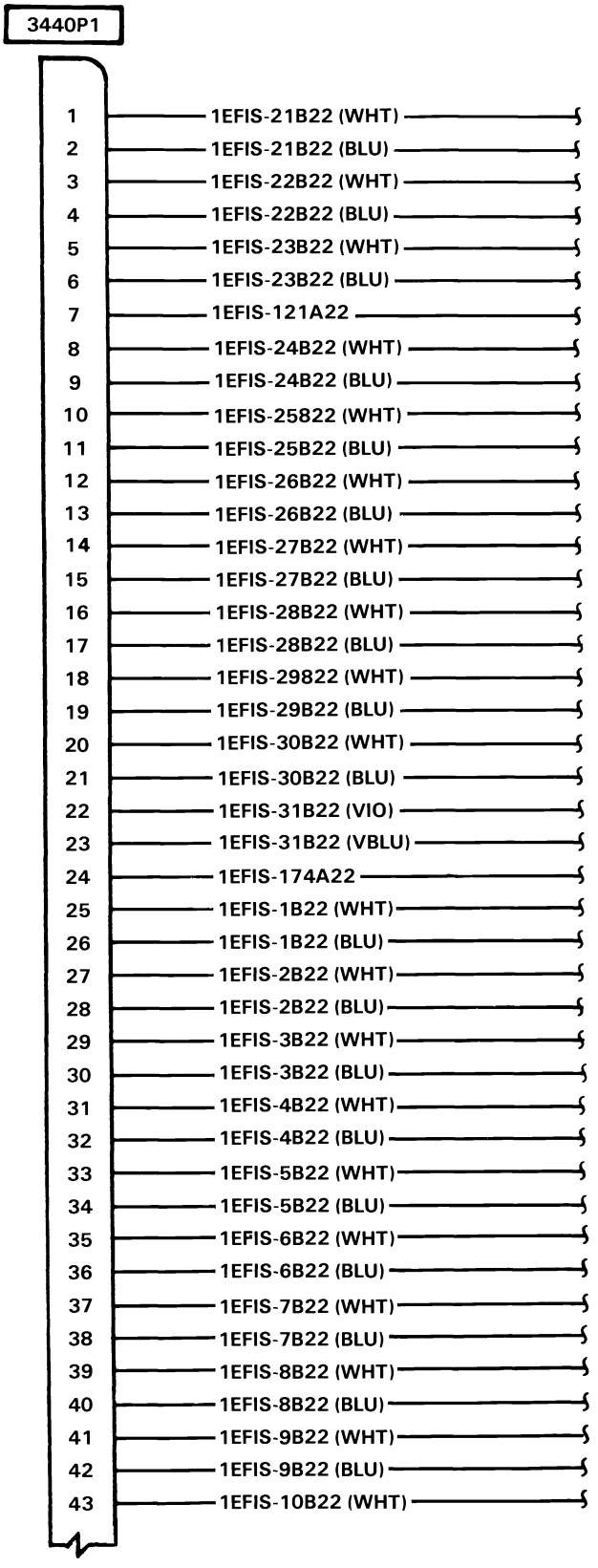

412-SAR-98-7-41

**Figure 98-7. Component replacement (Sheet 41)**



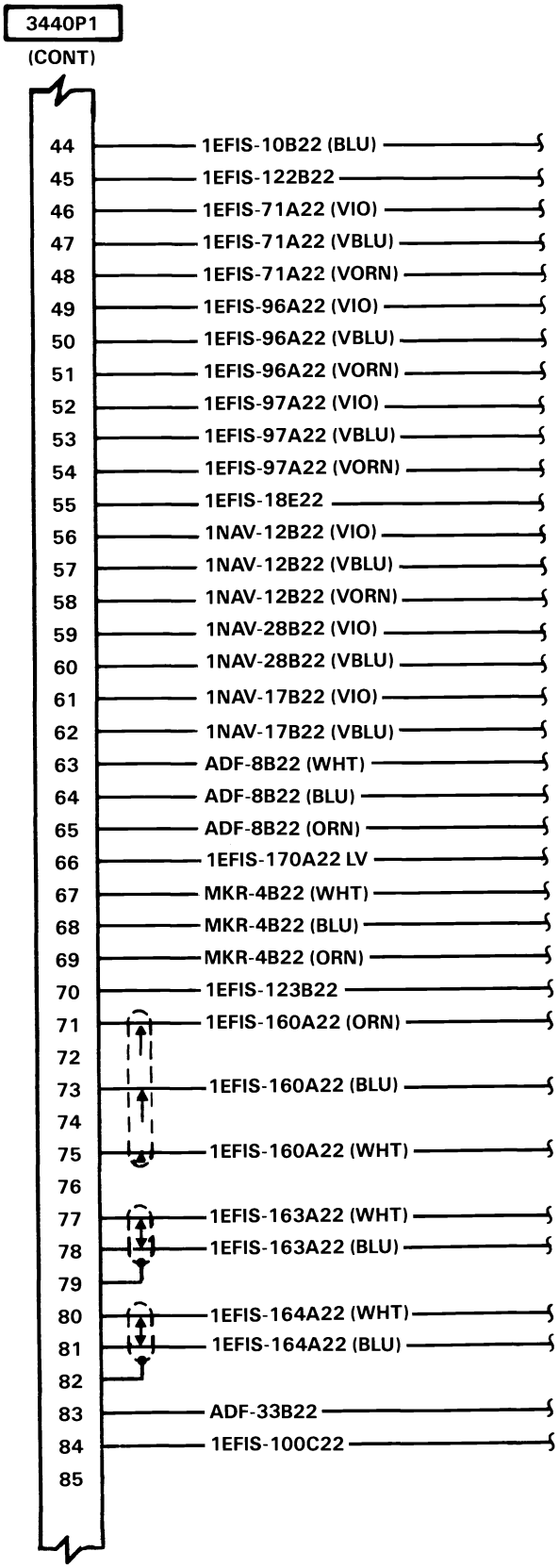

412-SAR-98-7-42

**Figure 98-7. Component replacement (Sheet 42)**



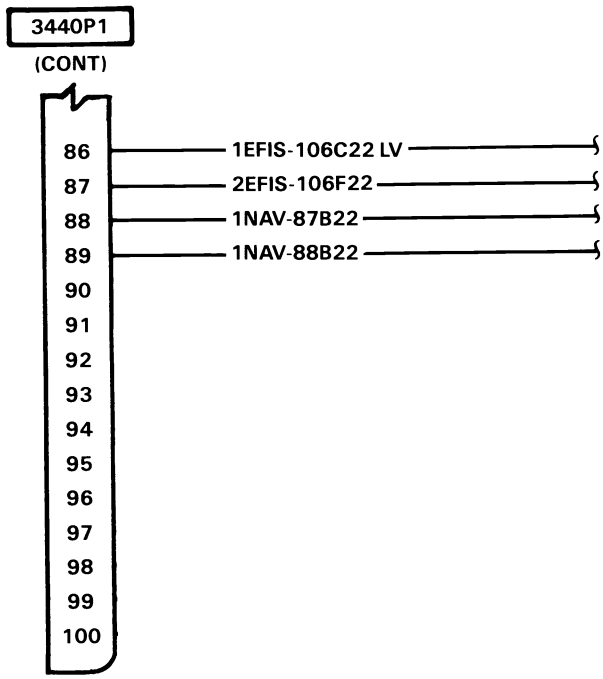

412-SAR-98-7-43

**Figure 98-7. Component replacement (Sheet 43)**



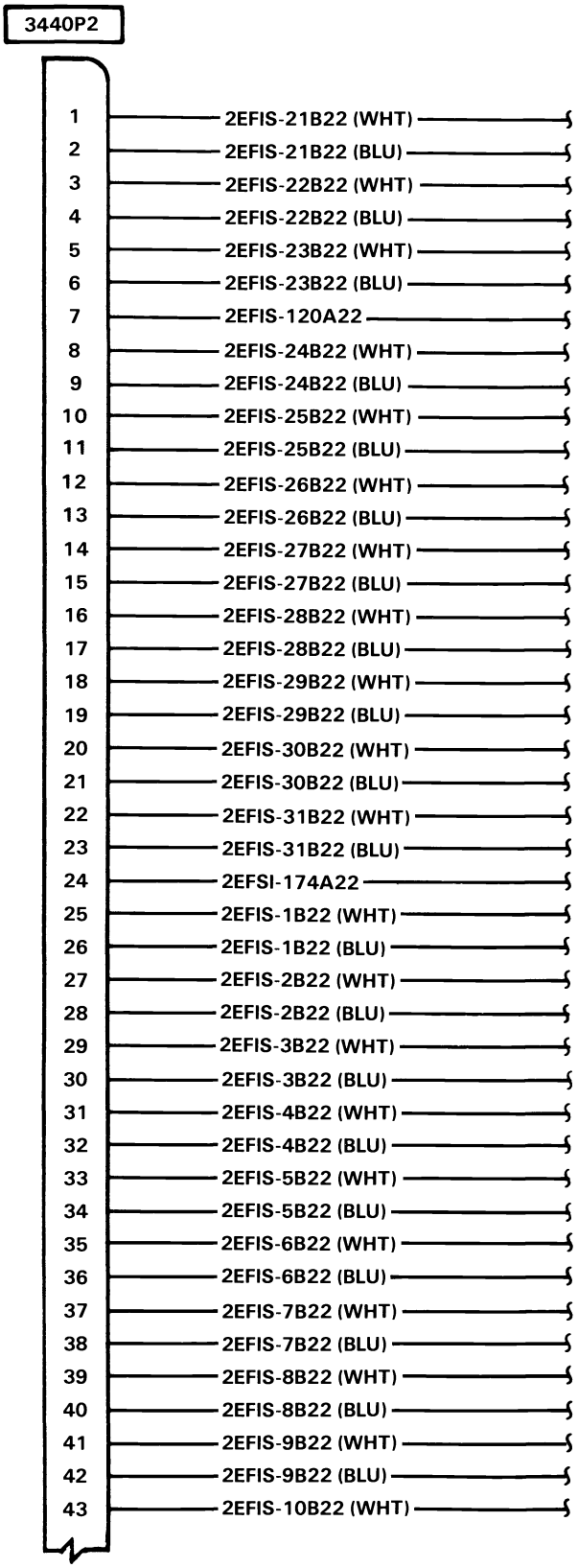

412-SAR-98-7-44

**Figure 98-7. Component replacement (Sheet 44)**



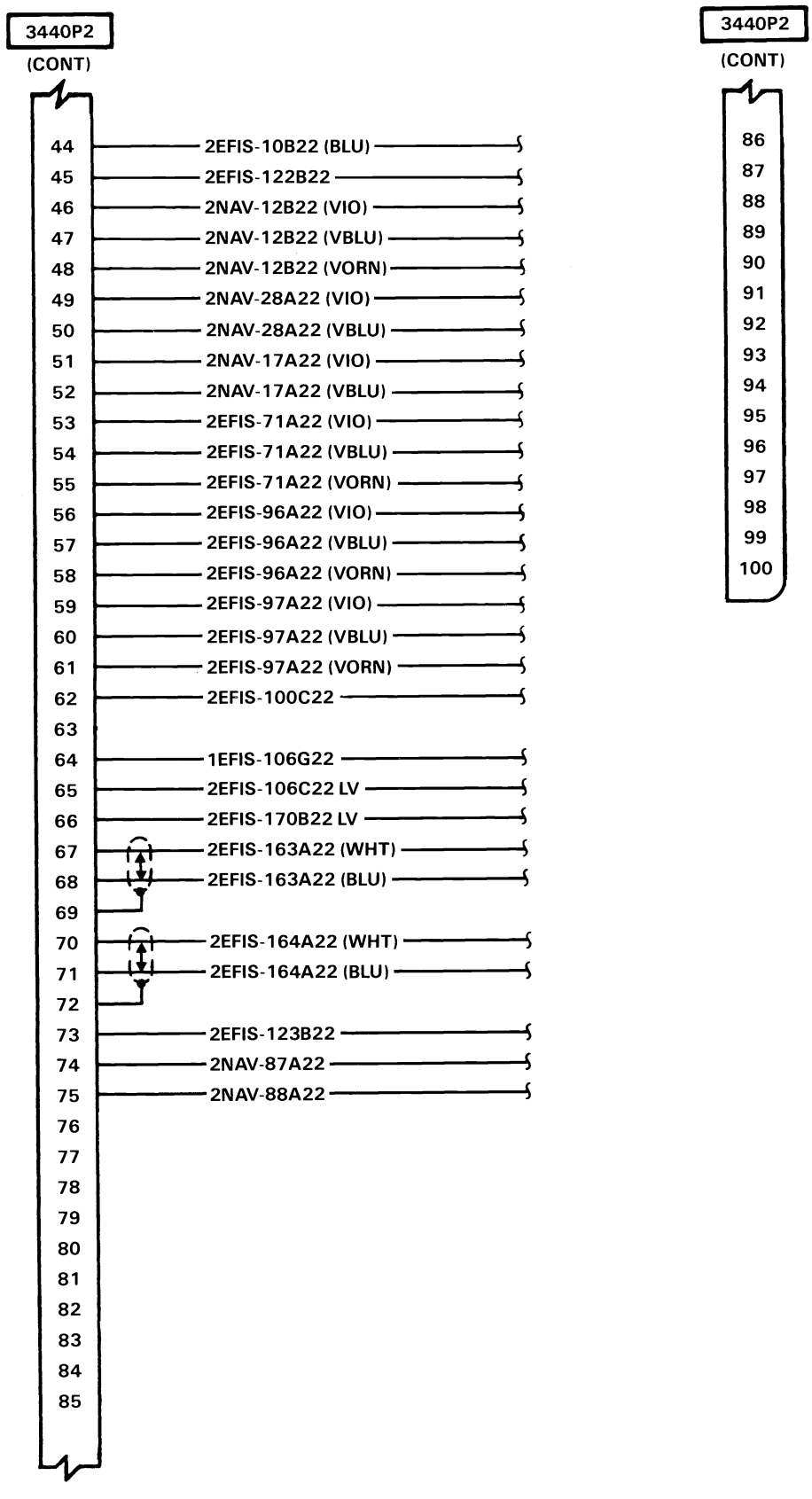

412-SAR-98-7-45

**Figure 98-7. Component replacement (Sheet 45)**



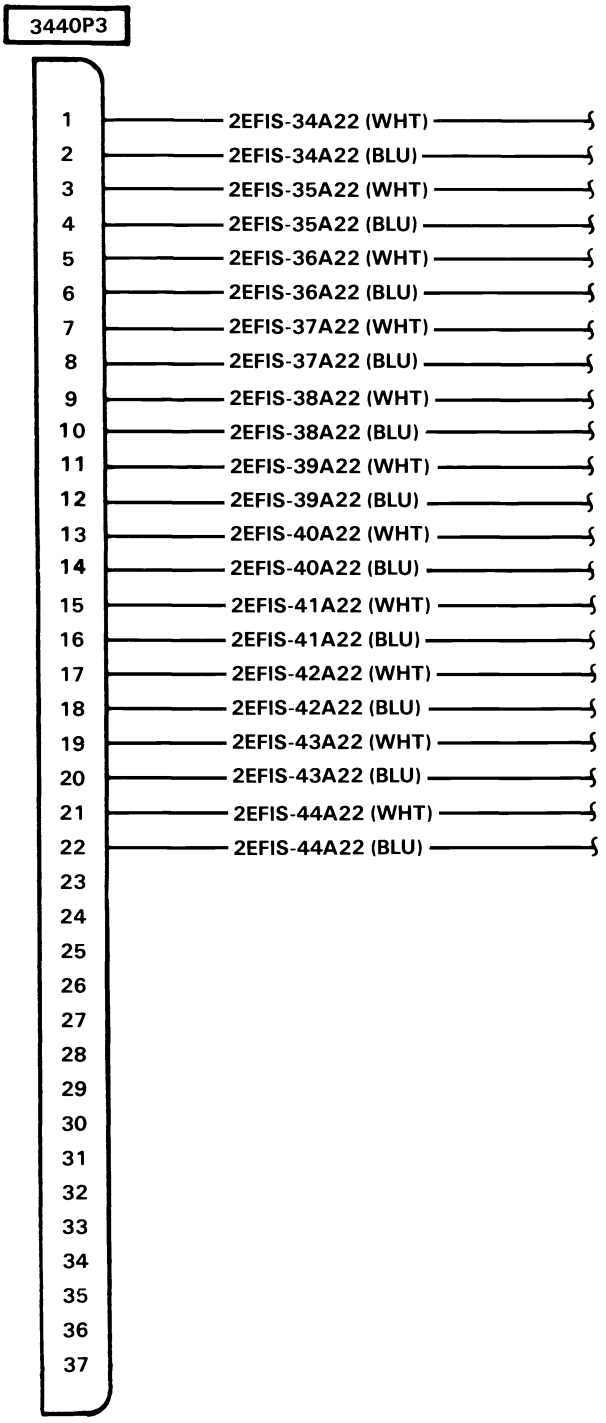

412-SAR-98-7-46

**Figure 98-7. Component replacement (Sheet 46)**



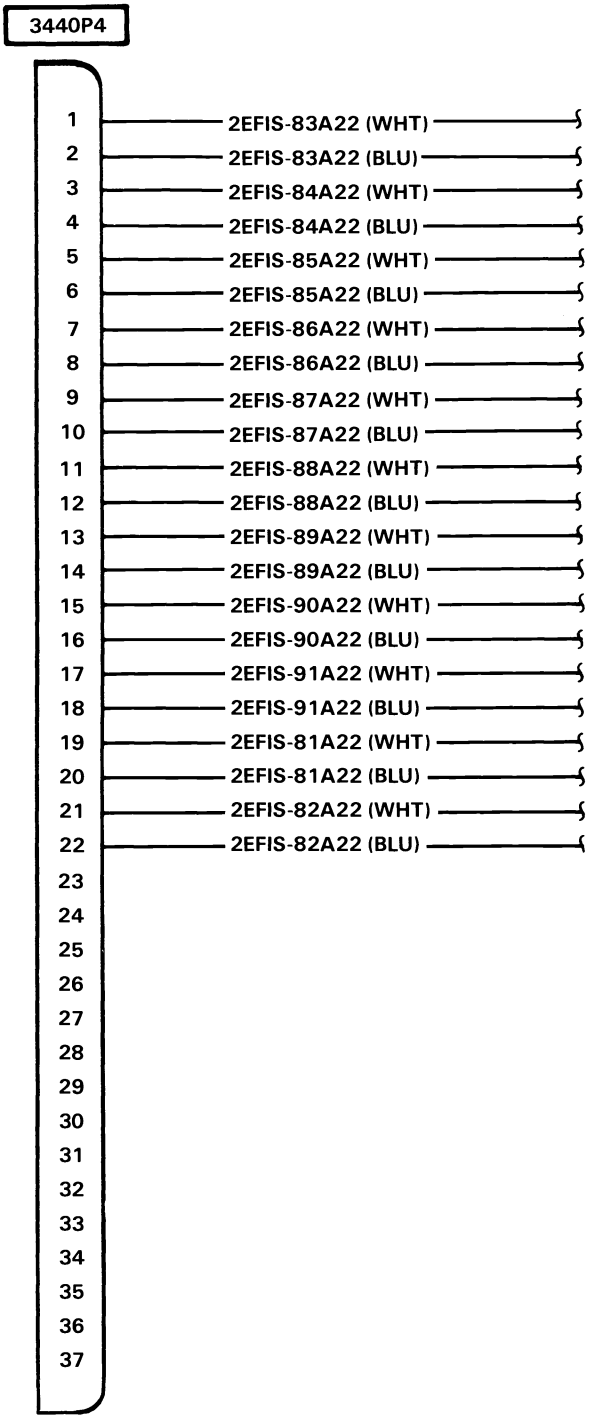

412-SAR-98-7-47

**Figure 98-7. Component replacement (Sheet 47)**



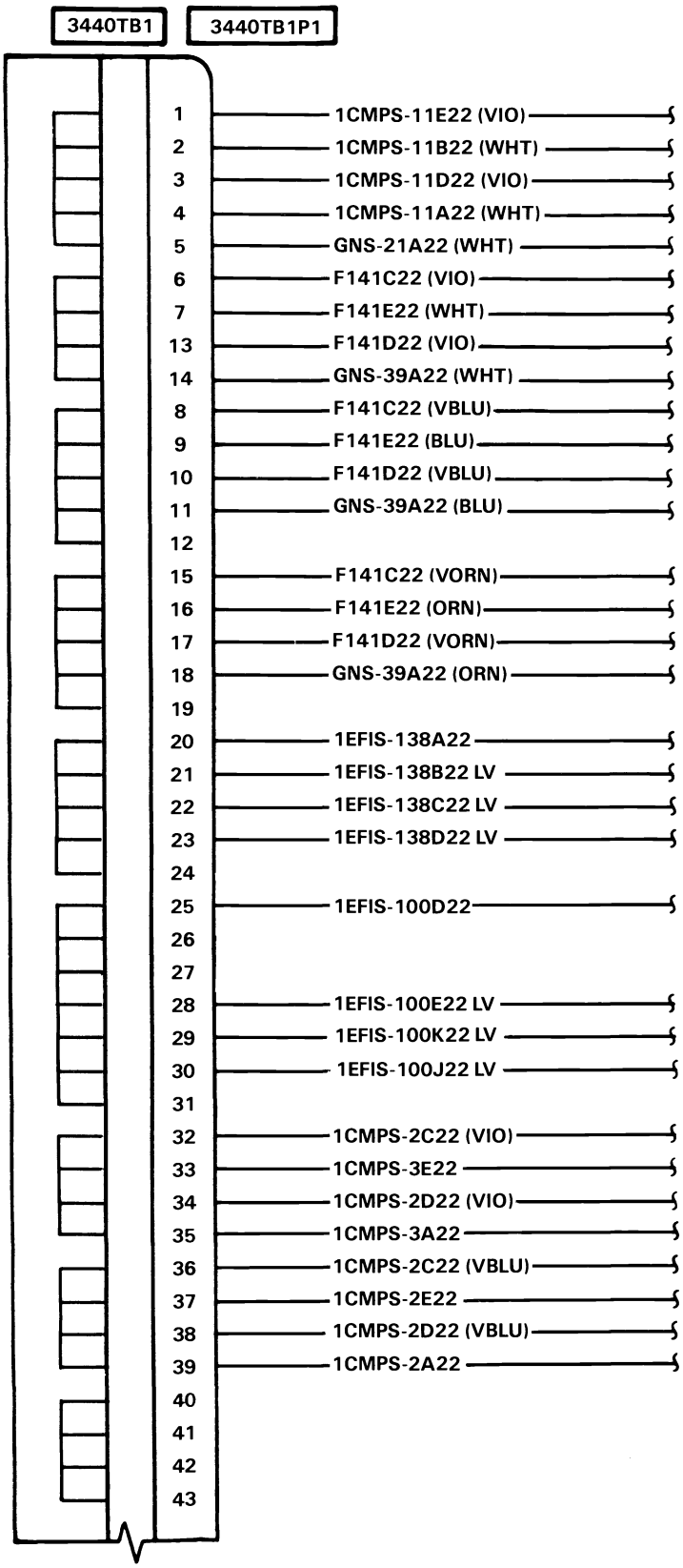

412-SAR-98-7-48

**Figure 98-7. Component replacement (Sheet 48)**



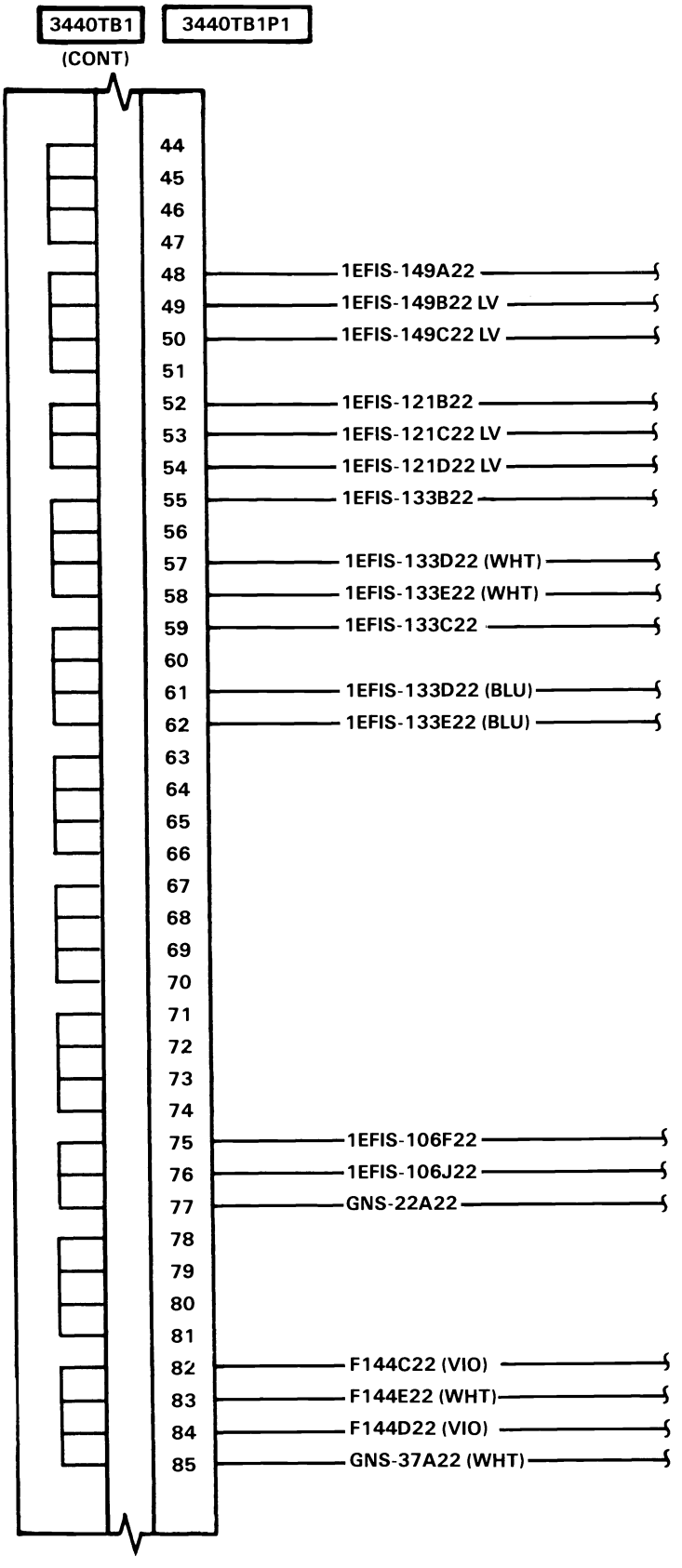

412-SAR-98-7-49

**Figure 98-7. Component replacement (Sheet 49)**



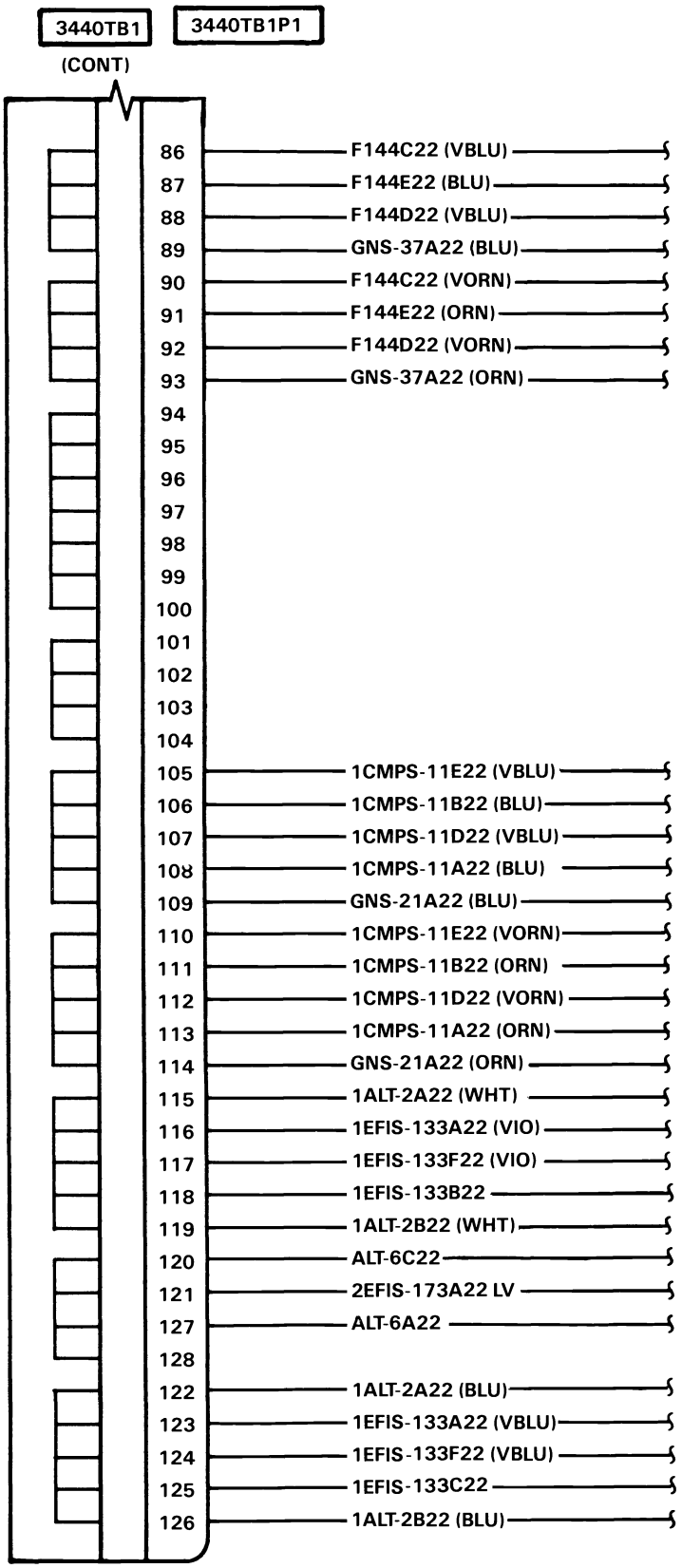

412-SAR-98-7-50

**Figure 98-7. Component replacement (Sheet 50)**



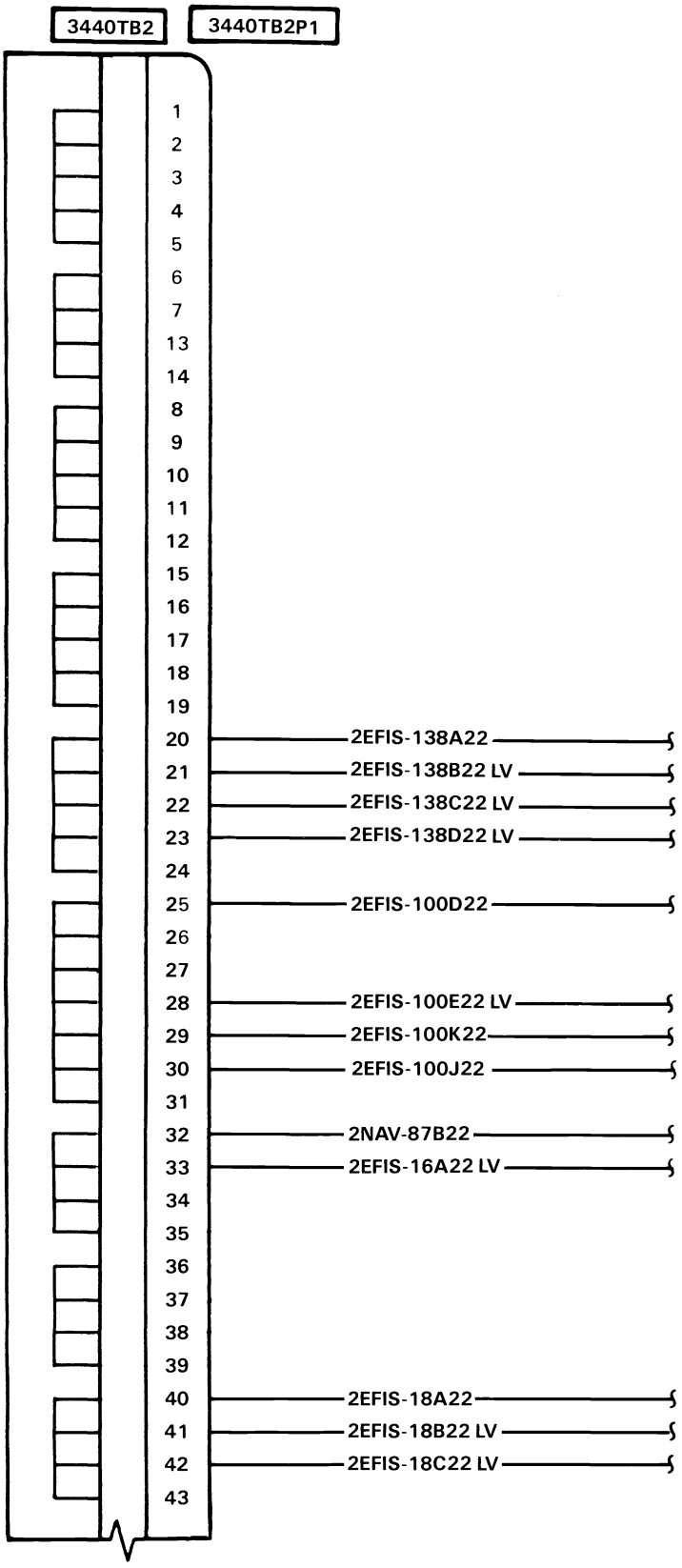

412-SAR-98-7-51

**Figure 98-7. Component replacement (Sheet 51)**



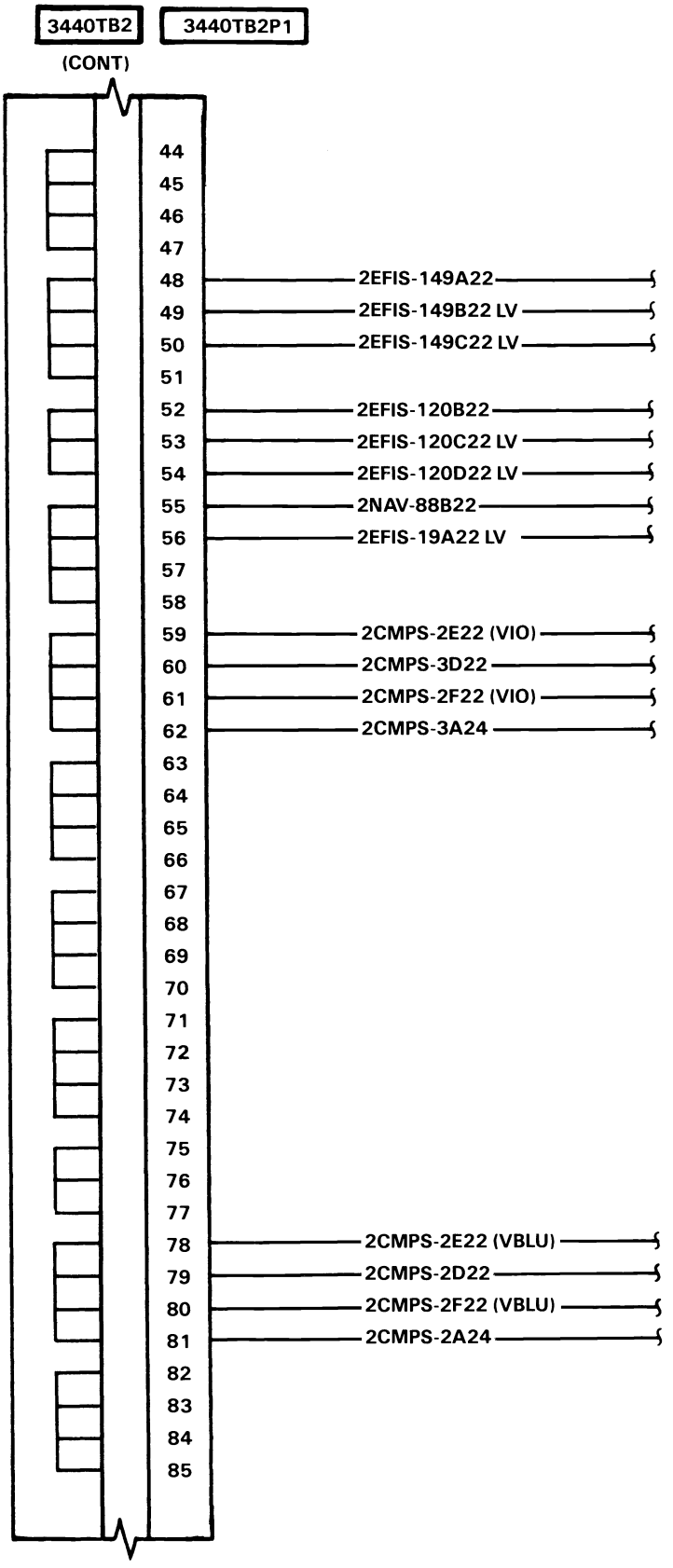

412-SAR-98-7-52

**Figure 98-7. Component replacement (Sheet 52)**



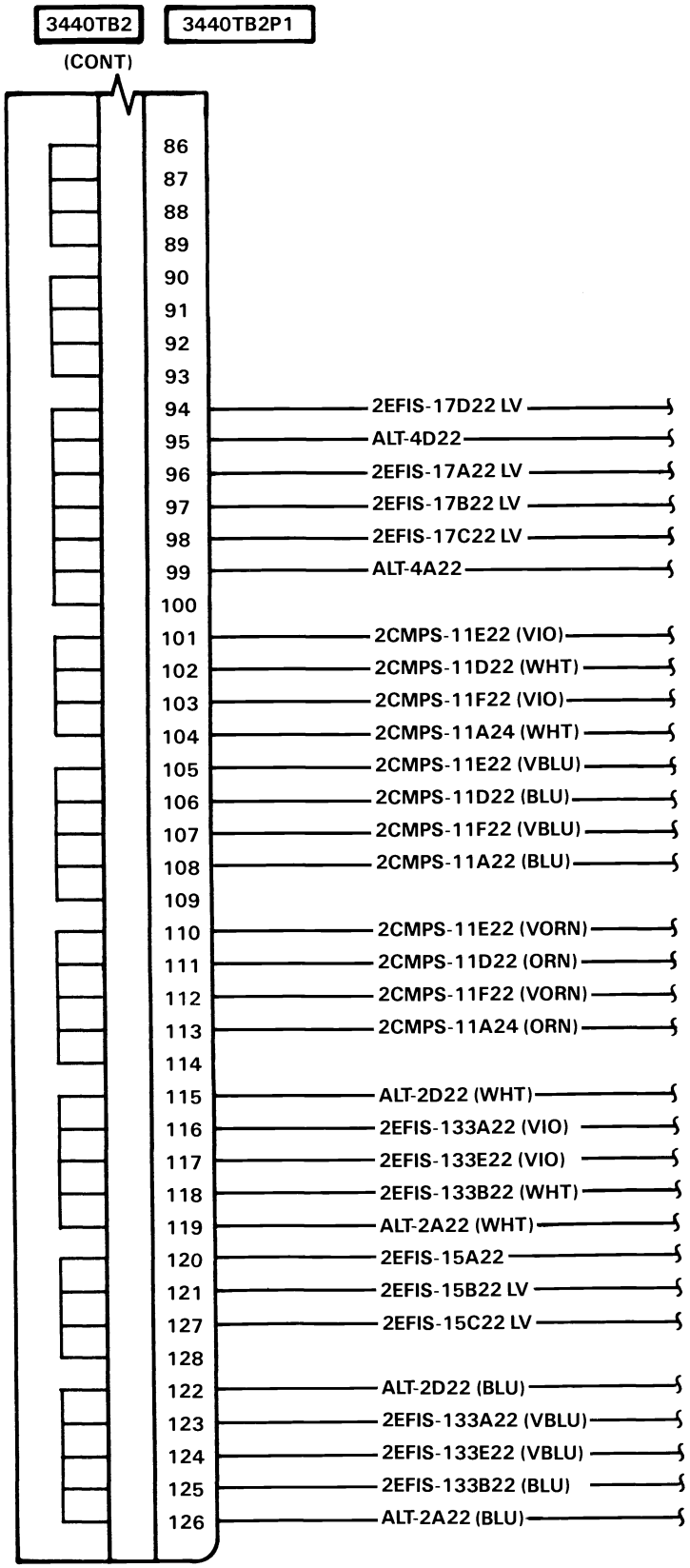

412-SAR-98-7-53

**Figure 98-7. Component replacement (Sheet 53)**



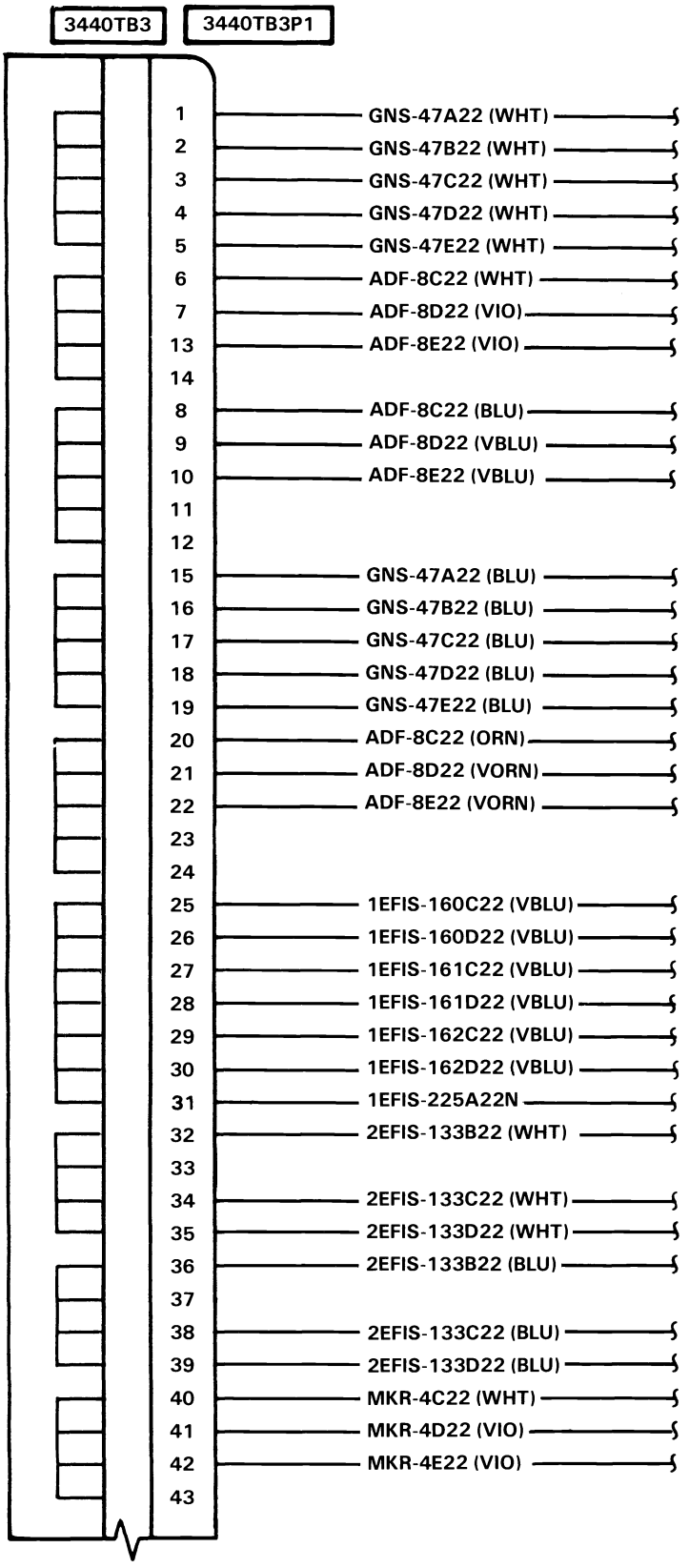

412-SAR-98-7-54

**Figure 98-7. Component replacement (Sheet 54)**



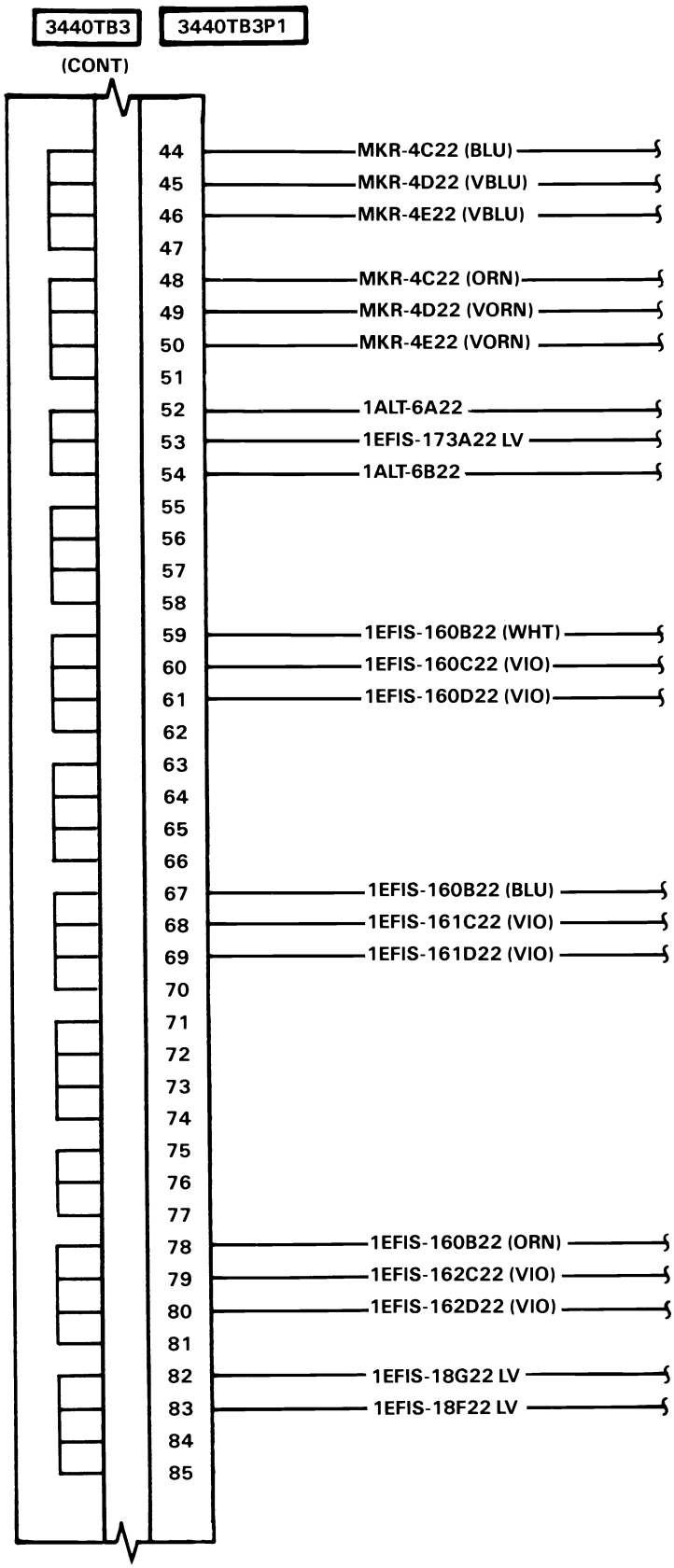

412-SAR-98-7-55

**Figure 98-7. Component replacement (Sheet 55)**



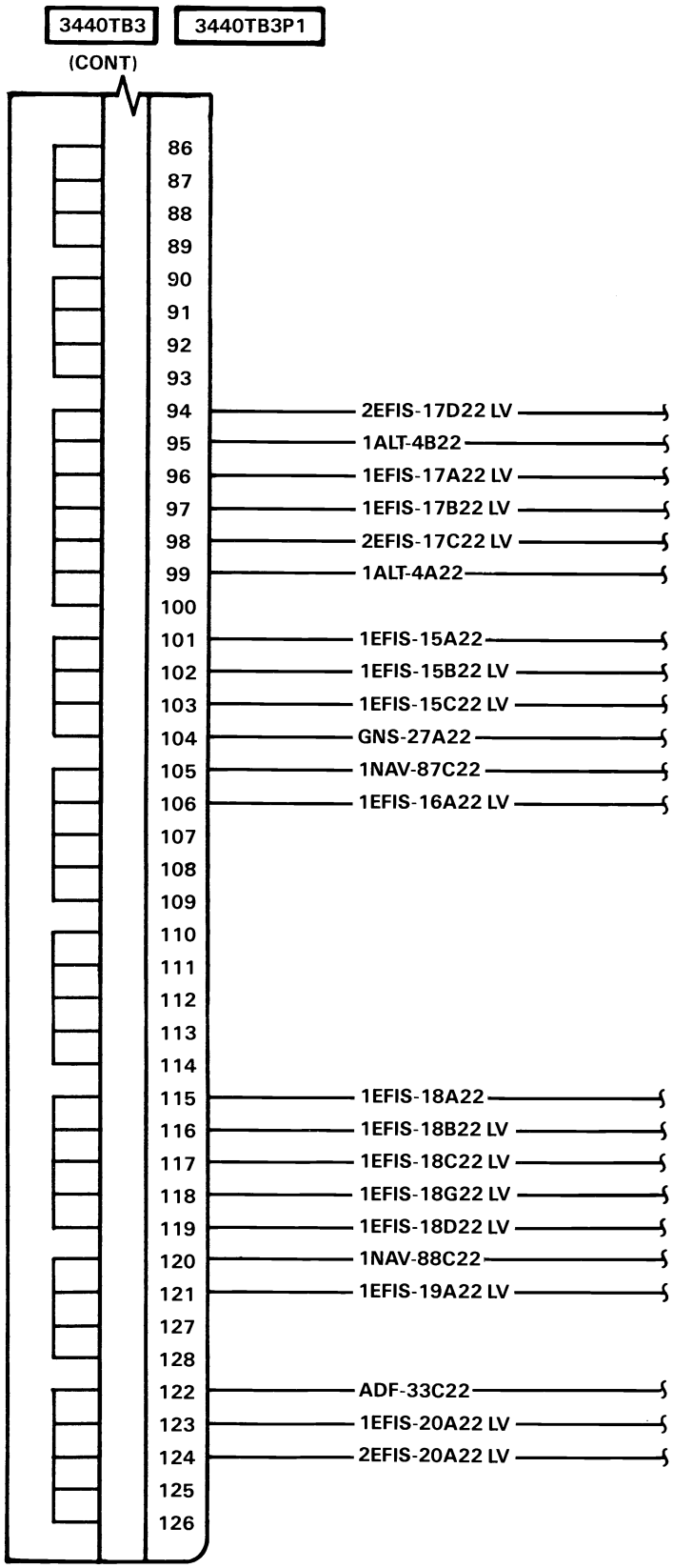

412-SAR-98-7-56

**Figure 98-7. Component replacement (Sheet 56)**



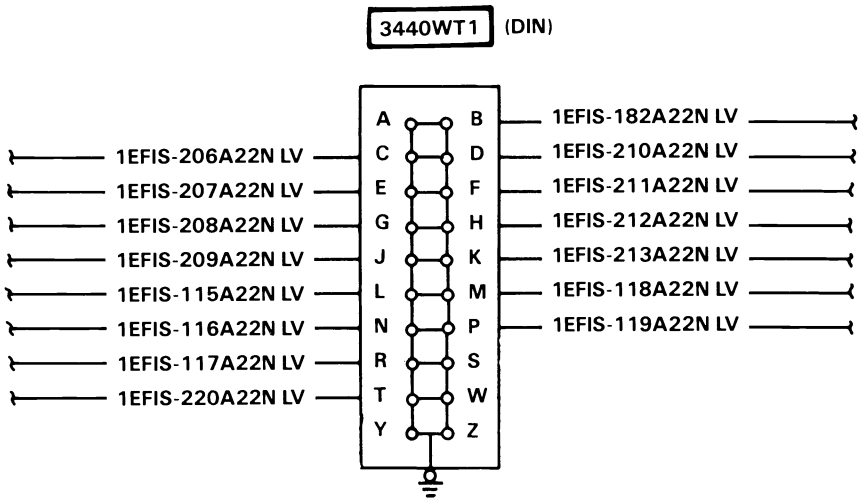

412-SAR-98-7-57

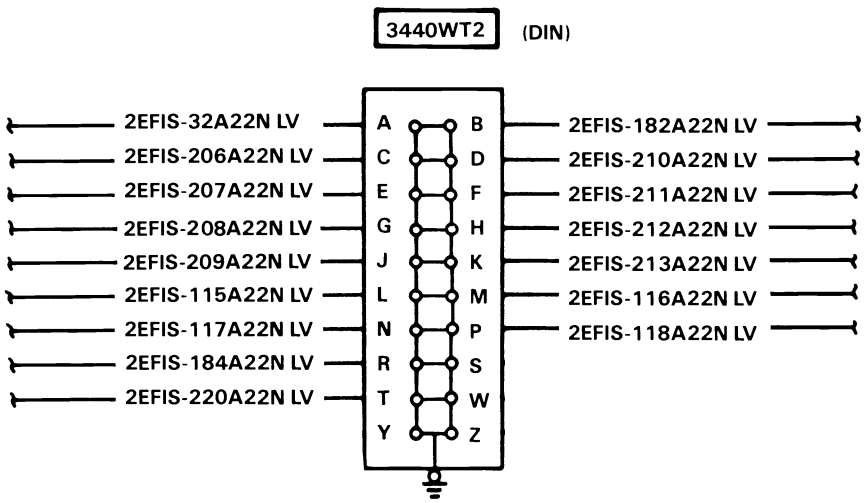
**Figure 98-7. Component replacement (Sheet 57)**




412-SAR-98-7-58

**Figure 98-7. Component replacement (Sheet 58)**




412-SAR-98-7-59

**Figure 98-7. Component replacement (Sheet 59)**